

System Overview

SW70-500

System Overview

**SW70-500
Release 520**

7/96

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The following standard document conventions may be used in this document. Check Section 1 for conventions unique to this document.

Item	Convention	Meaning
<KEYCAPS>	Pointed brackets around a term	A key on a computer keyboard
[TARGET]	Square brackets around a term	For touch screen monitors, an area that can be touched to invoke a system action
entry data	Courier bold text	Data you enter into the system
screen/program data	Courier text	Screen or program data

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System Overview

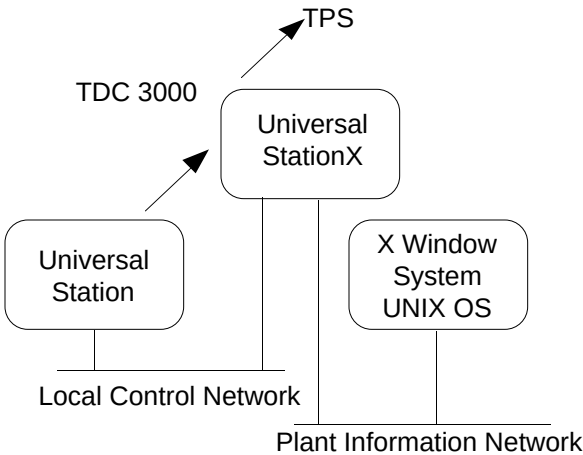
Section 1—Introduction

1.1 About This Document

Basics

Purpose	Provides an overview of the architecture, features, functions, and benefits of the TotalPlant Solution (TPS) System.
Intended audience	All users and prospective users of the TPS System
For product release	TPS R520

Product background



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For many years and for thousands of users Honeywell’s TDC 3000 control system has been synonymous with advanced data acquisition and control. Today Honeywell has raised the power of the TDC 3000 system to a new level—to TDC 3000^X, now known as TPS.

Concurrently with TDC 3000^X, Honeywell had introduced a new operator station known as Universal Station^X. As the preceding figure shows, the Universal Station^X is capable of a dual role. It performs all functions of a regular Universal Station, plus it interconnects over a standard communication network with computers and other devices operating in MIT’s X Window system environment.

Section focus

The following are brief descriptions of each of the sections in this document:

- 1— Introduction, page 1—Introduction to this document.

Introduction

1.1 About This Document

- 2— System Characteristics, page 3—Gives a brief overview of the system's architecture, its major benefits, and a brief description of available special software packages.
- 3— System Dependability, page 31—Describes the factors that help to ensure system availability for performance of the tasks for which it was designed.
- 4— Data Acquisition and Control, page 45—Describes what functional components of TPS are available to move the user towards optimized process control.
- 5— Human Interface, page 91—Describes the Universal Station, Micro TDC 3000, Universal StationX, and the Universal Work Station, and introduces the software functions that govern system engineering, system operation, and system maintenance.
- 6— System Operation, page 113—Describes the human interface functions that allow the user to efficiently monitor and manipulate the process using the TPS system.
- 7— System Implementation, page 155—Describes what system building functions the engineer can use to define the process to the TPS system.
- 8— System Maintenance, page 173—Describes what system services and troubleshooting functions are available to the maintenance technician.

Section 2—System Characteristics

2.1 TPS System Architecture

Overview

TotalPlant Solution (TPS) is a data acquisition and control system that can be tailored to meet your specific user requirements. It can be a small system with just a handful of devices, or it may be a highly complex system with hundreds of devices and several kinds of communications networks interconnecting these devices. It can begin as a small system and be easily expanded whenever the need arises.

The backbone of every TPS System is a communications network known as the Local Control Network. The LCN directly links those devices that provide the following functions:

- Human interface
- Interface to UNIX/X Window devices
- File server, history storage, and analysis
- Extensive advanced control
- Interface between multiple LCNs
- Interface to computers
- Interface to process networks
- Interface to UCN distributed control

TPS LCN architecture

The TPS LCN architecture is illustrated in the following figure.

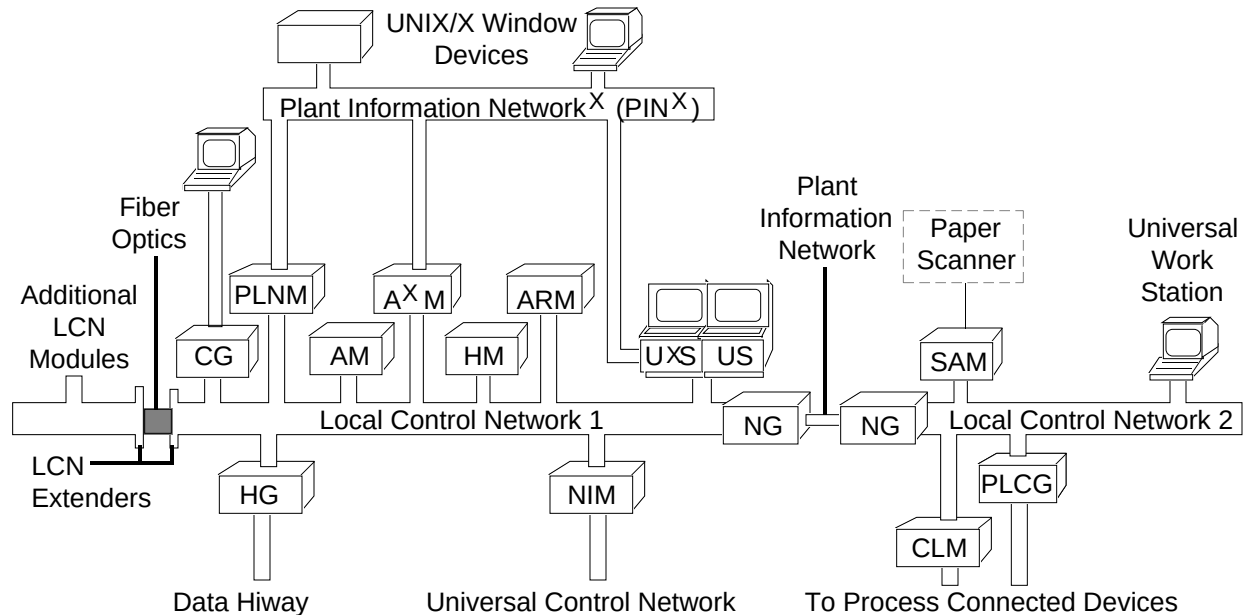
The devices associated with each function are described immediately following the figures:

- Human interfaces—Universal Station (US) and Universal Station^X (US^X)
- Interface to UNIX/X Window devices—Plant Network Module (PLNM, including CM50S and CM50N), Application Module^X (AM^X) or Universal Station^X (US^X). Any of these devices are connected up to the UNIX/X Window devices through a PIN^X cable.
- File server, history storage, and analysis—History Module (HM) and Archive Replay Module (ARM).
- Extensive advanced control—Application Module (AM) and Application Module^X (AM^X)
- Interface between LCNs—Network Gateways (NGs)
- Interface to computers—Computer Gateway (CG)

System Characteristics

2.1 TPS System Architecture

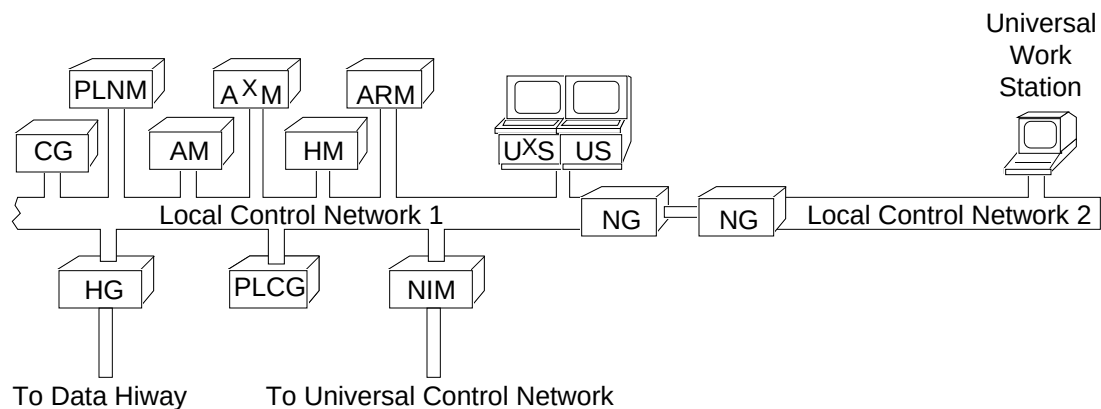
- Interface to process networks—Network Interface Module (NIM), Hiway Gateway (HG), Communication Link Module (CLM), Programmable Logic Controller Gateway (PLCG) and Enhanced Programmable Logic Controller Gateway (EPLCG)
- Interface to UCN distributed control—Process Manager (PM), Advanced Process Manager (APM), High-performance Process Manager (HPM), Logic Manager (LM), and Safety Manager (SM).



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LCN modules

The devices connected directly to the Local Control Network are called LCN “modules.” Following is a discussion of the standard modules, which are grouped according to their function listed previously.



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2.2 Human Interface Devices

Universal Station The US is the primary TPS human interface. It provides a single window to the entire system, whether the data is resident in one of the LCN modules or in one of the devices on the UCN or hiway. A single station can be used by an *operator*, by the *process engineer*, and by the *maintenance technician* to accomplish the different set of tasks for which each is responsible.

System and process *operation* functions include the following:

- Monitoring and manipulating continuous and discontinuous operations.
- Annunciating and handling alarms.
- Displaying and printing trends, logs, journals, and reports.
- Monitoring and controlling system status and diagnostics.

Process *engineering* functions include the following:

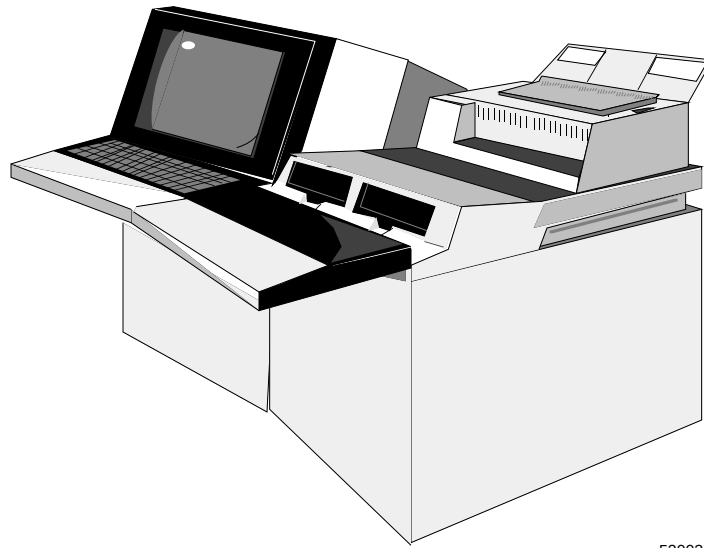
- Network Configuration.
- Building the process database.
- Building custom graphics.
- Designing reports.
- Preparing Control Language programs.

Maintenance functions include the following:

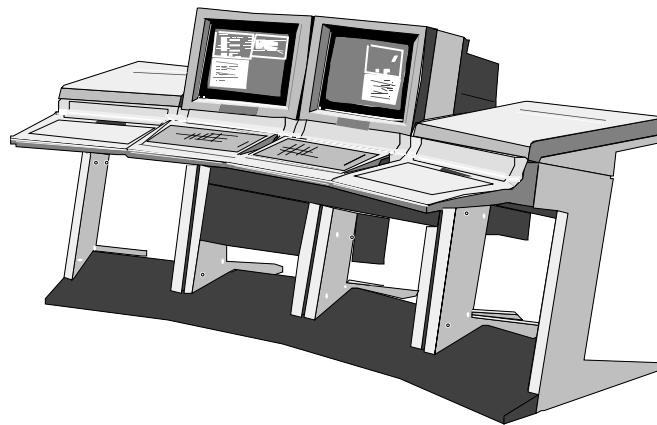
- Diagnosing system problems.
- Displaying and printing information required during troubleshooting.

System Characteristics

2.2 Human Interface Devices



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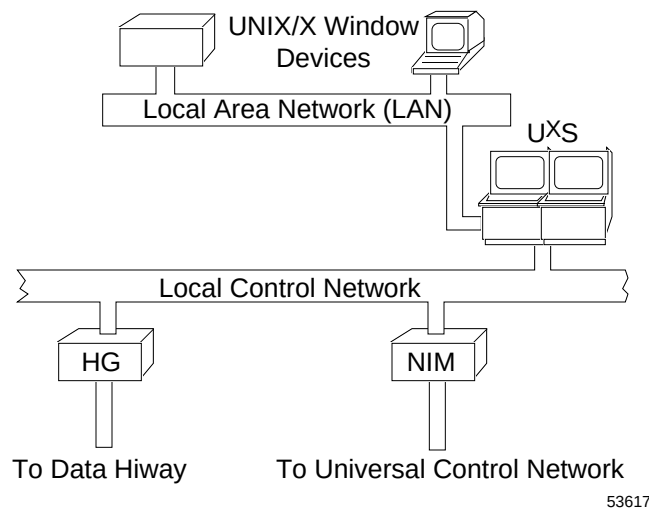
Universal Station^X The Universal Station^X is a Universal Station that is augmented with an industry-standard coprocessor. Universal Station^X actually serves two basic functions—as a human interface to the process and as an interface to UNIX-based devices that exist on a user's plant-wide communications network.

The result is a station that, in addition to performing all functions of the standard US, assumes the functionality of a workstation operating under UNIX in the X Windows software environment.

Using improved performance and higher resolution CRTs, Universal Station^X is capable of

- concurrent presentation of the LCN window along with windows displaying external data from multiple X Window-compliant devices,
- resizing and moving the windows in accordance with Motif Window Manager, and
- performing other special operations such as magnifying a selected area of a reduced LCN display.

In the event of UNIX/X Windows failure, Universal Station^X is designed to revert to standard TPS operation.

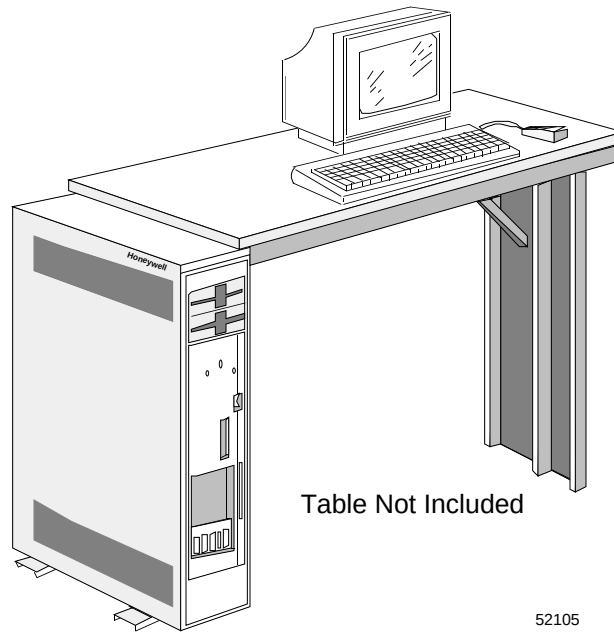


Universal Work Station

Another alternate human interface module, the UWS, is also available for process engineering and maintenance functions, and for monitoring by the process supervisor. It is capable of virtually all of the same functions as the Universal Station, but is designed for an office environment.

System Characteristics

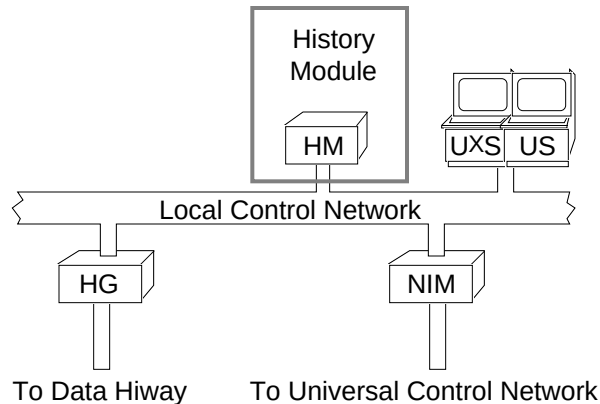
2.2 Human Interface Devices



2.3 Mass Data Storage and Analysis

History Module

The HM, available with redundant drives and different storage capacities, makes possible storage of, and quick access to, large quantities of data:



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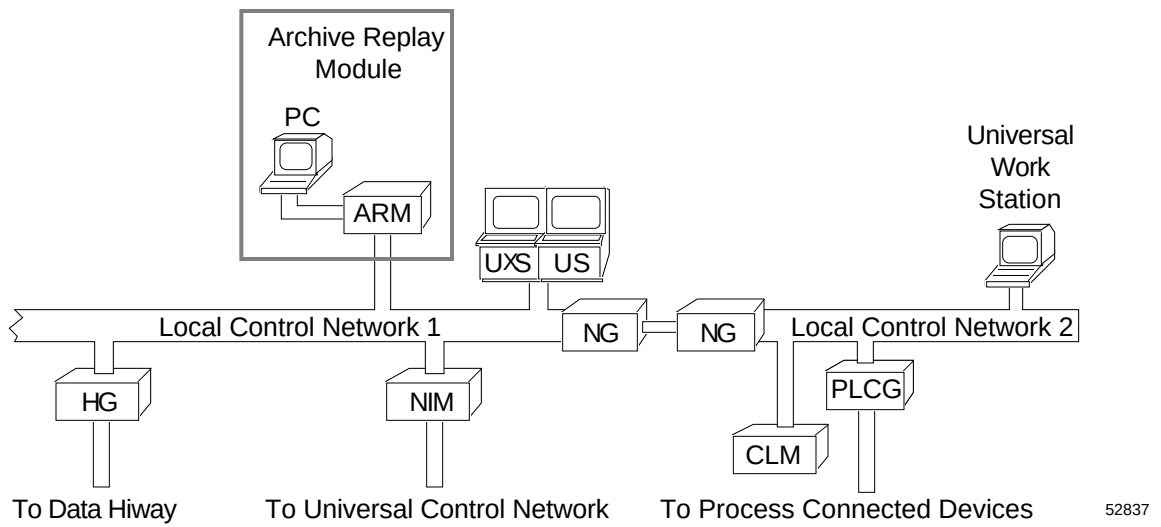
- History of process alarms, operator changes, operator messages, system status changes, system errors, and system maintenance recommendations; also continuous process history to support logs and trends.
- System Files of all types, load images, and other data required anytime modules are reloaded or personalities are changed.
- Checkpoint data for maintaining up-to-date device settings in the event a device is taken out of service.
- On-process maintenance information and analysis.

Archive Replay Module

The Archive Replay Module (ARM) integrates a personal computer into the LCN for the specific purpose of collecting and analyzing large quantities of historical data. Optical disks serve as the mass data storage devices. The historical data can be retrieved at any time and, depending on the type of data, displayed as a graphic trend or formatted for analysis by the ARM or by third party personal computer software packages.

System Characteristics

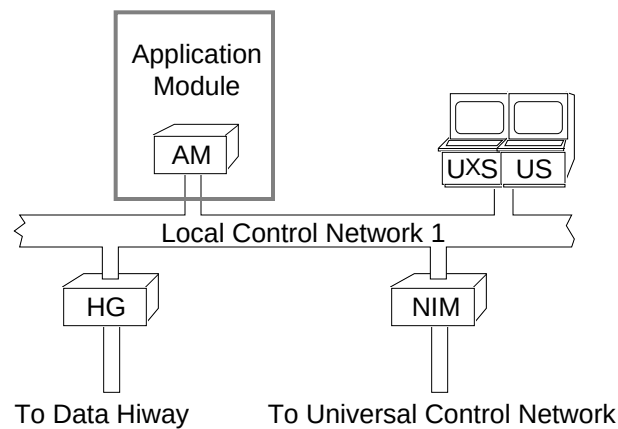
2.3 Mass Data Storage and Analysis



2.4 Extensive Advanced Control

Application Module

The AM permits the implementation of certain complex control calculations and strategies that may not be practical or possible when using only process-connected devices. A set of standard advanced control algorithms is included, and custom algorithms and packages can be developed by using a process engineer-oriented Control Language (CL/AM). A number of standard software control packages for specific applications are available for the AM, a few of which are briefly described later in this section.



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For calculations and data handling that is even beyond the capability of the Application Module, see the discussion of computer interfaces that follows.

A^XM

Honeywell's Application Module^X (A^XM) is a high-performance, high-integrity application platform that can host advanced control, optimization, plant history and information systems, expert systems, and general purpose applications. The A^XM includes embedded UNIX workstation technology tightly integrated with the process control technology of the Application Module (AM) which provides an open environment for these types of applications on the TPS. Open Data Definition and Access (OpenDDA) is the tool used for building A^XM applications that need access to real-time data. The key advantages of A^XM and OpenDDA 200 are Application Investment Protection, Flexible Data Access, High Performance, and a Robust Application Platform. Existing Application Module (AM) functions—point processing, broad Proportional, Integral, Derivative (PID) algorithm handling, and Control Language (CL) capabilities—are expanded in the Application Module^X with the addition of a RISC-based UNIX coprocessor that can accommodate a wide range of software applications from a variety of suppliers.

OpenDDA

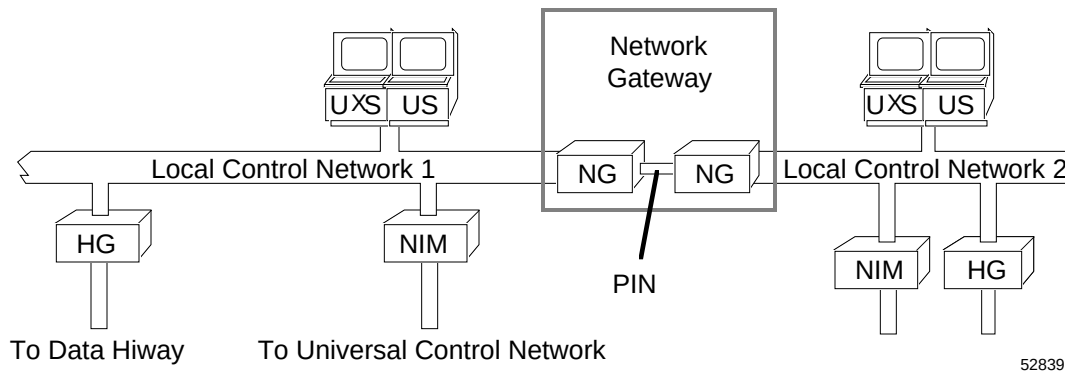
OpenDDA provides an application access to TPS data (both read and write access) in a user-friendly format. Programming to the OpenDDA interface will migrate your application source code to Honeywell's future platform. OpenDDA applications will not be able to see the difference if the data comes from TPS or from some other data source in the future (such as the next generation Honeywell system or some competitors DCS). This means applications are protected from changes in the data source.

Because applications are a major source of investment today, keeping that investment protected is key to Honeywell and its customers.

The Application Module^X dual hardware interface allows it to serve as a module on the TPS Local Control Network (LCN) and a node on the ethernet Plant Information Network (PIN). Accordingly, it communicates with other modules on single or multiple LCNs; with process-connected devices on Universal Control Networks and Data Hiways; and with PCs, workstations and other devices that reside on the PIN and communicate using industry standard network communications.

2.5 Interface Between Multiple LCNs

Network Gateway The NG connects the LCN to a Plant Information Network (PIN), using fiber optic cable or carrier band technology on coaxial.

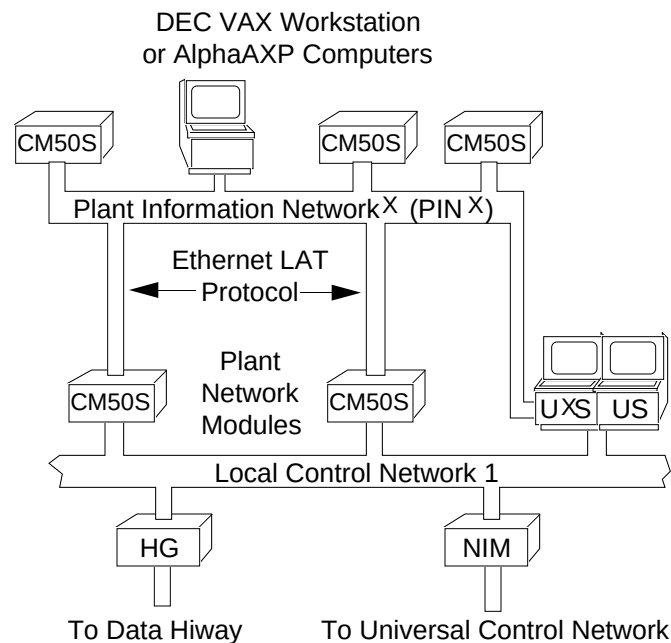


Plant information network By connecting two or more LCNs with a common PIN, a single TPS System can be extended to include many control rooms within a large complex, essentially networking hundreds of nodes. Data can be easily and securely exchanged across systems. This provides a single window to the entire plant.

2.6 Interfaces to Computers

Plant Network Module

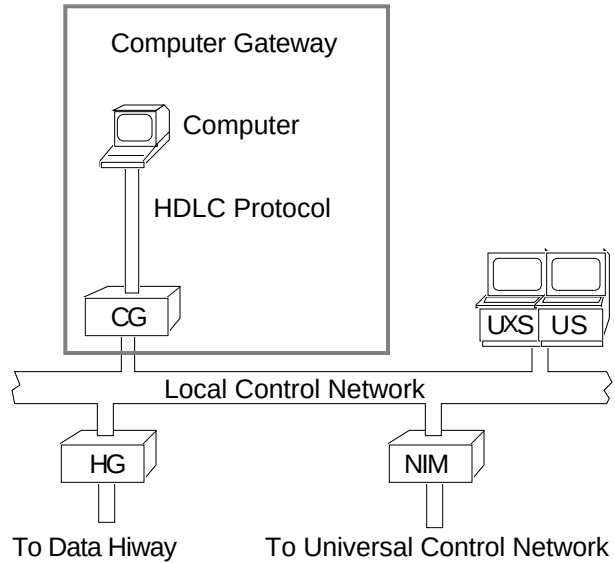
The PLNM provides an interface between the LCN and DEC VAX or Alpha AXP computers. Connection of computer systems makes it possible for computer-level functions to be integrated into the TPS System. The VAX provides an environment where the user can develop, debug, and execute programs to perform such functions as running complex calculations, process optimization, advanced custom report generation, long-term data storage, scheduling, and plant management. Because the computer is completely integrated into the TPS System, no data conversion software needs to be written. The CM50S software package is available for use with the PLNM. It uses an Ethernet LAT communication channel between a VAX or Alpha AXP computers and up to 4 PLNMs.



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Computer Gateway

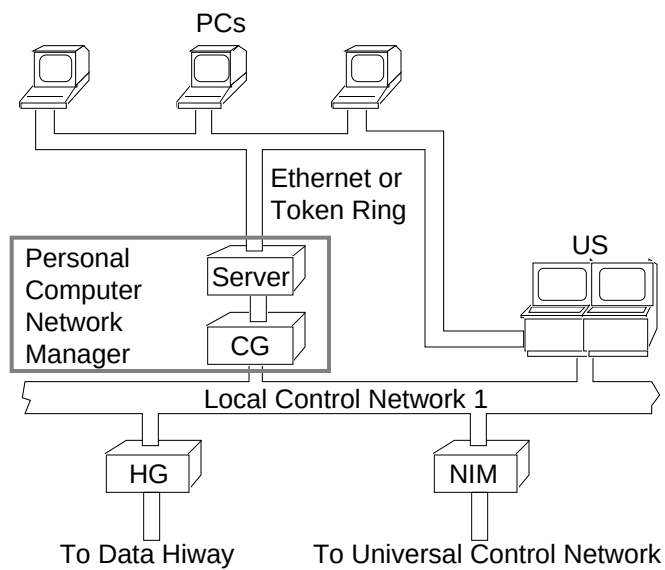
The CG provides a standard communication link between the LCN and host computers that use HDLC protocol. Through the CG, computers from other third party vendors can perform the same kinds of functions described above for DEC computers.



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Personal Computer Network Manager

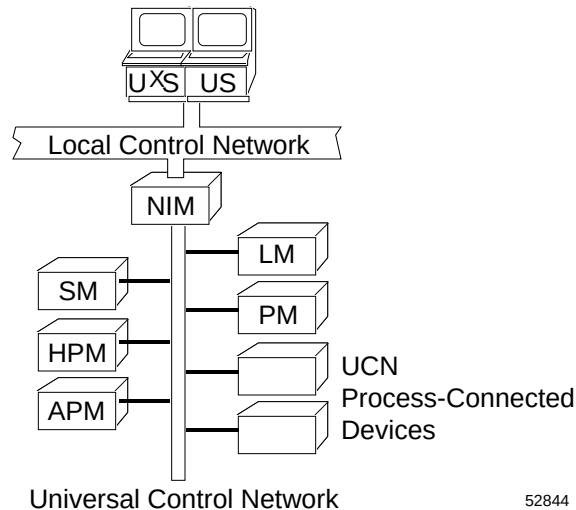
The PCNM consists of hardware and software that serves as a secure interface to multiple personal computers over an Ethernet or Token Ring Local Area Network. Users on the network have real-time access to LCN data for viewing in graphic displays, on spreadsheets, or in other third party applications.



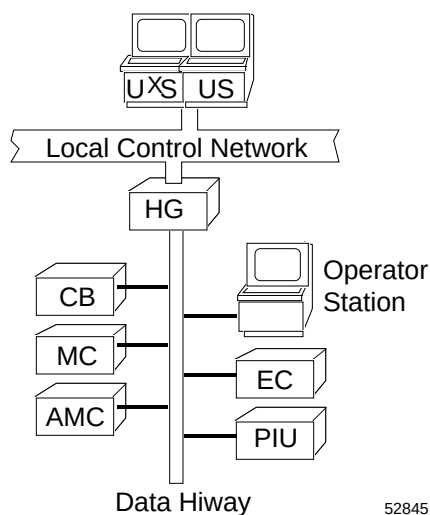
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2.7 Interfaces to Process Networks

Network Interface Module The NIM interconnects the LCN with the Universal Control Network. The UCN integrates Honeywell's most advanced process-connected devices into TPS. These products are discussed in the following section.

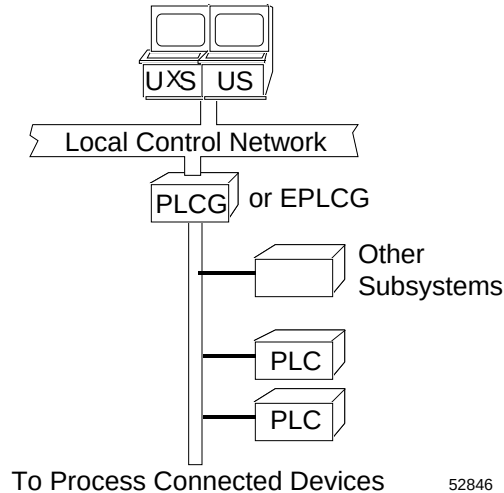


Hiway Gateway The HG interconnects Honeywell's Data Hiway with the LCN, making the transition from the transmission technique and protocol of the LCN to the transmission technique and protocol of the hiway. It also provides other functions such as polling, alarm scanning, and time synchronization for hiway-based devices. The devices that can reside on the Data Hiway are discussed following the UCN device discussion.



Programmable Logic Controller Gateways

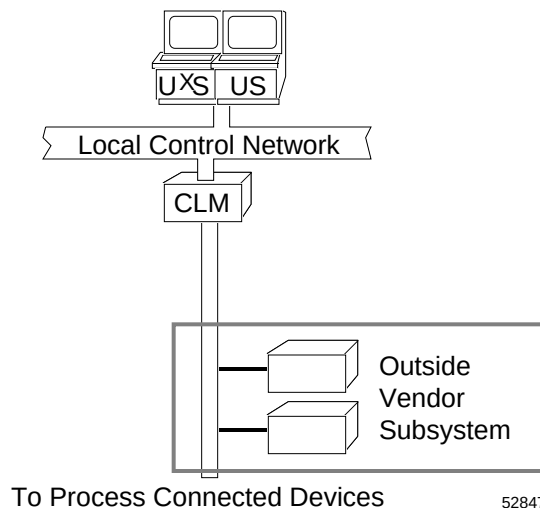
The PLC Gateway and Enhanced PLC Gateway provide an efficient LCN interface for non-Honeywell programmable controllers. They do so by supporting the de facto industry subsystem protocols of Allen-Bradley and Modbus.



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Communications Link Module

The Communications Link Module provides the Universal Station operator a window of access to points located on a non-TPS network. Designed to reduce the number of different windows to the process, the CLM interfaces to remote subsystems in a transparent manner that simulates their direct connection to the Local Control Network.

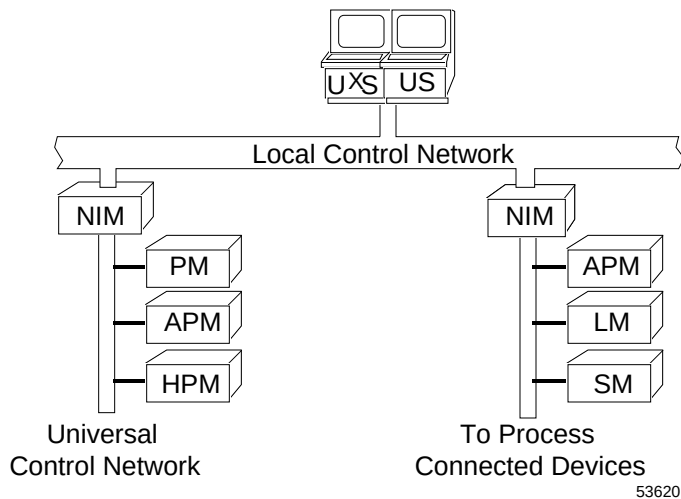


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2.8 UCN-Based Distributed Control

Universal Control Network

Designed to be compatible with IEEE and ISO standards, the UCN is a high-speed, high-security process network. Its peer-to-peer communication capability allows for easy sharing of process data, making it convenient to implement sophisticated control schemes involving more than one Process Manager, Advanced Process Manager, High-Performance Process Manager, Logic Manager, Safety Manager, or any combination of these devices.

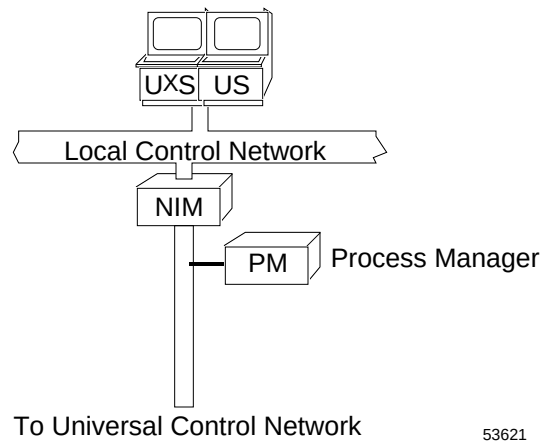


Process Manager

The Process Manager (PM) is a highly flexible data acquisition and control device that provides a complete range of capabilities. It consists of a powerful control processor that performs regulatory, logic, and sequencing functions, coupled with up to 40 selectable intelligent I/O processors.

In addition, separate communication and I/O interface processors assure high-speed flow of information both internal and external to the PM. The number and types of control functions to be implemented, along with the processing rate, are configurable by the user. It is fully integrated with smart transmitters and supports bidirectional digital communication. It is also digitally interfaced to Honeywell's UDC 6000 single-loop, high speed controller.

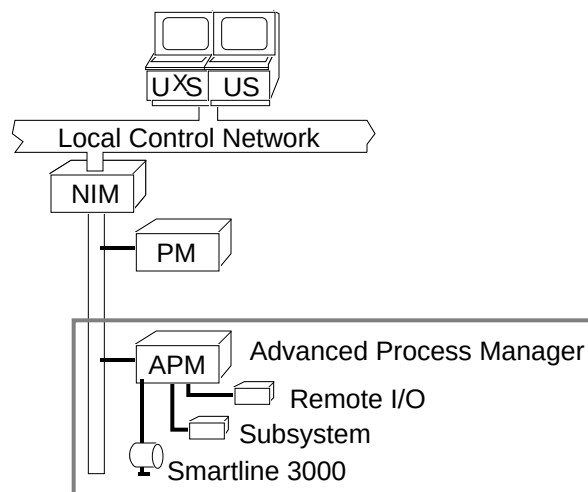
A process-oriented programming language is available for implementation of batch and hybrid applications as well as the computational capability needed for some continuous control tasks.



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Advanced Process Manager

The Advanced Process Manager (APM) adds new functions and capacity to those available with the Process Manager. It offers increased integration of discrete control with the device control point. It also supports flexible subsystem integration with the array point and serial I/O processor. There are additional I/O Processors, and new types of control points. The programming language sequence statement capacity and flexibility are increased with support of times and strings. Also there are increased maximums for flag and numeric points over the PM.



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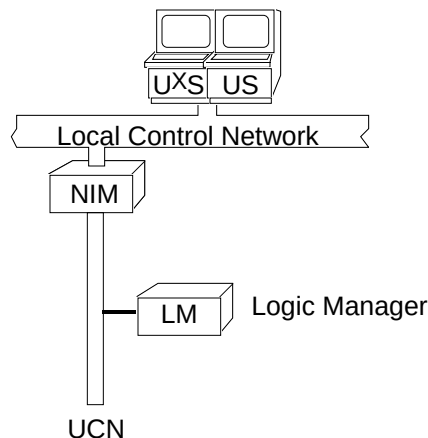
High-Performance Process Manager

The High-Performance Process Manager (HPM) represents an evolution of the Process Manager (PM) and Advanced Process Manager (APM). Its dual 68040 platform offers a wide range of capabilities for present and future process requirements. HPM control performance is significantly greater, with five times the processing power of PM and AMP. The HPM offers flexible I/O functions for both data monitoring and control. Control functions, including regulatory, logic, and sequencing control are provided for continuous, batch, sequencing, and hybrid applications.

A toolbox of functions can be configured and programmed to meet the needs of data acquisition and advanced control requirements in a highly secure and performance-intensive manner. HPMs capabilities include peer-to-peer communications and compatibility with industry-standard communications protocols.

Logic Manager

The Logic Manager (LM) integrates the power of high-speed logic functions into the architecture of the Universal Control Network. Its Control Processor is programmed using ladder logic, which is continuously and rapidly executed as long as the Control Processor is in the RUN mode. A Logic Manager Module provides the interface for the rest of the TPS System to access a variety of data point types, and also provides data to be processed by the ladder logic program. Both serial and parallel I/O systems are available, and each can accommodate a wide variety of I/O modules.



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Safety Manager

Safety Manager (SM) describes a family of nodes on the UCN that consist of a Safety Manager Module (SMM) interface card and a Safety Manager that provides the safety functions. There are two Safety Managers available. The R500 Safety Manager is for the Triconex TRICON Version 8. For R510, Honeywell offers the Safety Manager for the Fail Safe Control (FSC) system.

Safety Manager for TRICON Version 8

The R500 Safety Manager (SM) Module for Triconex TRICON Version 8 provides a Triple Modular Redundant (TMR) fault-tolerant controller for safety and shutdown applications on the TPS Universal Control Network (UCN). The R500 Safety Manager (TRICON SM) consists of a Honeywell R500 Safety Manager Module (SMM) and a Triconex TRICON Version 8 controller.

The R500 Safety Manager adds TMR logic functions to the architecture of the Universal Control Network to support integrated operations and control. This gives a TPS-based window into this safety system that has superior uptime and safety-related performance for applications such as:

- Emergency Safety Shutdown (ESD) in refineries, petrochemical/chemical plants, and other industrial processes
- Boiler Flame Safety
- Turbine Control Systems
- Offshore Fire & Gas Protection

Safety Manager-Fail Safe Control system

The R510 Safety Manager (SM) Module for the Honeywell Fail Safe Control (FSC) System, referred to as SM-FSC, is a microprocessor-based safety system that performs safety-related functions in process control applications. It provides a seamless integration of the FSC as a new UCN node with the TPS System. The R510 Safety Manager consists of a redundant Fail Safe Control (FSC) system and an R510 Safety Manager Module (SMM-FSC). Its safety design includes self-diagnostics and hardware/software component monitoring. It is approved in all configurations by UL and TUV, and is compliant with emerging international standards (IEC1508).

The connection of the FSC to the UCN via the R510 SMM offers notable advantages to Honeywell customers, including:

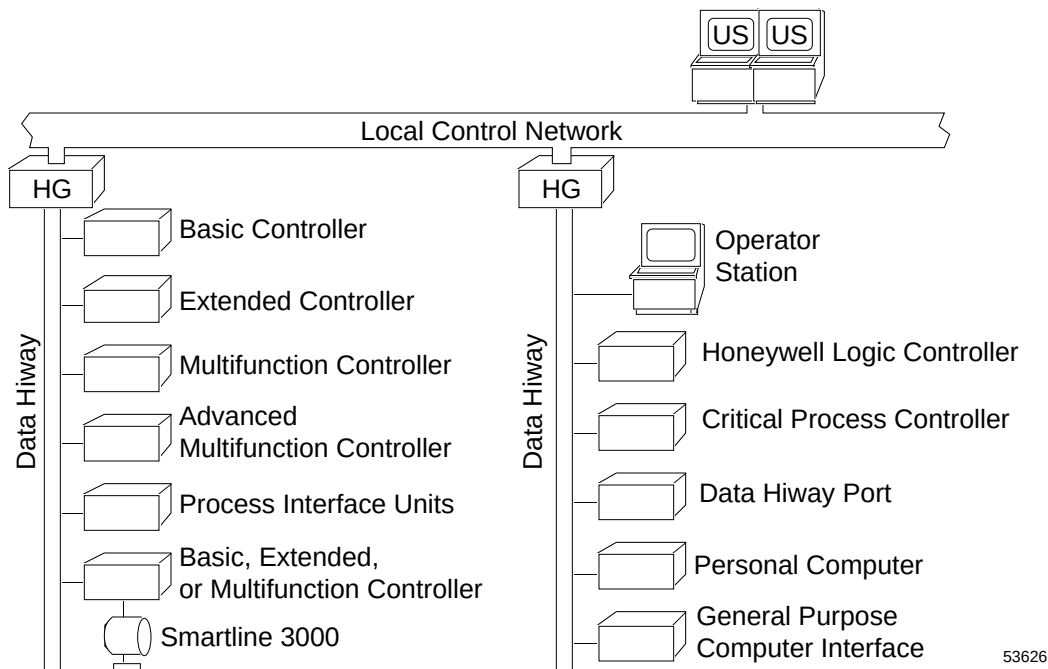
- Peer-to-peer communication on the UCN for shorter response time
- Time synchronization for simplified “first out” capabilities of the two systems
- FSC sequence of events logging and display at LCN level (HM, UXS, printer)
- Events Initiated Processing (EIP)
- FSC diagnostic displayed on UXS improving the MTTR
- FSC write protection to ensure high safety integrity level
- UL and TUV-approved integration
- Reduced interfacing hardware
- Reduced configuration time

2.9 Hiway-Based Distributed Control

Data Hiway

Honeywell pioneered industry's transition to distributed digital process control systems in 1975 with the introduction of TDC 2000 and the first Data Hiway-based process devices or boxes. During subsequent years, this network was expanded with many additional devices that provided new and more sophisticated functions. In 1984, this entire complement of TDC 2000 products was made an integral part of TDC 3000^X. In this way, all TDC 2000 users are enabled to take advantage of the advanced TDC 3000^X control capabilities without obsoleting their existing control systems. In 1996 TDC 3000^X evolved to TPS.

These hiway-based devices that, like the newer Process Manager/Advanced Process Manager, perform the first level of process control, consist of data-acquisition units, controllers, and other interfaces to specialized types of equipment. They are described briefly following.



Data acquisition units

Process Interface Units are used when a large number of process variables need to be monitored or fed to a higher level module. They exist in high-level (HLPIU), low-level (LLPIU), and low-energy (LEPIU) versions. The levels refer to the amplitude (voltage or current) of the signals with which the PIUs interface; the low-energy PIU is configured with mux boxes that can be distributed to different locations in the plant.

Controllers

The **Basic Controller (CB)** handles continuous I/O operations. It accommodates I/O for eight control loops, plus eight additional monitor-only inputs ("Remote Variables").

The **Extended Controller (EC)** handles continuous I/O and logical operations. It provides 16 control inputs, 8 outputs, 16 digital (status) outputs, and 16 optional digital (status) inputs.

The **Multifunction Controller (MC)** and Advanced Multifunction Controller (A-MC) handle continuous I/O Operation, logical operations, and sequential control for batch operations. Each has the capacity to control 16 analog loops, and the capability of handling up to 32 analog inputs, up to 72 analog outputs, up to 256 digital inputs, and up to 64 counter inputs. The primary differences between the two are packaging (the A-MC takes less space) and slot processing rate, which is once per second for the MC and twice per second for the A-MC.

All of the above controller types can be equipped with a digital interface to Honeywell's **ST3000 family of "smart transmitters" (Smartline 3000)**, making it possible for the controllers to take advantage of the advanced capabilities of these devices.

The Honeywell **620 Logic Controller system** is offered in several models that provide capabilities ranging from relay replacement to advanced function capabilities. A wide selection of I/O modules is available. Also available is a COP Industrial Microcomputer, with a number of powerful software packages to extend the usefulness of the system.

The **Triconex Fault Tolerant Controller** is a triple-redundant programmable controller, providing the ultimate in reliability for those applications where this is a primary consideration.

Other Hiway Devices

A **Data Hiway Port (DHP)** provides the vehicle for interfacing numerous types of third party process boxes with the Data Hiway. Included in this list of devices are programmable controllers, analyzers, tank gauging systems, machinery monitoring systems, emergency shutdown systems, data acquisition systems, and compressor control systems.

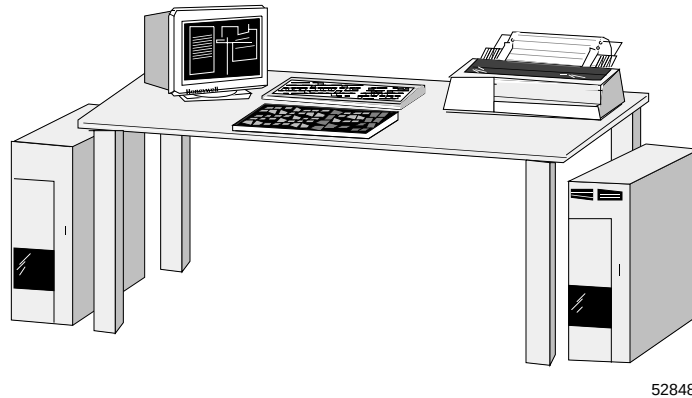
A **Personal Computer Serial Interface (PCSI)** provides an interface to any personal or minicomputer with an RS-232 C serial port.

A **General Purpose Computer Interface (GPCI)** serves as an interface to computers that have a general purpose I/O adapter.

2.10 Micro TDC 3000

Minimum-sized control system

For users with processes that require only a minimum-size control system, Honeywell offers Micro TDC 3000. One configuration of the Micro TDC 3000 Control system is illustrated here.



Compact but fully functional

Although Micro TDC 3000 is extremely compact, and only limited module options are supported, it is a fully functional control system in the Honeywell TPS family. This system communicates to the process through a Process Manager over a Universal Control Network.

2.11 System Design Benefits

Distributed Architecture, Integrated Functions

The TPS design philosophy of full distribution of architecture and full integration of functions makes possible important benefits to the user. Honeywell has taken advantage of these opportunities, to provide expanded capabilities beyond what is available with any other system.

Incremental Expandability

The functional specialization of the physical modules allows the purchaser to start with those devices, and only those devices, that match initial requirements and available funds. When the time comes to expand, as the result of either an increase in process capacity or of functional requirements, this can be accomplished by the simple addition of hardware modules, without significant impact on the existing system; that is, no reprogramming of software is necessary.

Anyone familiar with computer-based control systems will appreciate the savings in time and cost inherent in the above design approach. Furthermore, the range of expandability is extensive. An entry configuration might consist of one or two process-connected devices on a Universal Control Network, a redundant Network Interface Module, and two or three Universal Stations. An expanded system can include hundreds of process-connected devices, as many as 64 LCN modules per LCN, multiple LCNs, and can include numerous Universal Stations/Universal Stations^X/Universal Work Stations, History Modules, Archive Replay Modules, Application Modules, and DEC VAX or other computers. A VAX can itself be networked with other VAXes or another local area network.

Network File Management System

For large systems, with more than one process or application engineer, having frequent need for simultaneous access to the system in order to build and edit files, the on-line network file-management capability is invaluable.

For example, one engineer seated at a Universal Work Station in an office can design and implement a new control strategy in Control Language on an Application Module, while another engineer at a different Universal Work Station, or Universal Station, can be updating a graphic display for use by the operator. At the same time, a maintenance technician might be accessing data needed to analyze a problem that has been logged by the system. Of course, while all of this is taking place, the process is under control and the operators are performing their normal duties at the Operator Console.

Distributed Risk In a control system with centralized architecture, failure of a component can immediately bring down the entire system. With the distributed architecture of TPS, it is likely that only the device containing the failed component would be lost, leaving all other system devices to continue to perform their functions. Loss of system performance is therefore minimized, depending on the criticality of the failed device, and whether or not it is redundant. This feature, called “graceful degradation,” was introduced with Honeywell’s earliest Data Hiway-based systems.

2.12 Special Software Packages

Software for specific requirements

Several software packages are available to assist users in meeting specific requirements. A few of these are briefly described below. Numerous others are available for specific applications.

AM-based software

LOOPTUNE II — This is a TPS software package that optimally tunes PID control loops. It can tune control loops that are resident in Basic, Extended, Multifunction and Process Managers / Advanced Process Managers, as well as Application Modules. Almost any process that can be controlled by PID can be correctly tuned by LOOPTUNE II.

HORIZON PREDICTIVE CONTROL (HPC) — HPC is a multiple input/single output control algorithm designed primarily to control the 5% to 10% of control loops in a plant that cannot be adequately controlled with PID algorithms. It is easy to use, is automatically tuned, and no assumptions about the process dynamics are required. Special displays are included for configuration, control status, and trending.

Real-time SPQC-II — Real-Time SPQC uses statistical control methods to detect quality problems before they cause major upsets. This package provides a full range of control charting functions including X-BAR, range, s, CUSUM, HISTOGRAM, Cp, and Cpk calculations. Extensive statistical alarming is provided, and historization with data retrieval is available.

Batch Supervisor — Based on Honeywell's Modular Batch Automation methodology, Batch Supervisor greatly simplifies batch control implementation. It is easy to modify, uses standard modules, provides easy on-line modifications, and provides a comprehensive set of standard displays and reports.

Turbine Generator Management Program — This is a modular package of software and services that helps users to minimize thermal and mechanical stress on steam turbines during startup, load-changing, and shutdown. Users can meet their specific requirements by selecting from the seven available modules: Rotor Stress Indication, Turbine Supervisory Monitor, Turbine Generator Startup Sequence Monitor, Generator Supervisory Monitor, Turbine Generator Shutdown Sequence Monitor, Simulators (Startup, Load Change, Shutdown), and Turbine Generator Sequential Control.

**DEC VAX
Computer-based
software**

Event History 800 — Event History 800 is a history collection package with statistical process control capability. Collection of process information from TPS is automatic, with collection being triggered in a variety of ways. It is valuable for the retention and recall of information required by regulatory agencies, environmental agencies, and the user's customers. It can analyze the data and provide statistical displays at both the Universal Station and VAX terminals. The user interface is menu driven and subroutine interfaces for data access and manipulation are provided. Externally-generated data can be input to the database.

Plant Information System (PI) — The Plant Information system provides tools to collect, archive, and view data, and distribute it over a network for engineering and management data analysis and decision support. Using sophisticated color graphics at a PI Terminal, the user can display the processing conditions in any unit in the plant for any time frame since system installation. Analysis of data can be performed using a variety of analytical techniques. PI can be customized for specific manufacturing operations.

Application Builder and Executive (ABE) — The Application Builder and Executive is a software system that provides the tools to build on-line applications in a real-time VAX/VMS environment. ABE provides a menu driven interface, real-time scheduling of applications, automatic access of current and historical process data, integration of Fortran, modeling and optimization packages, an off-line testing environment, and data capture/playback features.

Process Modeling and Optimization — This is a software system for real-time modeling, optimization and analysis of process and plant operations. Included are regression to historical data, on-line optimization of operating parameters, custom reports, and advanced control.

TPS Expert — an advisory expert system shell that allows a user to capture operator or engineer expertise, and makes that knowledge available to less experienced personnel 24 hours a day. Applications where it is useful include those where early operator recognition of potential problems can be important, where there is a desire for more standardization, where startup and shutdown of a process unit is highly complicated, and where there are patterns for recognizing desirable and undesirable situations.

PC-based software

WORKBOOK — Customers who want to get an early start on TPS configuration activities (establishing the database for their system) can make use of Honeywell's PC-based software package, called WORKBOOK. It is menu-driven, and the user enters configuration information into TPS-like forms displayed on the Personal Computer CRT. System or user-defined default values are available on the forms. Range checks and duplicate-record checks are performed. Following entry and storage of the data into PC memory, the Text File Converter (see below) is used to generate a TPS format cartridge or floppy disk.

Text File Converter — This product translates back and forth between the two formats used by an MS-DOS PC and the TPS System. Text files in TPS can be imported to the PC for editing, and text files in the PC (including those generated by WORKBOOK described above) can in turn be translated and stored onto disks that are compatible with TPS.

**Macintosh
Computer-based
software**

Universal Station Recorder — This software tool makes it possible for a user to transfer images and reports from the Universal Station screen to an Apple Macintosh computer. Accordingly, it is ideal for generating printed material for operator training. Text files transported to Macintosh can be exported to spreadsheets or other analysis programs.

Universal Station

RULA (Remote User LCN Access) is an optional software package for a Universal Station, UXS, Universal Workstation, or MicroTDC 3000 that allows you to connect a local Station to a remote TPS System anywhere in the world, via phone lines or IEEE 802.3 standard protocol-compliant bridge or router. RULA is for users who wish to perform remote engineering, diagnostics, troubleshooting, and system administration on a distant TPS System from a Universal Station at their current location. RULA also allows Honeywell's Technical Assistance Center (TAC) to perform optional online diagnostic service for TPS System support. TAC can view and troubleshoot your system's problems, providing quick and direct analysis and resolution of difficult system problems without having to make a site visit.

System Characteristics

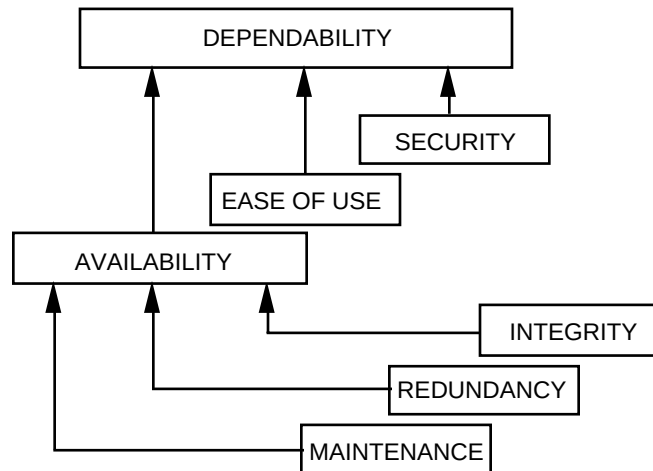
2.12 Special Software Packages

Section 3—System Dependability

3.1 "Dependability" — It Works When You Need It

Dependability factors

A dependable product is one that is available to qualified users, does useful work without confusion and false steps, and is worthy of the confidence that it is correctly performing its tasks. The major factors that contribute to dependability are availability, security, and ease of use. These three factors are conceptually shown in the following figure and are described in succeeding sections.



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3.2 Availability

Definitions	Availability is the proportion of time the system performs to specifications. Superior system availability is provided by the following factors:
Integrity	In addition to the architecture, integrity is achieved by self-checking, failure detection, fault containment, and the inherent safety designed into the system to ensure correct operation.
Redundancy	The configuration of multiple elements that can produce correct output when one or more of the elements is not functioning correctly.
Maintenance	Activity that keeps the equipment in satisfactory working order, including tests, diagnoses, measurements, replacements, adjustments, and repairs.

3.3 Integrity

Architecture's contribution to integrity

The Local Control Network links together a series of independent modules. Self-checking by the **TotalPlant** Solution (TPS) System hardware and software is facilitated by the independence of each LCN module. Because each module contains its own set of hardware and software, self-checking readily pinpoints a problem that occurs within. The independence of the modules also eliminates or limits the effect of a fault on the rest of the system, so this architecture contributes to fault containment.

Self-checking and failure detection

Self-checking and failure detection are built into the hardware and the software of all TPS products, including the process-connected devices on the UCN or Data Hiway, and the modules on the Local Control Network. Power-supply voltages, data paths, and control logic are monitored so that failure information is captured at optimum check points. All errors are recorded in a system error journal and all device failures cause an alarm.

All LCN modules have extensive checking and failure-detecting circuitry and software that ensure that these modules correctly process information; if not, these mechanisms ensure that the use of incorrect information is prohibited and that errors that can affect correct system operation are reported. This circuitry and software provide the following features:

- **Data Path Integrity** — All major and critical data paths carry a parity code that is checked at all physical interfaces. Other verification takes place to support integrity-coverage and error-isolation goals.
- **Register Integrity** — Key data and control registers have parity checking.
- **Memory and Firmware Integrity** — All dynamic and static memories detect all single-bit errors and some detect multiple-bit errors. The main memory in each module, except the Universal Work Station, has error detection and correction (EDAC)—featuring single-bit error correction and double-bit error detection. All main-memory addresses are checked.
- **Software Integrity** — Several major protection features are built into the software. Among the most important are database naming and typing rules that allow the software to verify that requests for access are legal before they are granted.
- **Local Control Network Integrity** — Thorough error checking and the LCN message protocol assure exceptional integrity in LCN communications.

In addition to these integrity features, each module on the LCN has a series of built-in test programs that contribute to integrity:

- **Startup Self-Tests** — run when power is applied, or when a module is restarted.

- **Quality Logic Tests** — run after the startup self-test.
- **On-Process Tests** — run periodically when a module is on-process in an active or backup mode.

These tests are described in more detail in 8— System Maintenance, page 173.

Fault containment

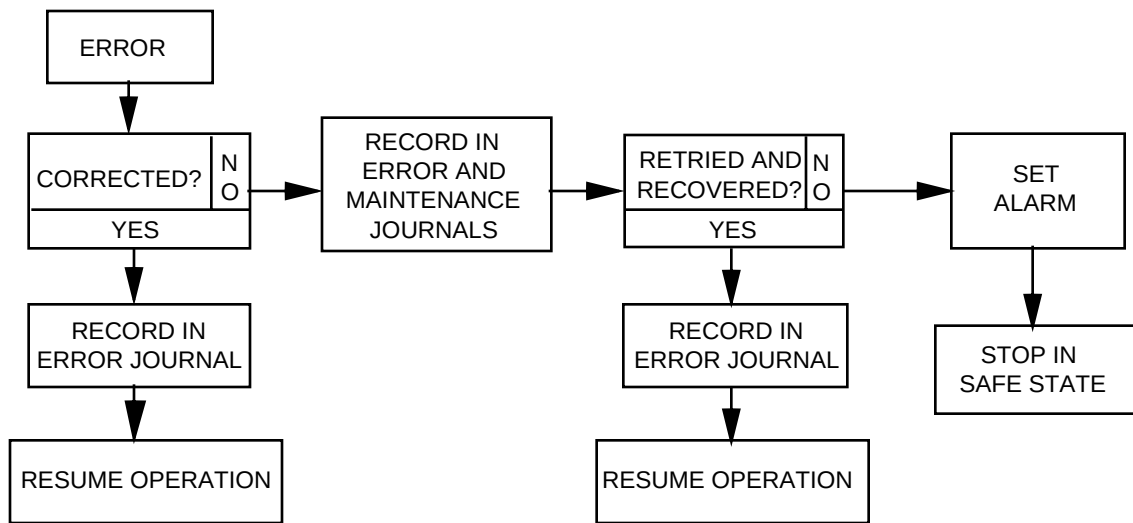
TPS System design has taken unprecedented steps to prevent faults and errors, and minimize the effects of any errors that do occur. This is accomplished in TPS through the following features and techniques:

- Most modules have an error detection and correction (EDAC) feature to assist the operator.
- Transfers on the LCN, to and from floppy diskettes, cartridge disks, and the HM's Winchester disk are retried, if a potentially recoverable error occurs.
- If neither error correction nor recovery is possible, limit the effect to the optimum replaceable unit usually a circuit board or a peripheral device such as a Floppy Disk Drive). If that is not possible, limit the effect to the module or device. The TPS System architecture is the principal contributor to this feature. If a module on the LCN fails and there is a redundant backup, it takes over. If not, only those system functions resident in the failed module are lost.

As illustrated in the following figure, any module on the LCN that detects an unrecoverable error does the following:

- Records the failure in a local error journal in systems without an HM. With an HM, records the error in the system error journal for use by the system maintenance journal.
- Generates a maintenance recommendation that is printed in the real-time journal and is entered in the system maintenance journal.
- If recovery from the error is not possible, stops operating in a safe state.
- Sets a system alarm.

No failure or sequence of related failures has any effect on another control room area.



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Inherent safety

This is probably the most important integrity factor and, therefore, one of the most important contributors to availability. Most of the other availability factors take care of situations where something goes wrong. Design for inherent safety minimizes the chance that anything can go wrong.

TPS hardware design follows a set of stringent Honeywell standards. These standards assure that the design is conservative and correct, that reliable components are used, and that the hardware can be manufactured under high-quality standards with a minimal chance of undetected manufacturing errors.

It is important in the hardware design to use highly reliable components and to operate them well within their ratings. Components are preconditioned, using heat-stress test cycles, before they are used to build the TPS hardware. This eliminates nearly all early-life component failures.

Before release for production, each hardware module is subjected to stringent design-verification tests. These test programs verify that each module meets or exceeds its operating and performance specifications.

3.4 Redundancy

Architecture's contribution to redundancy

The architecture of TPS provides for redundancy at several levels. First, the most critical system elements are fully redundant with automatic switchover. The Local Control Network, the Universal Control Network, and the Data Hiway are always redundant. Because the Network Interface Modules and Hiway Gateways are the critical elements joining the LCN and their respective communication networks, they are usually redundant.

Also the TPS architecture provides for redundant disks in the History Module and for redundant Application Modules.

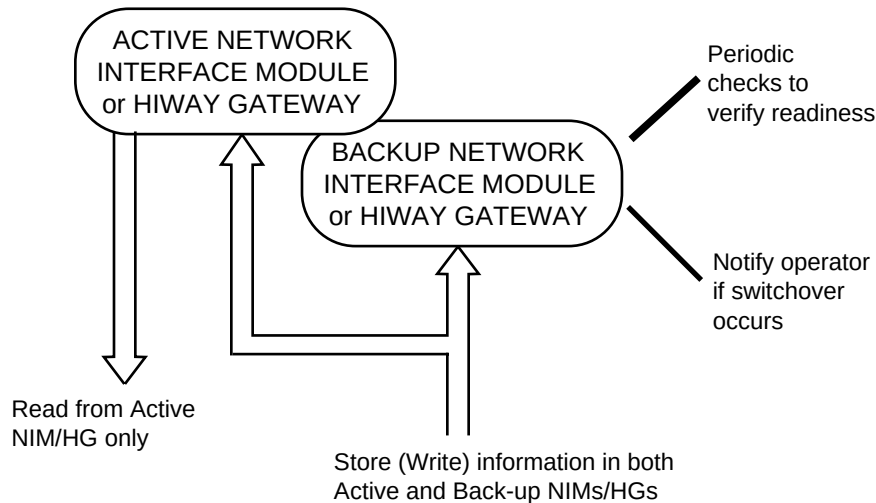
In addition, the Operator Console is typically configured with at least three Universal Stations. Redundancy is provided by the ability of any US in an Operator Console to assume the role of any other.

At the process-connected control level, redundancy is available for the Process Manager Module/Advanced Process Manager Module/High-Performance Process Manager Module, Safety Manager, and some of the I/O Processors in the PM/APM/HPM and SM; redundancy is available for Logic Manager Processor Card Files. All UCN cabling is redundant. Redundancy for critical elements is provided by the Uninterrupted Automatic Control (UAC) feature of TPS Basic, Extended, and Multifunction Controllers. Triple redundancy is standard in the Fault Tolerant Controller.

Network Interface Modules and Hiway Gateways

As illustrated in the following figure, the state of the back-up Network Interface Module or Hiway Gateway is kept as close as practical to the state of the active device by storing all new information in both the active module/gateway and its backup. Information is read from only the active device. Error checks and other exercises are periodically performed to determine the ability of the backup device to function if the active one fails. If the active module/gateway fails, the operator is notified and the backup one becomes active. The switch-over time is less than 5 seconds. The switch-over is not considered a system failure and does not "bump" the process, but it may be noticed by the operator. System performance is not degraded.

Redundancy functions



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Application Modules

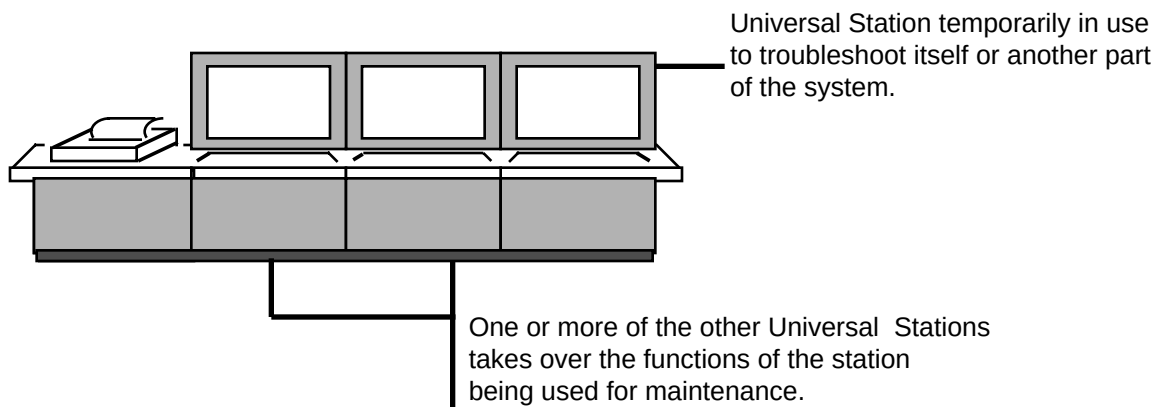
Like the gateways, the secondary Application Module recovers within 5 seconds of a primary failure. Upon recovery, the first cycle processed by the secondary is the one following the last cycle completely processed by the failed primary. Actions and messages will not be lost.

History Module

When History Module redundancy is implemented, backup is provided separately by data volume. Accordingly, a single disk error affects only the data volume where a fault occurs. The remaining volumes on that disk, as well as all error-free volumes on the second disk, continue to be accessible. This greatly enhances the capability of the system to perform normally even though both disks have suffered read/write failures.

Universal Stations

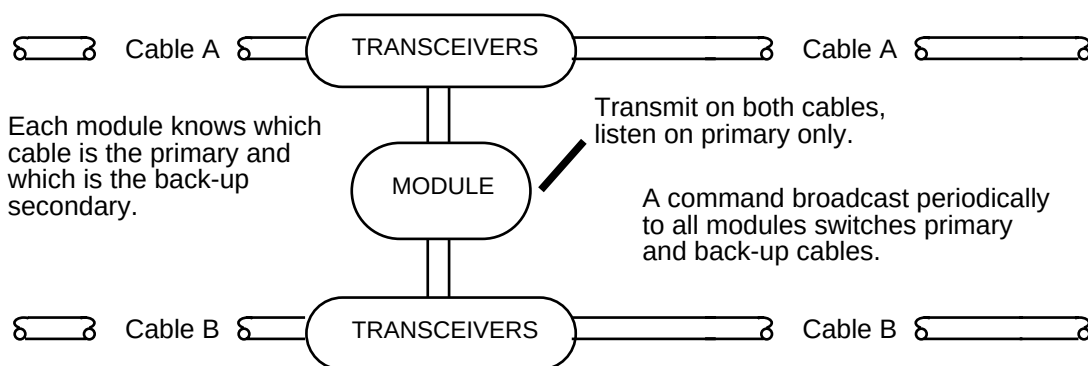
All Universal Station functions can operate on any station, and all stations in a console have access to the same information. So any one of them can take over the functions of another, whenever it is necessary. As illustrated in the following figure, if one of the Universal Stations in a console is needed for use by the maintenance technician to troubleshoot some part of the system, the remaining station(s) can take over the on-process or process-engineering functions of the converted station.



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Local Control Network

As illustrated in the following figure, one cable in the Local Control Network is designated as primary and the other is designated as backup. Information is transmitted on both cables but is received from only the primary cable. A command is periodically broadcast to all modules to switch the primary and backup cables. If a module cannot receive information from the primary cable, it "listens" to the backup cable, and sends an error message to the error journal and to the operator.

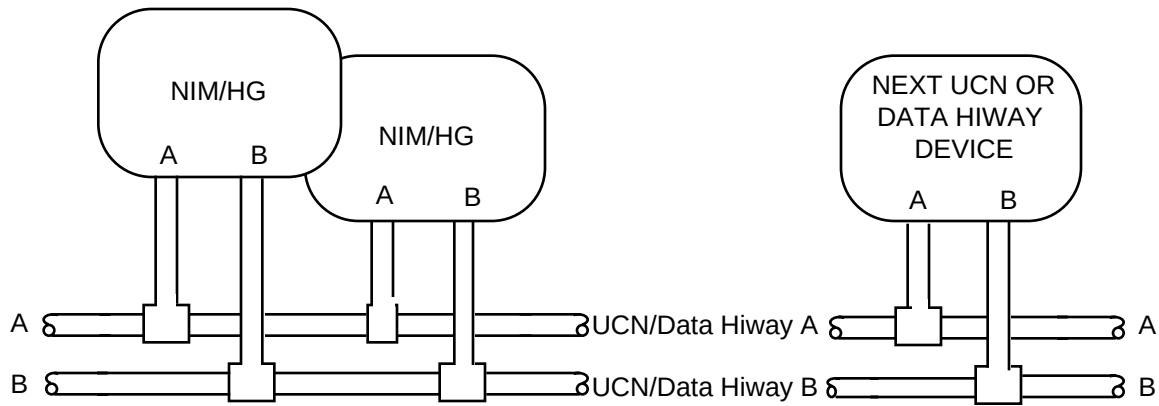


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UCN and Data Hiway

As shown in the following figure, a similar arrangement is used for the Universal Control Network and the Data Hiway. One cable is the primary and the other is the backup. The TPS System periodically switches cables to make sure both are working. Should a cable fail or the error rate become excessive, the associated module/gateway automatically switches to the backup cable.

The status of the UCN and Data Hiway cables and the devices connected to them can be observed on the TPS Universal Stations. The operator can switch to the backup cable, if there is a need to do so; for example, to allow one cable to be temporarily disconnected or serviced.



Network Status displays on TPS Universal Station show UCN/hiway status and can be used to switch cables.

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3.5 Maintainability

Design for maintainability

The TPS System is designed to be maintainable. This means that the time needed to diagnose troubles and to correct them is minimized.

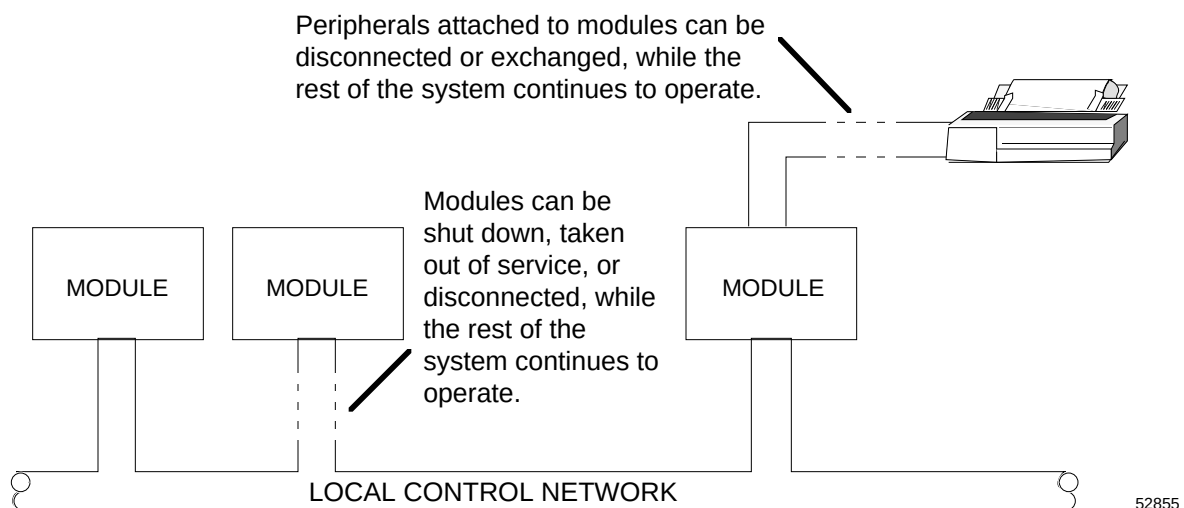
The discussion of TPS's maintainability features in this section shows how they contribute to availability and dependability.

Additional information about TPS System maintenance is in 8— System Maintenance, page 173.

Architecture's contribution to maintainability

As illustrated in the following figure, the TPS architecture makes it possible to remove modules from service or to place them in service without changing the functions of the remainder of the system. It is also possible to remove from service peripheral devices that are attached to the modules, or to place them back in service, without disturbing the remainder of the system. In this way, the architecture allows independent modules and devices to be maintained without disturbing the rest of the system.

Because of this architecture, many maintenance activities can take place while the system remains on-line, in control of the process. Most maintenance activities reduce system availability little, if at all.



Maintenance features' contribution to availability

All of the maintenance features contribute to availability and dependability. Reference: 8— System Maintenance, page 173

The following two features make very significant contributions:

- **On-Process tests** — These tests run while the system is on-line and controlling the process. They don't require that the system or any part of it be taken off-line, so availability is greater than for systems in which a unit or the whole system has to be taken off-line to be tested.

- ***Diagnosis, not just tests*** — The built-in test programs and other maintenance support software actually diagnose problems, rather than just report failures. This software makes specific recommendations to test, verify, or replace failed ORUs. This reduces the time spent in analyzing test results, so failed units are quickly returned to operation.

3.6 Security

Qualified users only

If unqualified users of a process management system are able to make improper changes to the process information or control information stored in the system, it will, at the very least, reduce the user's level of confidence that the available information is correct. Furthermore, improper changes might cause damage to the process or plant, and injury to people. To keep this from happening, the TPS System provides these four levels of access:

- **View Only Level** — No entry is allowed that can change the process database.
- **Operator Level** — Certain changes to the process database are allowed. For example, the operator might be allowed to change setpoints but not alarm limits or grouping of display points.
- **Supervisor Level** — Certain changes to the process database are allowed, in addition to those allowed at the Operator Level. For example, the supervisory operator might be allowed to change alarm limits or display groups but not tuning constants or history collection.
- **Engineer Level** — All database changes are allowed.

When the process database is built at a Universal Station/Universal Station^X, the user specifies the level of access for each position of the Universal Station's keyswitch. The keyswitch accepts a supervisor's key and an engineer's key, and selects access levels according to the type of key inserted and the position the key is turned to. If only qualified people are given keys for these switches, and if they remove their key from the Universal Station when they leave it, unauthorized changes to the process database should not occur.

A Universal Work Station, on the other hand, is intended for location in an enclosed office that can be locked, if necessary, to prevent access by unauthorized persons. When first brought on line, the UWS comes up at the Operator Level. A configured keyboard key must be pushed to change level of access.

Data naming and type checking

Several security provisions are built into the TPS software. These include the use of unique names for all data objects in the system and the identification of a type (such as alpha, numeric, floating point, integer) for each. Before data is transferred, validity checks are made to ensure that inappropriate data or data of a wrong type has not been requested.

3.7 Ease of Use

Design for ease of use Time spent in finding how to do something with the system or in retrying an operating procedure is unproductive. The TPS human interfaces, training courses, and user publications minimize such unproductive time. Both the training courses and the user publications are oriented to how to use the system, rather than how it works.

Displays guide users A person who has received appropriate training will find that the human interface easily guides the user through the operating procedures. In addition, Help Displays can be custom-defined to assist the operator when infrequent or difficult operational conditions arise. These displays can then be associated with operating displays and are available to the operator by a single request for help. When reference to operating instructions is needed, the information is easy to find and easy to understand.

System Dependability

3.7 Ease of Use

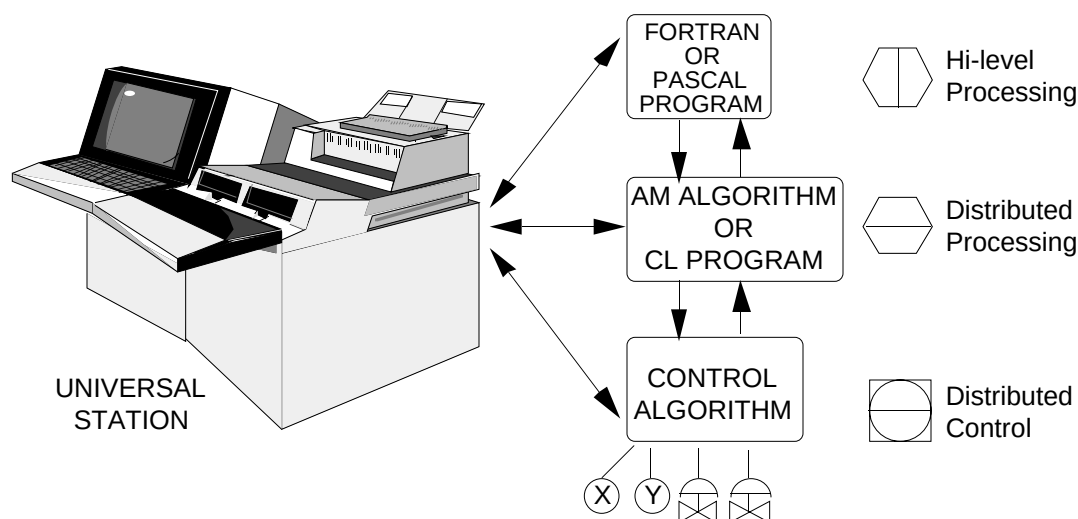
Section 4—Data Acquisition and Control

4.1 Control Devices and Levels of Control

Multiple control levels

The TotalPlant Solution (TPS) System offers the capability to implement the complete range of control strategies, from the simplest to the most complex, as conceptualized in the following diagram.

Multiple control levels diagram



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Process Manager, APM, or HPM

Many plant control strategies can now be accomplished using a single process-connected product, such as Honeywell's Process Manager (PM), Advanced Process Manager (APM), or High-Performance Process Manager (HPM). While the PM, APM, and HPM can be configured for only data acquisition, it also is capable of executing powerful control functions, including regulatory, logic, and sequencing control to accommodate continuous, batch, sequencing, or hybrid applications.

Data Hiway

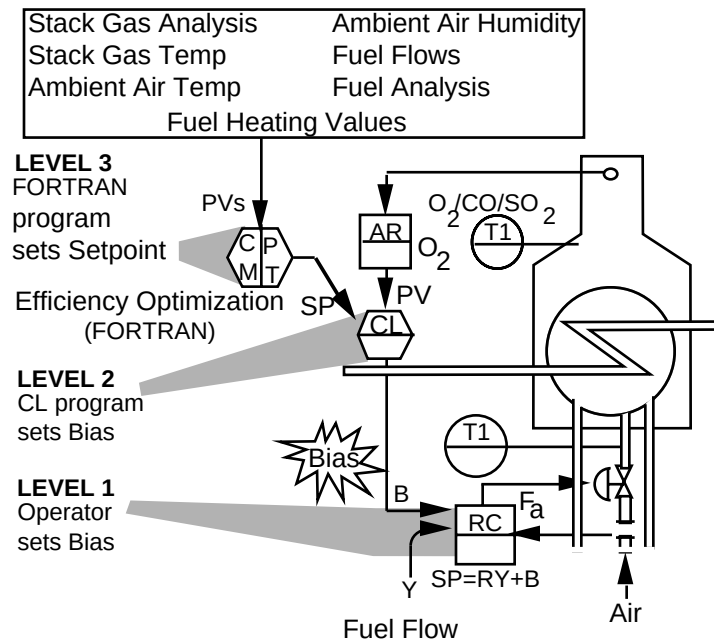
Most of the same functions have been implemented over many years by numerous users of Honeywell's Data Hiway-based systems, by employing a combination of available data acquisition and control devices. These products continue to be available, and can coexist on a TPS System with the newer Process Manager, Advanced Process Manager, High-Performance Process Manager, Logic Manager, and Universal Control Network.

Application Module	Where certain types of advanced control and compute power are required, or where additional memory is needed to accommodate custom control logic, the LCN-connected Application Module is available. The AM derives its inputs and sends its outputs to other devices connected to the system, such as the Process Manager, Advanced Process Manager, High-Performance Process Manager, or Multifunction Controller.
Host computers	An even more powerful computational capability is available using a host computer. A DEC VAX workstation or Alpha AXP computers can be connected to the LCN through a Plant Network Module or Computer Gateway and programmed in standard programming languages such as Pascal or Fortran to accomplish such tasks as advanced optimization or history collection.
Control component selection	Control system engineers select from the above three components—process-connected device, AM, and CM—based on the most cost effective combinations to fit their present needs. As their needs grow, more of these devices are easily added to expand strategies or to increase the number of points to be monitored or controlled.

4.2 Simple Control Example

Description

Using the control of air and fuel flow to a typical process furnace, the following figure shows an example of continuous control implemented by using a standard algorithm plus custom control calculations written in Control Language and in Fortran. This example is expanded later in this section.



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Level 1

At Level 1, the air flow is maintained at a ratio of the fuel flow by a ratio controller in a process-connected device. The setpoint for that controller (SP) is a ratio of the fuel-flow input (Y), and B is an air-flow bias constant. The fuel flow is measured, the bias is calculated at another level, and the operator enters the ratio.

Level 2

At Level 2, proper oxygen trim is maintained. A user-written Control Language (CL) program calculates the bias for the ratio controller, taking into account the O₂ content of the stack gas (shown), as well as the air-and-feed-flow rates and the total heat of combustion of two fuels (not shown). The operator could enter the O₂ setpoint, but in this example it is calculated by another program.

Level 3

At Level 3 in this example, a user-written Fortran program minimizes fuel consumption. The program calculates the combustion efficiency as a function of the process variables shown. The program then uses a process model to determine the optimum O₂ and sends the O₂ setpoint to the CL program. The CL program calculates the bias and delivers to the hiway box a bias value that is most representative of actual; process conditions required.

4.3 Data Points: The Basis for Data Acquisition and Control

Definition and structure

All of the process data and control information in the TPS System is organized into "data points."

A **data point** is a named collection of related parameters and processing information.

As shown in the following table, a data point has three parts: a **tag name**, **parameters**, and **functions**. Depending on the type of data point, the content of these three parts can vary; but these basic parts remain the same. Every piece of data in the system is identified by a combined data-point name with parameter identifier.

Table 4-1 Structure of a Sample Data Point

Parts	Sample Contents
Tag Name	TIC_100
Tag Name plus Parameters	TIC_100.PV = 532° F TIC_100.SP = 535° F TIC_100.OP = 55% etc.
Function	OP = Function of (PV, SP, other parameters)

Tag name

A unique alphanumeric name identifies each data point. Each name is from 1 to 16 characters long and the first character must always be an alpha.

Parameters

Each parameter consists of a single value. Parameters can be boolean (flags), integers (numerics), real numbers, strings, time, dates, and arrays of any of these. Examples of values include:

- **Input data** — flow, temperature, pressure, etc.
- **Tuning parameters** — gain, time constants, bias, recipe data, etc.
- **Output data** — valve outputs, alarm states, etc.

A task that needs a parameter in a data point asks for it by name. For example, if a task needs the output of data point TIC100, it simply asks for TIC100.OP. The request is internally converted to a local routing code for direction to the owner of the data. The owner verifies that the request is legitimate and returns the parameter to the requestor. The TPS bookset includes parameter reference dictionaries which list and define every user-accessible parameter associated with specified system devices. (Some parameters exist for internal system use only.)

Data point functions

The functions that are to be performed when the data point is processed are specified in the function field. Some data points have only fixed functions, already built in, that are pertinent to the type of task the data point is intended to accomplish. Other data points have both fixed functions and user-specified functions. The functions specified can be:

- Algorithms (standard or user-written)
- A sequence of procedures specified by the user
- A user-written program

Data point types

To simplify the engineer's job of building the data points, the TPS System labels them as types that accomplish specific kinds of tasks. The fixed functions mentioned earlier vary according to data point type. A regulatory data point, for example, in addition to user-selectable functions has many fixed functions that allow it to accept inputs, perform appropriate checks on them, manipulate them as instructed, and produce an output. A digital input data point, on the other hand, has fixed functions that allow it to convert a raw digital state to a PV value appropriate for display, recording, or transfer to another data point.

In the following subsections are lists of most user-visible types of data points, a brief summary of the primary functions they perform, and indicators showing where in the TPS System they can be implemented—in a UCN device, in a Data Hiway box, in an Application Module, or in a Computing Module. To assist in understanding the role of each data point type, they are grouped according to how they are typically used:

- Process Input
- Standard Control
- Custom Control
- Process Output
- Utility
- Miscellaneous

ATTENTION

In the following point descriptions:

PV = process variable

OP = control output

EUs = engineering units

U = Universal Control Network

H = Data Hiway

A = Application Module

C = Host Computer

Process input data points

Data points in the Process Input group represent process variables from process hardware, such as transmitters, contacts, etc. These PVs can be directly transmitted raw variables, or indirectly accumulated events and values.

Table 4-2 Data points in the Process Input group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Analog Input	√	√			Processes raw variable and converts it to PV in EUs.
Digital Input	√	√			Filters raw state and converts it to PV.
Counter	√(1)	√	√		Counts pulses or transitions and converts the count value to values in engineering units. Used to accumulate values such as: • Flow • Speed • Volume • Event counts
(1) Implemented as a Digital Input					

Standard control data points

Data points in the standard control group represent those elements of a control strategy that are necessary to produce an effective control output. Functions of these points include standard algorithms, user-written algorithms, and user-written CL programs.

Table 4-3 Data points in the Standard Control group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Regulatory	√	√	√		Configured algorithm or CL algorithm calculates PV and/or OP.
Analog I/O Composite		√			Displays PV and accepts command to output to a final control element.
Digital I/O Composite	√	√			Displays PV and accepts command to output to a final control element.
Logic	√	√			Executes logic algorithm and outputs parameters to configured connections.
Device Control	√				Executes discrete control logic with digital composite and graphical logic functions.

Custom control data points

Data points in the custom control group represent user-supplied variables and/or programs necessary to form custom control strategies. They can also be used to link custom control strategies to standard control strategies.

Table 4-4 Data points in the Custom Control group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Custom			√		Supports user-written CL programs.
Switch			√		Conditionally switches from one strategy to another.
Array	√				Groups multiple data point types (flags, timers, numerics, and strings) under a single tag name.
Process Module	√	√			Executes a sequence program, written by the user in CL for the Process Manager or Multifunction Controller.
Advanced Control Interface (ACIDP)				√	Interfaces Fortran or Pascal program in a Computing Module with the system database.
Calculated Results (CRDP)				√	Stores calculated results of an ACIDP.

Process output data points

Data points in the process output group convert a control output to a raw value or state before sending it to the final control element.

Table 4-5 Data points in the Process Input group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Analog Output	√	√			OP in EUs is converted to a 4 mA to 20 mA output value.
Digital Output	√	√			OP is converted to a contact output state.

Utility data points

Utility data points hold internal variables and functions that are useful in implementing sequential and logic functions. They also provide a mechanism through which the operator can input and output data to such functions.

Table 4-6 Data Points in the Process Input Group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Flag	√	√	√		Allows the operator to communicate with sequential functions, or can be used as a digital "scratchpad" variable.
Timer	√	√	√		Times process actions such as soaking, charging, etc.
Numeric	√	√	√		Stores a numeric value that can be set by or displayed to the operator and read or set by the sequence functions. Allows the operator to communicate with sequential functions, or can be used as a numeric "scratchpad."

Miscellaneous data points

Data points in the miscellaneous group represent control-system information not included in any other group of data points.

Table 4-7 Data points in the Process Input group

Data Point Type	Where Implemented				Data Point Function(s)
	U	H	A	C	
Linkage	√				Supports peer-to-peer communication from a Logic Manager to another Logic Manager, Process Manager, APM, HPM, or SM.
Processor Status			√	√	Captures performance data for an LCN-based module.
Box Parameters (PM or MC)	√	√			Provides access to Process Manager, APM, HPM, LM, SM, or Multifunction Controller flags, numeric, and timer variables.

Data point residency

A data point resides in one, and only one, of several TPS System components; the engineer specifies the process-connected device or LCN module when the network is configured. Where it resides determines the scope of the information the point has access to, as shown here.

Table 4-8 Residence Effect on Scope of Information

Data Point Residency	Scope of Information Accessible to the Data Point
Hiway-Based Box	Box and information from its process connections.
UCN Device	Internal, information from the device's process connections, and information from other Process Manager, APM, HPM, SM, or Logic Managers on the same Universal Control Network.
Application Module	On-line database (current data from all nodes in the system).
Host Computer	On-line database + HM history.

The type of device or module in which a data point resides may restrict the function set that can be performed on that point. For example, a regulatory data point in a Process Manager or Application Module has available a larger selection of alarm functions than a regulatory point in the Basic Controller.

In the following table, abbreviations for the device names are as follows:

- Data Hiway devices; CB=Basic Controller, EC=Extended Controller, MC=Multifunction Controller, PIU=Process Interface Unit, and DHP=Data Hiway Port.
- UCN Devices; PM=Process Manager, APM=Advanced Process Manager, LM=Logic Manager, SM=Safety Manager, and HPM=High-performance Process Manager.

- LCN Modules; HG=Hiway Gateway, AM=Application Module, and CG/PLMN=Computer Gateway/Plant Network Module.

Table 4-9 Data Point Residences

	CB	EC	MC	PIU	DHP	PM	APM/ HPM	LM/ SM	HG	AM	CG/PLN M
Advanced Control Interface											√
Analog I/O composite									√		
Analog Input	√		√	√	√	√	√	√			
Analog Output			√	√	√	√	√	√			
Array							√				
Calculated Results											√
Counter			√	√	√	√(1)	√(1)			√	
(1) Counters in PM, APM, and HPM are implemented as Accumulated Digital Input points.											
(2)Box String point type for HPM only											
Custom										√	√
Device Control							√				
Digital I/O Composite						√	√	√	√		
Digital Input		√	√	√	√	√	√	√			
Digital Output			√	√	√	√	√	√			
Flag			√			√	√	√		√	
Linkage								√			
Logic Block (Slot)			√			√	√				
Numeric			√			√	√	√		√	
Process Module			√			√	√				
Processor Status									√	√	√
Regulatory	√	√	√			√	√			√	
String							√ (2)				
Switch										√	
Timer			√			√	√	√		√	

Parameter residency

Certain parameters of data points that are resident in process-connected devices are maintained in the associated Network Interface Module (NIM) or Hiway Gateway (HG). The NIM or HG is considered to be the "data-point owner" for process-device communication with other modules on the LCN.

Because each parameter is identified by its data point parameter name, a requestor (any system task needing that data) does not need to know where the data is physically located. This contributes to very efficient communication throughout the local communication networks (the Local Control Network, the UCN, and the Data Hiway).

Data point building

Data points can be built and entered into the system database using a Universal Station (or Universal Work Station) or they can be built externally on personal computers and loaded into the system when convenient.

A special "Data Entity Builder (DEB)" function in the TPS System is used to set up data points in the process database and enter the information they need. To begin, the Builder requires selection of the point type and entry of a tag name. Based on this initial information, the Builder presents the engineer with sets of questions (forms) that must be answered to completely define the point.

After the data point is built, its information is loaded into the device where it will reside. It is then available to be called up by a requesting function anywhere in the system, including graphic displays that are designed and built by the user.

Honeywell's PC-based software packages, WORKBOOK and Text File Converter, allow data points to be built on a personal computer and stored on cartridge or floppy disk. They can subsequently be loaded into the TPS System.

In addition, Honeywell offers The Engineering Bridge; a networked, personal-computer-based package. The Engineering Bridge automatically brings information from intelligent drawings of the process, instrumentation, and control schemes into the TPS database.

4.4 Using Data Points for Control

Fixed and selectable functions

Data points that have fixed functions are used, of course, for their stated purpose: in the simple case of obtaining a PV in engineering units, for example, the engineer simply configures an Analog Input data point in a process-connected device.

Using data points that have selectable functions, on the other hand, can be as simple or as complex as the engineer needs. The following sections describe the tools available to the engineer for use in defining a data point's function: algorithms, Control Language, and application programs.

Algorithms

The TPS System provides standard preprogrammed routines, called algorithms, to handle a wide variety of regulatory control situations. Based on the control action desired, the engineer makes a choice from the list of algorithms displayed for the data point being configured.

The following table shows the algorithm functions available in TPS devices. Depending on the type of algorithm selected, the system then asks the engineer to enter the PV range, the SP source, alarm limits, gain, scaling factors, etc. When the data point is processed, the selected algorithm is executed, using these parameters.

Algorithm function availability

Table 4-10

Algorithm	CB	EC	MC	PM/APM	HPM	AM
PIDs	√	√	√	√	√	√
Summer	√	√	√	√	√	√
Multiplier	√	√	√	√	√	√
Divider	√	√	√	√	√	√
Selectors	√	√	√	√	√	√
Dead Time		√		√	√	√
Lead/Lag	√	√	√	√	√	√
Flow Compensation		√		√	√	√
Switch	√	√	√	√	√	√
Logic		√	√	√	√	√
Characterization		√		√	√	√
Auto-Manual	√	√	√	√	√	√
Data Acquisition	√	√	√	√	√	√

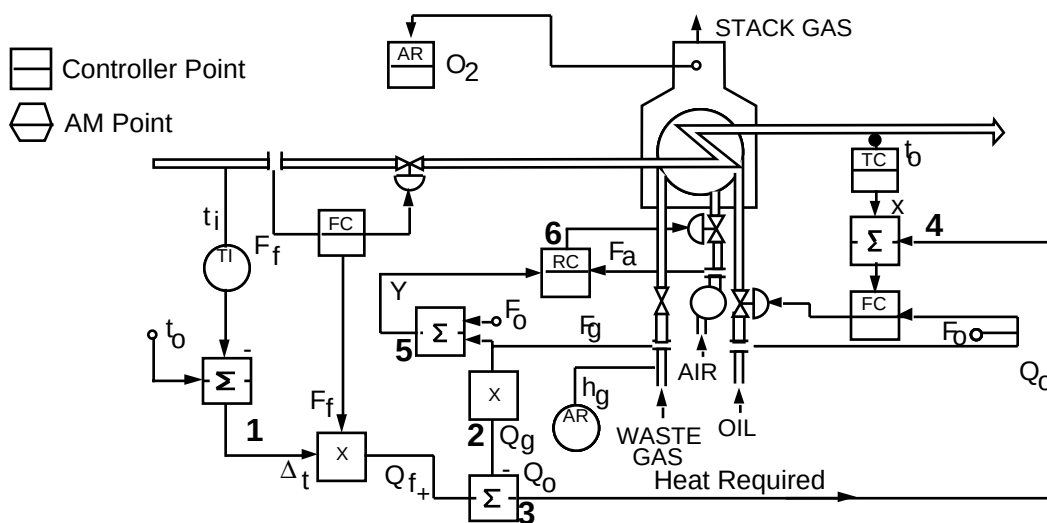
Table 4-10

Algorithm	CB	EC	MC	PM/APM	HPM	AM
Ratio	√	√	√	√	√	√
Multiply/Divide w/ Ramping Bias					√	
Summer w/ Ramping Bias					√	

Because of the evolution of the TPS product line, process-connected devices and modules may have algorithms with similar functions but different names or the same names but with differences in form.

**Example,
continuous
control with
standard
algorithms**

Using only data points with standard algorithms in a process-connected device, the following figure illustrates feedforward control of feed temperature and fuel/air ratio control with a dual-fuel furnace.



In this feedforward example, an increase (or decrease) in the feed-flow rate must trigger an increase (or decrease) in the total amount of heat supplied by the two fuels. So three caloric flows (heat inputs) must be calculated, using summer and multiplier algorithms:

In the equations to follow:

Table 4-11

B	= air-flow bias	K ₂	= Btu constant for fuel oil
C _f	= specific heat of process feed	K ₃	= outlet temperature constant
F _g	= mass flow rate of waste gas	K ₄	= heat-required constant
F _f	= feed mass flow rate	R	= air/fuel ratio

Table 4-11

F_o	= mass flow rate of fuel oil	t_i	= feed inlet temperature
h_g	= waste-gas heating value	t_o	= feed outlet temperature
K_1	= Btu constant for waste gas	X	= temperature controller output

The following are the feedforward control calculations for the furnace example. The bold numbers (for example, **1**) refer to the bold numbers in the preceding figure.

- **1**— Q_f , the total heat required to raise the temperature of the feed from the inlet value to the output valve requires two algorithms.
 - $\Delta_t = t_o - t_i$ (summer)
 - $Q_f = F_f \times C_f \times \Delta_t$ (multiplier)
 C_f is Operator entered
- **2**— Q_g , the heat supplied by waste gas, is calculated from:
 - $Q_g = h_g \times F_g$ (multiplier)
 h_g is operator-entered
- **3**—Then Q_o , the heat to be supplied by the fuel oil, is calculated from:
 - $Q_o = Q_f - Q_g$ (summer)
- **4**—Then the fuel oil setpoint is a function both of outlet temperature (feedback control) and feed-flow rate (feedforward control):
 - $SP = K_3X + K_4Q_o$

This control scheme also controls the airflow, based on the amount of oil and waste gas being used and the desired stack O_2 content.

- **5**—A summer algorithm first calculates the total Btu content:
 - $Y = K_1F_g + K_2F_o$
 K_1 and K_2 are operator-entered.
- **6**—A second summer algorithm sets an efficient air-flow setpoint:
 - $SP = RY + B$
 B and R are operator-entered.

With the proper R and B values, the stack gas O_2 content can be held nearly constant over the operating range.

Control Language The standard algorithms provided with the TPS System cover most frequently encountered control situations. Control strategies do exist, however, that require custom calculations or processing routines. For these situations, the TPS System provides a set of high-level Control Languages (CL).

All CL programs are written from, and compiled in, the TPS Universal Station or Universal Work Station.

- CL programs for continuous operations reside and execute in the Application Module (CL/AM);
- CL programs for discontinuous operations reside and execute in the Process Manager (CL/PM), Advanced Process Manager (CL/APM) and Multifunction Controller (CL/MC).

Each version of CL is a process-oriented language designed specifically for process and application engineers to use to implement special control schemes. With CL, process problems are easily solved, even though the engineer may not be an experienced programmer. Engineers who are experienced programmers, on the other hand, will find that CL is powerful enough to accomplish quite complex control strategies.

In addition to arithmetic and logical operators, and arithmetic functions usually provided in a programming language, CL provides a variety of general and process-oriented statements for use in both continuous and discontinuous environments.

The following three tables list the CL operators, functions, and statements:

Table 4-12 CL Arithmetic and Logical Operators

Operator Type	Operator	Meaning
Arithmetic	**	Exponentiation
	*	Multiplication
	/	Division
	mod	Remainder
	+	Addition
	-	Difference or negation
Logical	not	Logical complement
	and	Logical and
	or	Logical or
	xor	Logical exclusive or

Table 4-13 CL Arithmetic Functions

Function	Meaning
ABS (x)	absolute value
ATAN (x)	arc tangent
AVG (x, y, ...)	average
COS (x)	cosine
EXP (x)	exponential
INT (x)	truncate to integer

Table 4-13 CL Arithmetic Functions (continued)

Function	Meaning
LN (x)	natural logarithm
LOG (x)	common logarithm
MAX (x, y, ...)	maximum
MIN (x, y, ...)	minimum
ROUND (x)	round to integer
SIN (x)	sine
SQRT (x)	square root
TAN (x)	tangent

CL programs and data points

CL programs in the Application Module can be associated with several types of data points—regulatory, custom, and switch. In the Process Manager, Advanced Process Manager, and Multifunction Controller environment, CL programs are associated with Process Module data points.

Table 4-14 CL Statements

Type	Statements
Assignment statements—access or change the value(s) of variables.	read write set change state
Control statements—direct or redirect the flow of a CL program.	if then else goto loop repeat
Delaying statements—stop CL program execution for specified time or until some event occurs.	pause wait
Communication statements—allow a CL program to communicate with a subpart of itself, another CL program, or the operator.	call send fail restart initiate
Termination statements—indicate the end of a CL program or subroutine>	resume abort exit end

CL in continuous operations

For continuous operations, a CL program is referred to as a "CL block" and can be used in the AM to:

- Substitute a custom algorithm for a standard PV, control, or logic algorithm.
- Modify standard point-processing routines (See Modified point processing in the AM, page 74).
- Provide custom processing routines (See Custom and Switch data points, page 76).

There are two types of CL blocks:

- Standard blocks that can be inserted into a point's processing cycle at many different locations, and are guaranteed to execute without pauses or delays, each time the point executes.
- Background blocks that are inserted only at the end of a point's processing cycle, and can be interrupted during execution by higher priority demands.

The overall structure of a CL block is as follows:

```
BLOCK heading
  Data and function declarations
  Statements
END
```

```
SUBROUTINE heading
  Data declarations
  Statements
END
```

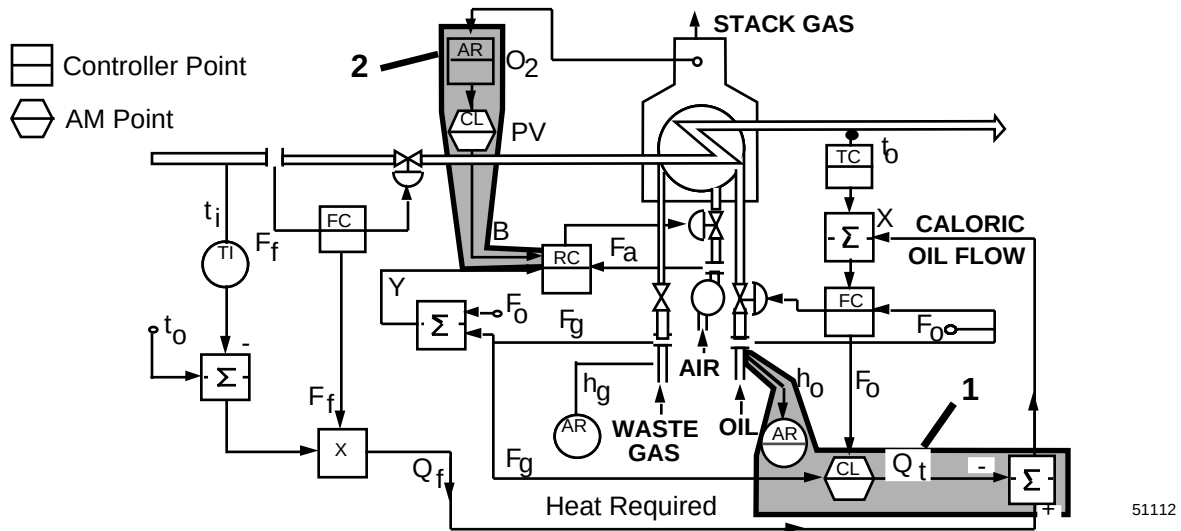
(More subroutines if necessary)

A data point in the Application Module to which CL blocks are attached contains data about these blocks. Most of this data is available at the Universal Station on the CL page of the Point Detail Display:

- Name of each "block" in a CL segment.
- Most severe error of each CL program in last processing cycle.
- Program location at which the error occurred.
- "On-process" or "test" status indicator for each CL program.
- Point-processing "insertion-point number" for each CL program.
- Point-processing "insertion-point order" for each CL program.

**Example,
continuous
control using CL**

The following figure continues the furnace example and describes the use of CL programs to accommodate fluctuating heat content of the two fuels and to stabilize the oxygen content of the stack gas.



In the preceding figure and the following equation:

Table 4-15

B	= air-flow bias
F_a	= mass flow rate of air
F_f	= mass flow rate of feed
F_g	= mass flow rate of waste gas
F_o	= mass flow rate of fuel oil
h_g	= waste-gas heating value
h_o	= fuel-oil heat of combustion
O_2	= oxygen content of stock gas
Q_t	= total heat of combustion
R	= air/fuel ratio

The previous example assumed that the heat content of the fuel oil was constant. In practice, this is often not so.

For this example, in the area marked **1** in the preceding figure, a second recording analyzer (AR) was added to monitor the heat of combustion (H_o) of the fuel oil. A CL-linked data point now reads and validates Btu analyzer data and calculates the total head supplied by the two fuels

- $Q_t = F_g H_g + f_o h_o$

and sends the value to a summer algorithm, to be compared against the total heat required (Q_f) to raise the current flow to the proper outlet temperature. Any variations in grades of fuel oil and/or waste gases are immediately reflected to the summer slot as deviations in the total heat of combustion.

If the output from the CL point becomes unavailable, feedforward control reverts to that of the simpler example, with the exception that waste-gas heat of combustion is not taken into account.

Because of nonlinearities in a furnace, keeping the ratio of air to fuel constant may not always provide the best amount of excess air for efficient combustion throughout the entire range of furnace operation.

In this example, in the area marked by 2 in the preceding figure, based on data from the analyzer, a CL-linked data point calculates a bias (B) for the ratio controller (RC) as a function of the air flow, feed flow, and total heat:

- $B = f(f_a, F_f, Q_t)$

This function could be a "learning model" of combustion in this furnace and could set either the bias (B) or ratio (R), leaving the other value for the operator to set. Bias is calculated in this example to improve "fall-back" control. If the output of this data point is not available, air/fuel ratio control continues as shown in the simpler example, using the last best bias value.

In both of these cases, the CL blocks could also conduct diagnostic tests on their respective analyzers to detect failure or questionable signals, and could take action to alert the operator.

CL in discontinuous operations

In a discontinuous environment, CL programs are called CL sequence programs and have a wide selection of statements that can be implemented. Executing in a Process Module data point, a sequence program acts as an automatic operator to direct the sequence of events in a batch operation. The sequence is divided into phases and steps that correspond to major process milestones. The program itself consists of:

- a main or primary program,
- zero or more subroutines,
- zero or more abnormal condition handlers

An "abnormal condition handler" is a special kind of subroutine. Unlike ordinary subroutines, an abnormal condition handler doesn't return control to the main body of the program when it ends. With this feature, the engineer can define abnormal conditions and the actions the system is to take if they occur, secure in the knowledge that the main sequence will not resume until the abnormal condition has been taken care of. CL provides three levels of response to abnormal conditions:

- HOLD

- SHUTDOWN
- EMERGENCY SHUTDOWN

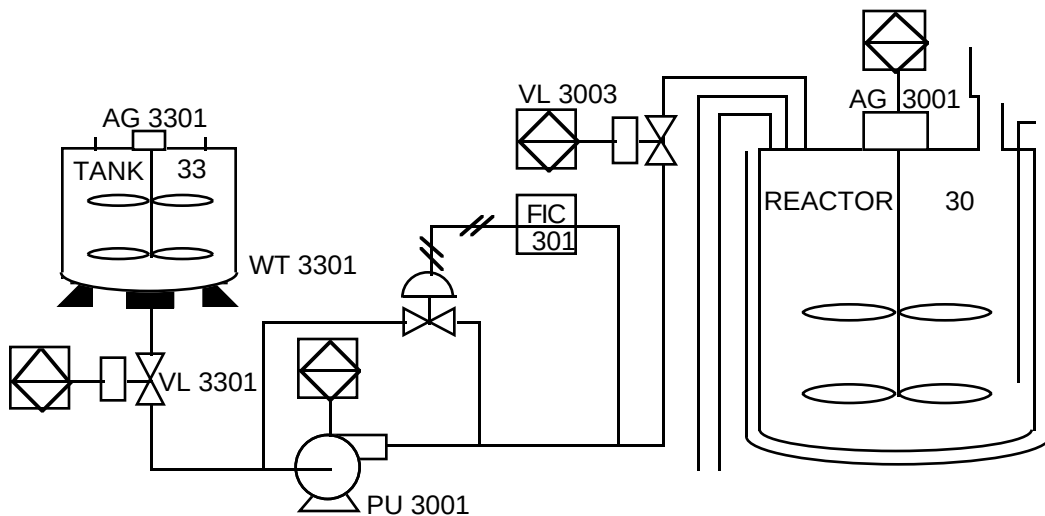
The overall structure of a CL sequence program is as follows:

```
PROGRAM heading
  Data and function declarations
  PHASE heading
    STEP heading
      Statements
      (Further steps)
    (Further phases)
  END
Anormal condition HANDLER heading
  STEP heading
    Statements
    (further steps)
RESTART (optional)
SUBROUTINE heading
  Data declarations
  Steps
END
  (Further subroutines)
```

A sequence program performs such actions as opening and closing valves; starting and stopping motors; and setting data point modes, setpoints, and outputs. Depending on the engineer's instructions, a CL sequence program can take these actions based on fixed values, operator-entered values, or the results of calculations it makes.

The following are a process schematic and portion of a very simple CL-sequence-program listing for the control of the polymerization reactor in the schematic. In the program, the monomer and catalyst are mixed and heated under controlled conditions.

In the program, lines that begin with "--" are comment lines and do not affect execution of the sequence.



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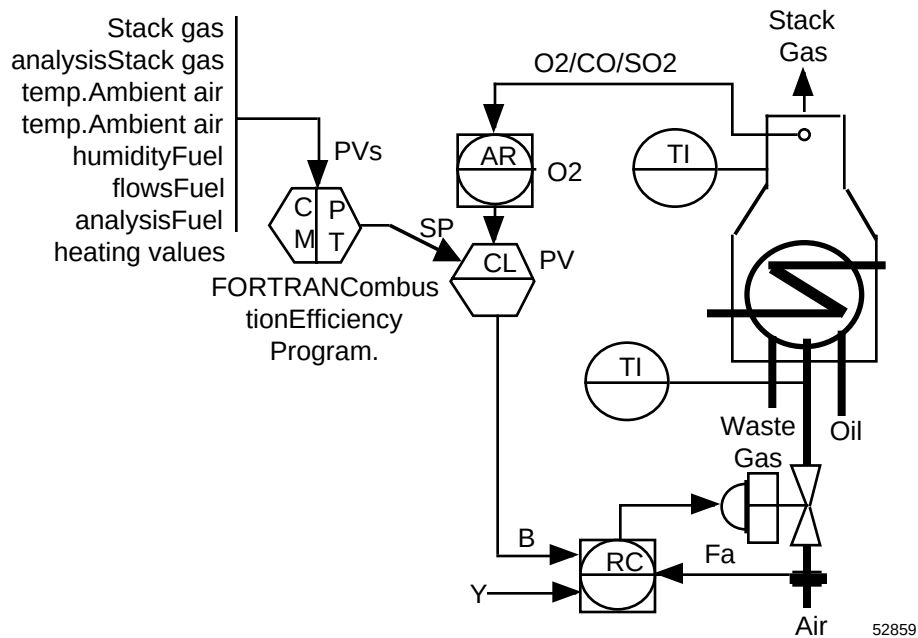
"- -" marks comment lines that do not affect the execution of the sequence.

```

phase 3_ch_R30
--Open valves and turn reactor agitator on.
  step chrg_mon
    open VL3301, VL3003
    on AG3001
--Set monomer flow to 50 gallons per minute.
  set FIC301.SP=50
  on PU3001
--When weight of monomer in Tank 30=zero, turn off pump and --agi-
tator and close valves.
  wait WT3301.PV=0
  off AG3301, PU3301
  close VL3301, VL3003
--Put monomer flow loop in MANual and close valve.
  set FIC301.MODE=MAN
  set FIC301.OP=0
--Send message to operator at Universal StationX.
  send (wait): "Momomer charge complete."
```


4.5 Large Application Programs

Host computers	CL and its environment are not suitable for large number-crunching tasks such as linear programs and complex performance calculations that might be required for some advanced types of control. To accomplish these tasks, the TPS System provides interfaces to host computers such as the DEC VAX and Alpha AXP computers that have high-level programming capability in languages such as Fortran, Pascal and C. Application programs written in these languages use data from the TPS modules, but run independently of data-point processing in the rest of the system. Because of their position in the TPS architecture, these programs have access to data throughout the entire system, including historical data.
Application programs and data points	These “computing modules,” like other devices and modules in the TPS System, process data points that were assigned during configuration. In most modules, data points contain information directly related to the process, such as the PV, SP, OP, and associated constants, ranges, and limits for a regulatory control loop. In the computing module, however, data points, which reside in the Computer Gateway or Plant Network Module, contain information relating to a computer program. The type of data point associated with the host computer is the Advanced Control Interface Data Point (ACIDP). Each ACIDP can be connected to one or more programs, and contains information about the program(s), such as the name, status, and execution schedule; these can be viewed at a Universal Station. The Computer Gateway or Plant Network Module has another type of data point, called the Calculated Results Data Point (CRDP). CRDPs are used to store the calculated results of a computer program, and make results available to other modules on the LCN. The contents of a CRDP can be used by multiple programs.
Example, higher level of control	The following figure shows the addition of a Fortran program to provide a still higher level of control to the furnace example: The calculation of combustion efficiency to minimize fuel consumption.



Notice that two temperature indicators (TI) have been added to supply some of the possible PVs to be processed by the Fortran program. The Fortran program calculates the optimum oxygen setpoint (SP) and makes it available as input to the Custom data point's CL program, which in turn calculates the required bias value (B) for trimming the oxygen supply.

The Fortran program calculates combustion efficiency to minimize fuel consumption. Notice that two temperature indicators (TI) have been added to supply some of the possible PVs to be processed by the program. The program calculates the optimum oxygen setpoint (SP) and makes it available as an input to the custom data point's CL program, which in turn calculates the required bias value (B) for trimming the oxygen supply.

4.6 Data Point Processing

Point residences As mentioned earlier, control in the TPS System revolves around the processing of data points. How and when a point's functions are executed—in other words, how and when it is processed—depends on where the point resides.

Point processing in process-connected devices Many types of data points are assigned by the engineer to reside in process-connected devices. Some of these data points—regulatory data points, for example—are scheduled and processed at whatever the scan rate established for the device in which they reside, as defined in the following table.

Table 4-16 Regulatory Point Scan Rates

Controller	Scan Rate
Process Manager, Advanced Process Manager, High-Performance Process Manager, and Safety Manager	1/4, 1/2, or 1 second (configurable)
Basic Controller	1/3 second
Extended Controller	1/2 second
Advanced Multifunction Controller	1/2 second
Multifunction Controller	1 second

Many control strategies can be formed using only process-connected devices. The Advanced Process Manager and the High-performance Process Manager, in particular, are capable of performing a wide range of control strategies—from the simplest standard algorithms to complex cascade strategies to sequencing control programs written in Control Language.

The types of control points that can be configured for an APM or HPM are:

- Regulatory PV
- Digital Composite
- Process Module
- Numeric
- Array
- String
- Regulatory Control
- Logic
- Flag
- Timer
- Device Control

For those installations, however, where the number of control data points is especially large, where processing of many points can be scheduled on a less frequent basis, where inputs are acquired from more than one UCN and/or Data Hiway, or where custom continuous-control strategies are required, TPS provides the Application Module. The capabilities of the AM are discussed in the following paragraphs.

4.7 AM Data Point Processing

Continuous data points

The Application Module (AM), like other devices in the TPS System, processes data points that the user builds and assigns to it during configuration. All points for a process unit are assigned to the same AM. An AM usually handles the points in several process units.

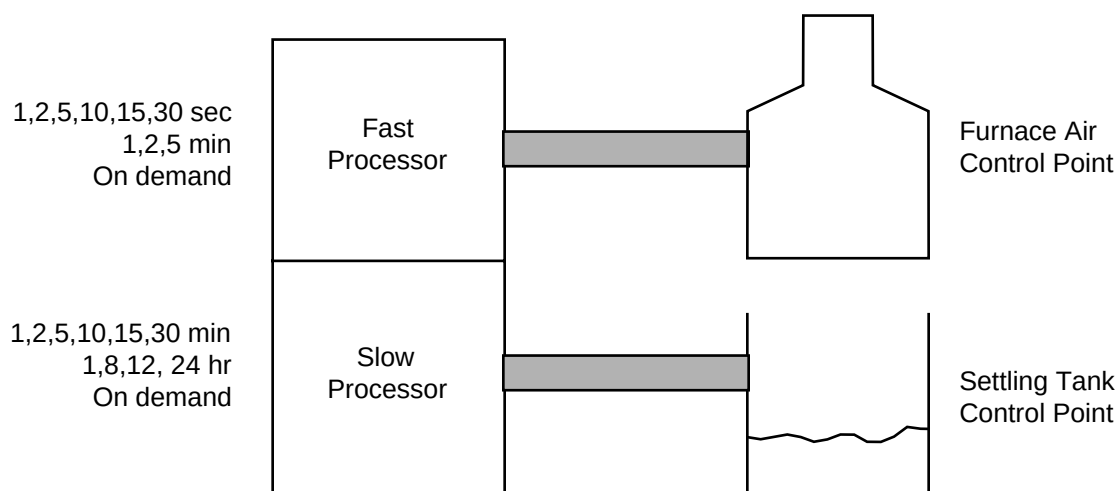
Point processing within an AM is carefully coordinated to give maximum control and flexibility. The processing of each continuous data point assigned to an AM is controlled by the following factors, selected by the engineer during configuration:

- The point's processing frequency (period)
- Specified cycle within the period (2X cycles per period)
- Before/after other specified points
- The point's connections to the system's database

Point scheduling

One of the factors involved in the processing of a data point is the point's processing schedule. During configuration of the AM, the engineer chooses for each point the interval at which it is to be processed. As shown in the following figure, a variety of time intervals is available. Choosing an interval assigns the point to either a "fast" processor or a "slow" processor. Scheduling a point to be processed every 5 seconds, for example, assigns that point to the fast processor.

Timer intervals on AM fast and slow processors



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When a time interval common to both processors is chosen, the system assigns the point to the slow processor; the engineer has the option, however, to change the assignment to the fast processor.

The "fast" processor has a higher priority, so points assigned to it have a smaller variation in sample time. Generally, noncritical points or points that take a long time to process are assigned to the longer intervals, so that they do not interfere with the processing of critical points.

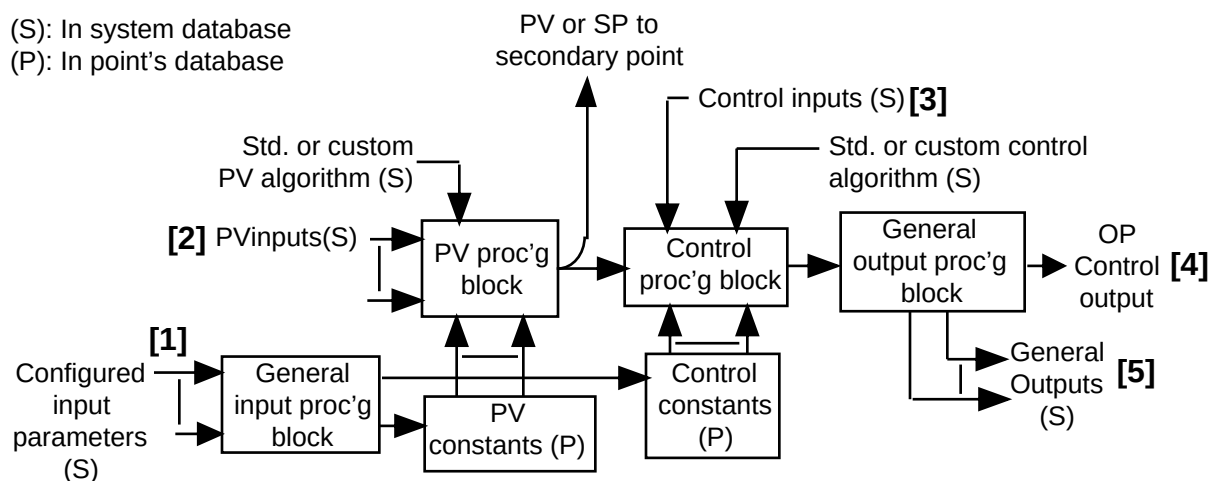
In addition to scheduling points at defined intervals and on demand, the engineer can schedule points to be processed before or after another point, or when some process event occurs.

Database connections

In a process-connected device, most of a data point's inputs are wired directly from a primary process-control element. In the AM, all of a point's inputs come from the process database, which contains information obtained not only from process-control elements but from system components as well. Processing of a data point must take these connections into account.

As an example, the following figure shows some of the connections that can be selected for an AM regulatory data point. In general, an AM point has function blocks for processing general inputs, PVs, control inputs and outputs, and general outputs.

In the following figure, the items like this **[1]** refer to descriptions that follow the figure.



AM regulatory points can be configured to

- do control processing to calculate an output,

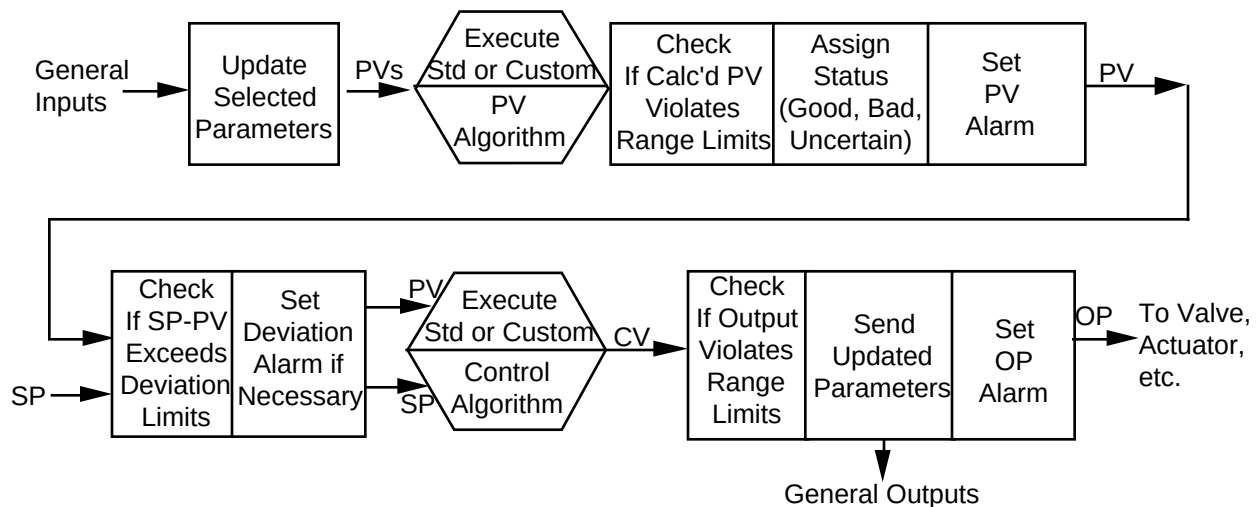
- use the results of its PV processing to calculate a control output, and
- send the results of PV processing to a secondary data point.

Further explanations of the concepts illustrated in the preceding figure are as follows:

- **General input connections [1]** are used to fetch data for constant-type parameters, such as constants in algorithms or alarm trip points.
- **PV input connections [2]** are used to fetch data for variable-type parameters in PV algorithms, such as flow rates or temperatures.
- **Control input connections [3]** are used to fetch data for variable-type parameters in control algorithms.
- **Control output connections [4]** are used to store data in variable-input parameters across initialized connections, such as setpoints for PID algorithms.
- **General output connections [5]** are used to store data in constant-type parameters.

Standard point processing in the AM

The third factor to be considered for a continuous data point is the order in which its parameters are processed. Each type of data point has its own standard processing order. As an example, the following figure shows the standard processing order for a regulatory data point in the AM.



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General Input Processing — At the user's option, an AM point can first fetch up to eight constant values from the system and update designated parameters within the point before proceeding with the remainder of its processing.

PV Processing — After processing any general inputs, the AM point fetches the designated PV inputs from a process-connected device or LCN module. Next it executes the specified PV algorithm and checks the resulting PV value against its range limits. Then it assigns a status (good, bad, uncertain) to the PV.

PV processing is completed by comparing the PV value with its alarm trip points. If the value exceeds the trip points, the system records, annunciates, and displays the appropriate alarms.

AM Regulatory PV algorithms are:

- Null
- Data Acquisition
- Flow Compensation
- Middle-of-Three Selector
- HI/LO/Average Selector
- Summer
- Multiplier/Divider
- Sum of Products
- Variable Dead Time (with Lead/Lag)
- Totalizer
- Statistical Calc
- General Linearization
- CL PV Algorithm

Control Processing — First, any secondary parameters necessary for control calculations are read or updated, and the incoming PV and SP are checked against deviation limits. The SP can also be compared against a target value specified for it. Then the PV value from the earlier PV processing steps is compared to the SP. If the PV lies outside some predetermined band around the SP, appropriate alarms are recorded, annunciates, and displayed. Next, the point executes the selected control algorithm, producing a control output (OP) value. The OP undergoes further processing steps, such as conversion to engineering units.

AM Regulatory control algorithms are:

- Null
- PID Normal
- PID External Reset Feedback
- PID with Feedforward
- Incremental Summer

- Lead/Lag
- Auto Manual
- Summer
- Multiplier/Divider
- Ratio
- Selector/Override
- Switch
- Ramp Soak
- CL Control Algorithm

General Output Processing — The point can be configured to send updated general parameters back to the system database.

Alarm Distribution Processing — Last, the point collects all changes in alarm states that have occurred during its processing cycle and notifies all appropriate system devices and functions, such as the Alarm Display.

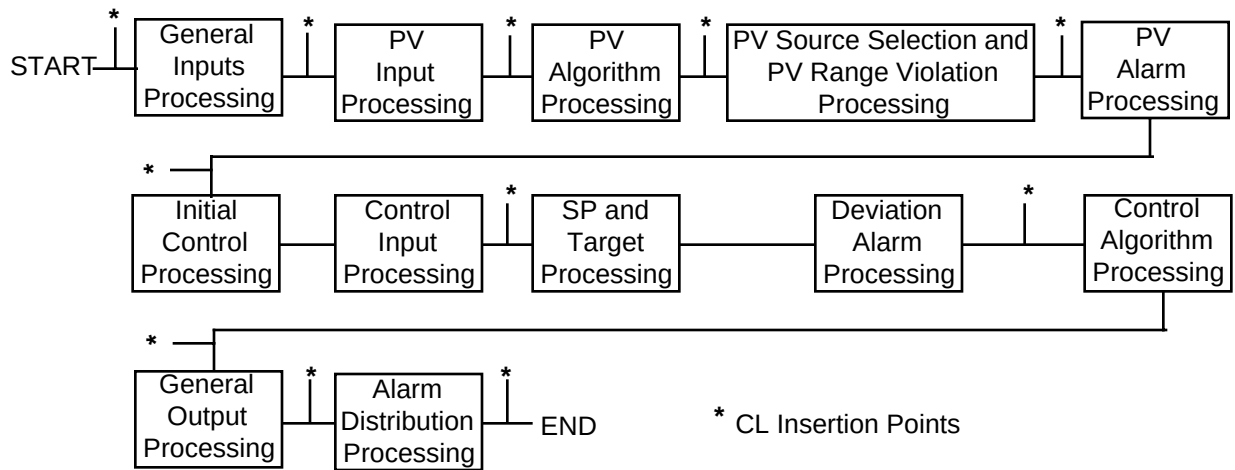
These fixed processing schemes cover many routine control needs. As discussed next, however, the engineer can insert custom processing steps to satisfy unique or complex process control needs.

Modified point processing in the AM

An unusual or proprietary control scheme may require a complex algorithm or nonstandard actions. TPS provides the capability within the AM for the engineer to insert custom processing blocks between, or in place of, the standard processing blocks. The engineer's tool for making these changes is CL/AM. Reference: Standard point processing in the AM, page 72

CL Algorithms — The engineer can use CL/AM to write custom algorithms, which can be used in place of any of the standard PV, control, or logic algorithms in the Application Module.

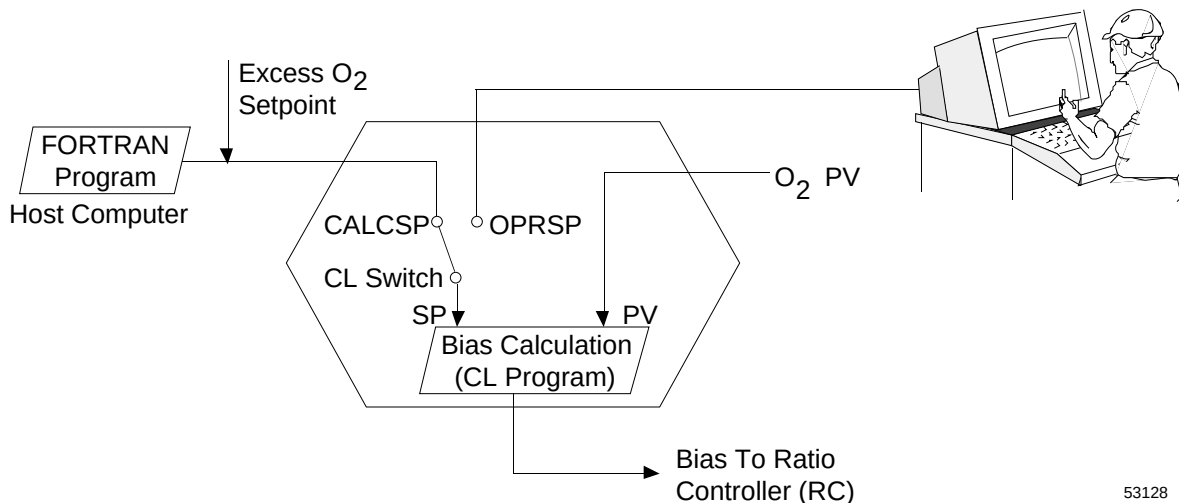
CL Insertion Points — Within the standard processing order of regulatory and logic data points, there are certain locations, called "CL insertion points," where the engineer can insert a block of CL code. The CL insertion point can be configured to execute every time the point is processed or only under specified conditions. As shown in this figure, CL routines may be inserted before or after almost any processing block.



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CL Switch — A "CL switch" is an optional set of parameters that allows an operator, an engineer, or a user-written program to command control strategy changes. The actual switching and/or monitoring operation is performed by user-written CL blocks. These blocks can be linked to any valid insertion point, and a point can have up to two switches. Each switch can have up to five different positions. A CL switch can be programmed to execute every time a point is processed or only when the CL switch position is changed.

The following figure uses the combustion optimization example in [Application programs and data points, page 67](#) to illustrate a simple 2-position CL switch. The CL switch is normally in the CALCSP position, and the Fortran program in the host computer calculates the excess oxygen setpoint. If the SP is not available from the computer, however, the operator changes the switch to the OPRSP position, and then supplies the excess oxygen (bias calculation) setpoint from the Universal Station.



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**Custom and
Switch data points**

The previous discussions explained how users can add custom data and CL blocks to standard regulatory data points to get enhanced and customized functions.

In many complex control strategies it is convenient to go even further—to have a data point consisting exclusively of custom parts, i.e., custom data and CL blocks. Typically this logic belongs more to the multipoint strategy than to any one regulatory point therein.

To serve this need, TPS provides in the Application Module a Custom Data Point consisting primarily of custom data and CL blocks with no standard processing functions (except identification and point scheduling). Where some standard MMI support is useful, the AM provides a switch data point, consisting of the Custom Data Point functionality and a CL switch, plus a custom process alarming facility.

4.8 Sequence Programs and Process Module Data Points

Description

A Process Module data point is the repository for information about its associated Control Language program. A CL program controls a set of equipment performing a batch operation. CL programs are compiled and loaded into a Process Manager, Advanced Process Manager, High-Performance Process Manager, or Multifunction Controller, where they are executed.

A Process Module data point itself is not scheduled and is not executed. It is, however, the vehicle through which the state and the mode of its sequence program are changed, by either a human operator or an "automatic" operator (another sequence program, a CL block, or an application program).

States and modes

A Process Module data point has the following states and modes to represent the condition of the associated sequence program.

- Execution states:

NOT LOADED	PAUSE
LOADING	FAIL
LOADED	ERROR
RUN	END

- Operational states:

OFF	SHUTDOWN
READY	EMERGENCY SHUTDOWN
NORMAL	RESTART
HOLD	

- Modes:

Single STEP	AUTOmatic
SEMI-AUTOmatic	

Program execution

The execution of a sequence program in a Process Manager/Advanced Process Manager or Multifunction Controller is interleaved with the execution of other sequence programs in the same device. A sequence may be suspended under certain conditions, to allow another sequence program to run. The other sequence can be an abnormal condition handler, a routine that is part of the same sequence, or an entirely different sequence. The suspension or "preemption" points occur at the beginning of each sequence program step, and at points in the sequence where the sequence is delayed.

The current execution state, operational state, and the mode are shown on the Process Module Display at the Universal Station.

4.9 Host Computer Program Data Points

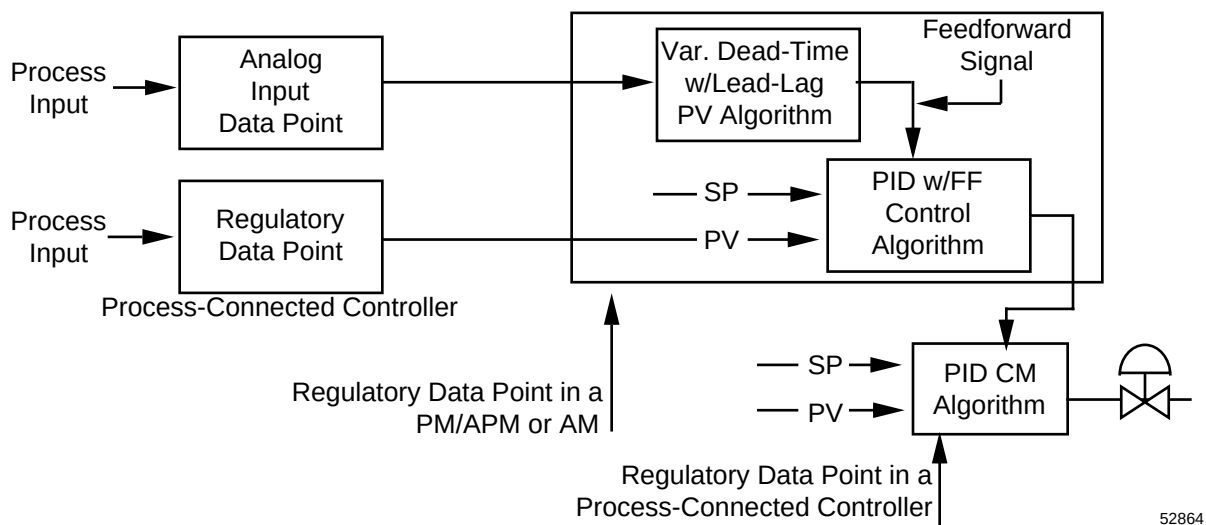
ACIDPs	An Advanced Control Interface Point (ACIDP) data point associated with a host computer is required when an application in the computer is to write to an LCN point. The ACIDP data points are not scheduled and never execute. The programs they represent, however, run according to schedules specified in the data point. Each program can be configured to run: • At selected times of day • At selected intervals • On demand from a Universal Station • When some process event occurs
Control input	If a computer program is providing control input to a continuous control point in a process-connected device, a parameter in the ACIDP can cause a timer to be set. If the computer program does not complete its write prior to the time expiration, the continuous control point is shed to its backup control strategy—auto or manual.
Custom data segment	An ACIDP can also have a custom data segment that can hold data for other data owners, such as a PM/APM, AM, and MC.

4.10 Forming and Manipulating Control Strategies

Forming control strategies

Many processes require a higher complexity of control than a single data point can provide. The TPS System provides mechanisms, discussed in the following sections, that allow complex strategies to be formed from data points and that allow these strategies to be manipulated.

To form desired control strategies, the engineer configures connections between the parameters of data points. Such configurations are strictly "soft-wired" connections, performed inside the process-connected device or module, by standard software. When two or more data points are interconnected, they are said to be members of a cascade control strategy. The following figure shows an example of interconnection between data points that form a flow-control strategy.

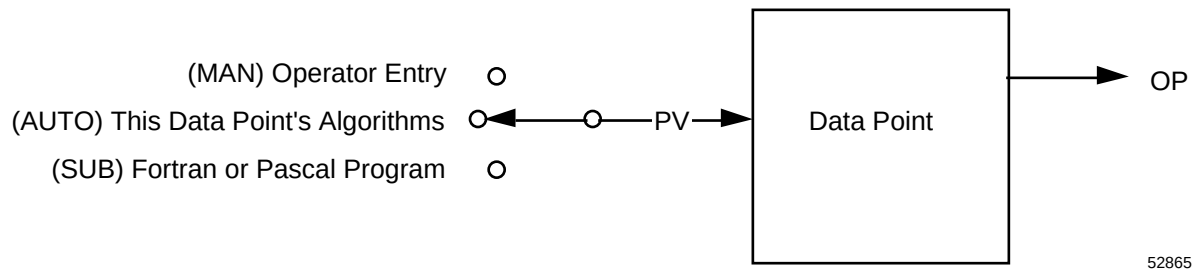


PV source selection

For almost every type of data point, the source of the data point's PV can be selected during configuration. If conditions warrant during process operations, the engineer can switch the PV source. As illustrated in the following figure, the PV sources available are:

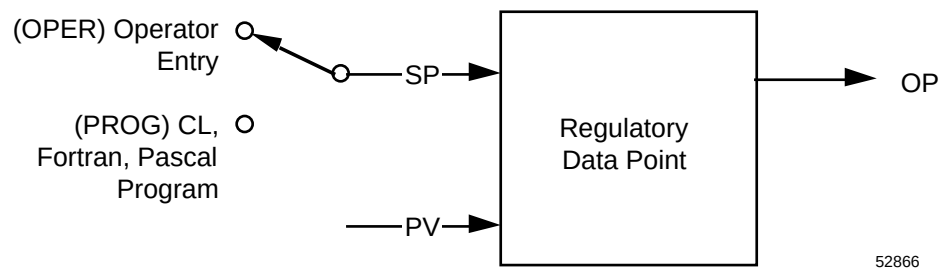
PV MANUAL	PV is entered at a Universal Station.
PV AUTO	PV comes from the configured PV algorithm (which may be CL).
PV SUBSTITUTED	PV is substituted, usually by a Fortran or Pascal program.

Selectable PV sources

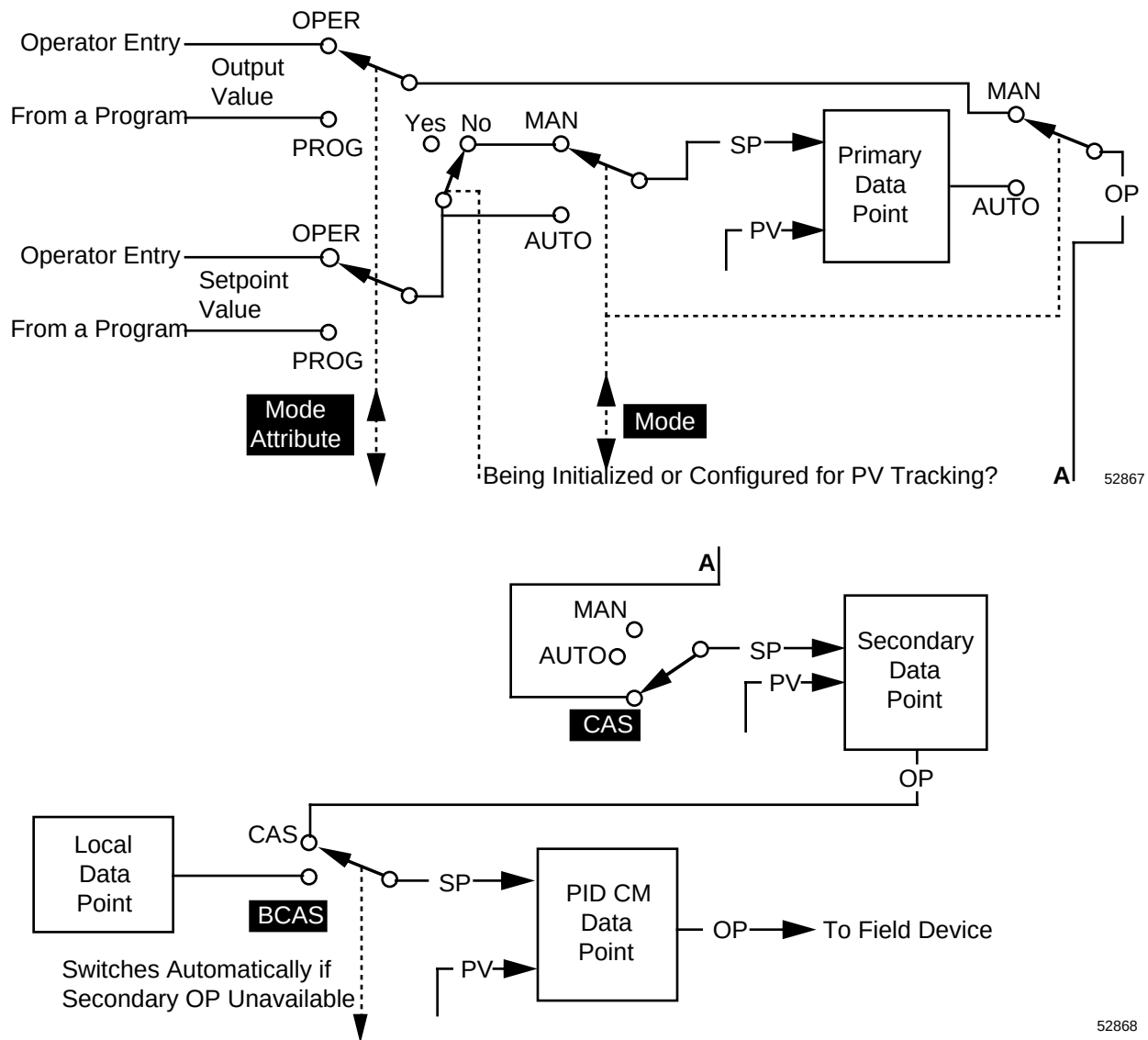


Mode attributes

As the following figure shows, regulatory and analog output data points also can have one of two mode attributes, which determines whether the operator (OPER) or a program (PROG) can change the point's mode, SP (or OP of an analog output data point), ratio, and bias value. If the attribute is OPER, the operator enters the required value(s) at the Universal Station. If the attribute is PROG, the value is supplied by a CL, Pascal, or Fortran program.



Data point modes Mode determines the role of a Regulatory or Analog Output data point in a control strategy. An operator, an engineer, internal logic, or a control program can change a data point's mode. The effects of the modes are shown in the following figure and descriptions of each mode follow the figure.



Manual (MAN) — MANual mode is typically used to temporarily override control strategies. When a data point is in MAN mode, automatic output is discontinued, and the output can be manipulated by an operator or engineer at a Universal Station, or by a user-written program. If the point uses a setpoint, and if the setpoint is not being initialized or is not configured for PV tracking, the setpoint can also be manipulated. As determined by the data point's attribute, either the Universal Station operator (OPER) or a user-written program (PROG) does the manipulating. This attribute will be OPER unless the data point is functioning under NORM mode with PROG attribute (as verified on the detail display).

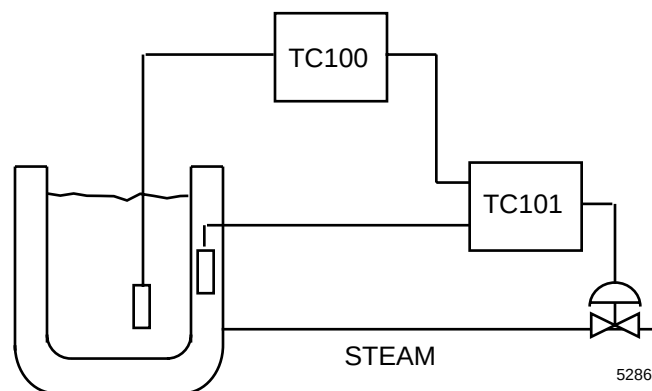
Automatic (AUTO) — AUTOMATIC mode is the normal mode for the first or ultimate, primary data point in a cascade-control strategy. It is also the normal mode for a data point that is the only data point in a control strategy. In AUTO mode, the setpoint is provided by an operator at a Universal Station (OPER) or by a user-written program (PROG).

Cascade (CAS) — CASCade mode permits the output of one data point to be connected (by configuration, not by wiring) to the input of another. This is the normal operating mode for all the secondary data points in a cascade. The data point that supplies the input to a secondary data point is referred to as the "primary" data point to the secondary. A cascade connection can be broken by switching from CAS to AUTO, and the data point just changed to AUTO mode then becomes the ultimate primary.

Backup Cascade (BCAS) — The Backup CASCade mode provides extra "insurance" for regulatory data points on the UCN or hiway that have inputs cascaded from a higher-level data point. While operating normally in CAS mode, another data point in the same process-connected device tracks the input (setpoint) from the higher-level data point. If the input from the higher-level data point becomes unavailable, the secondary data-point's mode automatically changes to BCAS, and the local data point takes over as the primary, supplying the setpoint to the secondary.

Selecting "normal" mode

With the TPS System, the engineer can make it easy for the operator to select the "normal" operating mode for a control loop. As an example, let's look at the cascade control scheme for a jacketed vessel, as shown in the following figure.



Heat-up of the vessel is controlled by a program as follows:

Table 4-17 Heat-Up Program

Stage	Description
1	Set mode of TC101 = MAN.
2	Set output of TC101 = 100%.
3	Wait until temperature as measured by TC100 = 200°F.
4	Set mode of TC101 = AUTO.
5	Set setpoint of TC101 = 300°F.
6	Wait 30 minutes.
7	Set mode of TC101 = CASC.

Because the program is supplying setpoint and output to point TC101, the point's attribute is PROGram. The program sets the normal mode of TC101: MANual in Stages 1-3, AUTOMATIC in Stages 4-6, and CASC in Stage 7.

Now suppose that sometime during the wait of Stage 6 the operator detects something abnormal and takes control of TC101 by setting its mode to MANual. If after correcting the problem, the operator wants to return heat-up control to the program but has forgotten the mode and attribute the point had, simply pushing the "NORMAL" key returns the mode to the last mode and attribute set by the program (AUTO and PROG).

In this way, the operator never needs to remember the mode the point would be in if normal operation had not been interrupted; the system remembers and responds when the NORMAL key is pressed. Most operator mode interactions involve only NORMAL and MANUAL keys, greatly simplifying the operator's task.

"Bumpless" mode changes

The goal of regulatory process control is to keep a process variable (PV) at the desired value (SP), so that:

$$SP - PV = 0$$

As discussed in the previous sections on modes in the TPS System, a setpoint can come from a variety of sources:

- the operator,
- another data point
- a CL block
- a sequence program
- a Pascal program
- a Fortran program

If the mode were changed and the "new" setpoint were different from the PV, the difference could cause the position of the valve to change, in an attempt to bring the PV to SP.

The TPS System provides three mechanisms for preventing this change in output that can result when a mode is changed: PV tracking, initialization, and antiwindup.

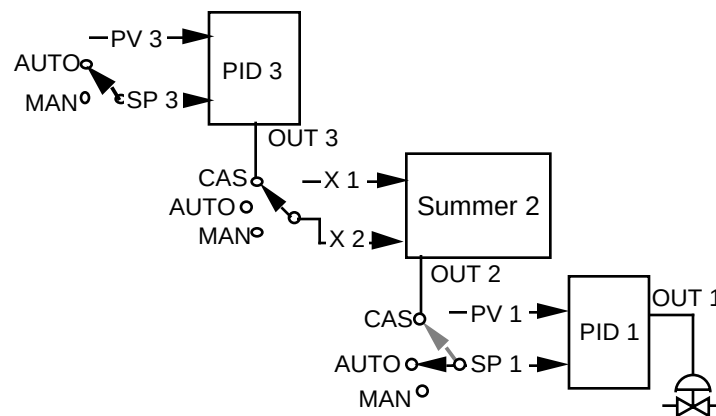
PV Tracking — When a data point is in MANual mode, its output (OP) is being controlled directly, not as the result of the difference between the SP and PV. Since the PV will change as the result of these OP manipulations, difference could exist between the SP and the PV when the point is put back into a higher-level mode. To prevent this difference, the TPS System provides PV tracking, which keeps the local setpoint of a data point equal to its PV while the point is in MANual mode.

PV tracking only applies to data points that have a PV input and that use a control algorithm with a floating output. If PV tracking is an option for a given data point, the system will so indicate during configuration; the engineer then selects PV tracking or not, depending on the role of the point in a control strategy.

Initialization — When data points are cascaded, the output of one becomes an input to another. When an open cascade connection is closed, if the output of the primary (OUT2) is not equal to the setpoint of the secondary, the difference can change the output to the valve. When configured to do so, the TPS System automatically calculates and adjusts the necessary parameters in a cascaded string of data points to prevent switching upsets; this is called initialization.

In the following figure, for example, when the mode of data point PID 1 is changed to CAScade, the system normally performs the following initialization calculations and parameter adjustment before the cascade connection is actually closed:

- Sets $OUT2 = SP1$, so that there will be no deviation, and therefore, no change in the output to the valve.
- Back-calculates the value of $X2$ necessary to yield that value of $OUT2$, and sets it to that value.
- Sets $OUT3 = X2$, completing the initialization, and puts the data points "on control."



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Antiwindup—If the PID1 data point in the preceding figure is taken out of CASCade mode, or reaches an output constraint, the data points "upstream" from it must be notified, so that they do not continue to calculate outputs that have no ultimate outlet. In the TPS System, this information is provided by a Windup Status parameter in the data points in a cascade. This Windup Status parameter is communicated "upstream" and used to inhibit further integral (windup) action of PID3.

Local Manual (LMAN) operation

Local MANual operation is unique to hardware analog outputs of regulatory data points configured in hiway boxes that have analog display units attached. When a point is set to LMAN, its output can be changed from only the analog display unit.

4.11 Alarming

Data point alarms Most data points provide for alarms of some type that are displayed and annunciated at the Universal Station. Alarms can be reported on an area or unit basis according to assigned priorities, and can be disabled or inhibited by point or unit.

Process alarms Listed below are typical process alarms that can be configured (depending on the device) by the engineer and detected by various components of the TPS System.

Table 4-18 PV and Deviation Alarms

(PV) Process Variable	(SP - PV) Deviations
Bad PV	High
PV High	High, high
PV High, high	High, high, high
PV High, high, high	Low
PV Low	Low, low
PV Low, low	Low, low, low
PV Low, low, low	
Significant change	
Rate of change (+)	
Rate of change (-)	

Table 4-19 Sequence and Digital Input Alarms

Sequence Alarms	Digital Inputs
Process Module	Off normal
State Change	Change of state
Sequence Executive	Command disagree
State Change	
Phase Time-Out	

Alarm detection Alarms are detected, reported, "journalled," and acknowledged. "Journalled" means recording the alarm in a History Module.

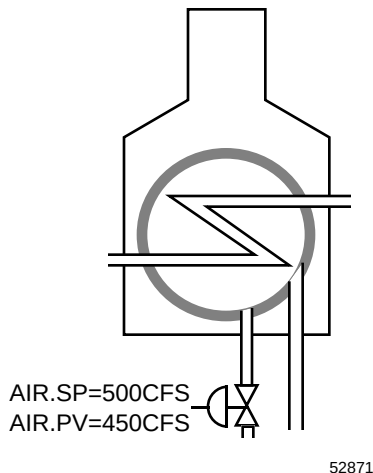
Individual data points detect alarm states by comparing values against limits, ranges, or other conditions specified by the engineer and report any changes in alarm states to the appropriate devices. Alarms are acknowledged at the Universal Station.

Alarms can be disabled, inhibited, or "cut out." Alarm disable and alarm inhibit are operator functions that filter out nuisance alarms during startup or shutdown of sections of the process. Alarm disable stops the reporting of detected alarms to the MMI; alarm inhibit bypasses the detection of an alarm.

In systems where one alarm will inevitably cause the occurrence of secondary alarms, the alarm cutout function can be used to suppress a number of configured secondary alarms.

Deviation alarm

In the following example, assume that the deviation trip point for air flow is 40 cubic feet per second. The (SP-PV) deviation is 50 cubic feet per second. Therefore, an alarm will be set for the air flow control group.



Alarm trip values

The engineer sets alarm trip values in engineering units when the point is built and can change trip values during operation. Trip values can also be set by a CL or Fortran program or can be determined by General Input connections. Trip values can have configured deadbands, usually expressed as some percentage of the PV range.

Alarm priorities

An engineer can configure alarms at one of five levels of priority:

5. Emergency
4. High
3. Low
2. Journal-only
1. No Action

Each priority is annunciated differently at the Universal Station.

**Process unit
alarms**

Alarms are reported, enabled, disabled, inhibited, and retrieved for entire process units as well as for individual data points.

4.12 History Collections

Description	In addition to the trends traditionally provided by TDC 2000 systems, components of the TPS System can be selected and configured to collect history on process variables and events for storage in the History Module. Such history can be viewed by the operator at the Universal Station or recorded on a printer. Data can also be collected for long-term volume storage through the Archive Replay Module.
Process variable history	The History Module (HM) collects history on process variables. The types of history that the engineer can specify to be collected for process variables are: • 5, 10, 20, or 60 second snapshots • Hourly averages • Shift averages • Daily averages • Monthly averages • User-specified averages
Event History	An event is defined as some action that causes a notification to be sent to some system component. History can be collected on the following types of events: • Process alarms • Digital input sequence of events • Operator process changes • Operator system changes • Operator messages • System status changes • System errors • System maintenance recommendations
Archive Replay Module data collection	The Archive Replay Module is capable of collecting four types of data from the LCN: • Fast history consisting of any real or enumerated point.parameter values. These can be collected at a sustained rate of approximately 20 parameters per second. • Real-time journal events.

- History Module continuous history snapshot values.
- Any ASCII file located on the History Module.

This data is ultimately stored on optical disks for retrieval and analysis as required.

Section 5—Human Interface

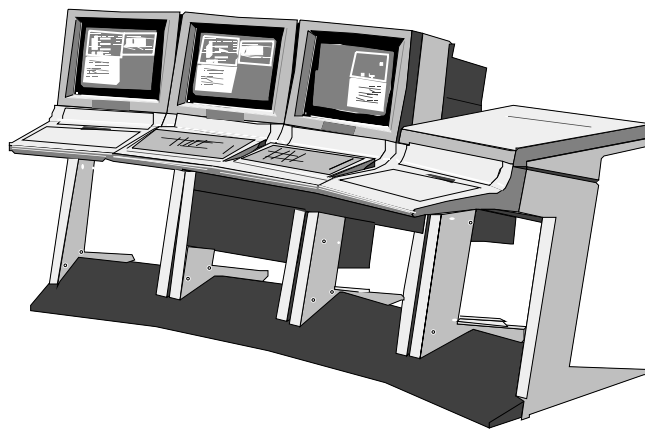
5.1 Introduction

Integrated interface

The TPS System provides an integrated interface between the process, the TPS hardware, and the users of the system. Standard hardware and standard software contribute to this interface, which is designed to meet the needs of those who interact with the system or who need access to system data.

Operator console

To satisfy a variety of user requirements, a number of user hardware station options are available. In each case, a color CRT screen and associated keyboard comprise the primary tools that connect the user to the process and to the TPS System itself. In a control room environment, several stations are typically combined into a process operator's console, as illustrated in the following figure.



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Integrated software

The integrating force that gives life to the console is the standardized TPS software. Software is provided to meet not only the requirements of the Process Operator, but also the very different functions needed by the Process Engineer and Maintenance Technician.

Since the operator's displays and associated targets are separate from those required for the engineering functions, the operators are not burdened with information outside their realm of activity. On the other hand, operators, engineers, and technicians can share the same primary hardware, thereby minimizing initial costs, maintenance costs, and spare parts requirements.

All the human interface software is designed to be easy to use. For example, keystrokes for individual operations are minimized. In fact, with a touchscreen option, a user can initiate a function or call up a display simply by touching a target on the screen with a finger or pencil.

Another ease-of-use feature is the on-screen guidance the standard TPS System provides to the process operator, process supervisor, process engineer, and maintenance technician as they carry out their normal operations. In addition, process engineers can build custom "Help" displays that contain supplemental information for operators.

Process engineering functions

The engineering software contains all the functions necessary to set up the process database (other tools using PCs are also available). This database is used by the TPS System to operate the process and to notify the operator of abnormal conditions, and by the operator to monitor and control the process.

Typical functions of the engineering software include building displays and reports; building and configuring data points; and writing and compiling Control Language (CL) programs.

System engineering activities are described in more detail in 4— Data Acquisition and Control, page 45, and 7— System Implementation, page 155.

Process operating functions

The Process Operator software contains all the functions necessary for normal operation of the process. With it, the operator has access to all the data needed, at several levels of detail, on current and past process operations, both continuous and discontinuous, regardless of its source in the system—process-connected boxes, Application Modules, computing modules, etc.

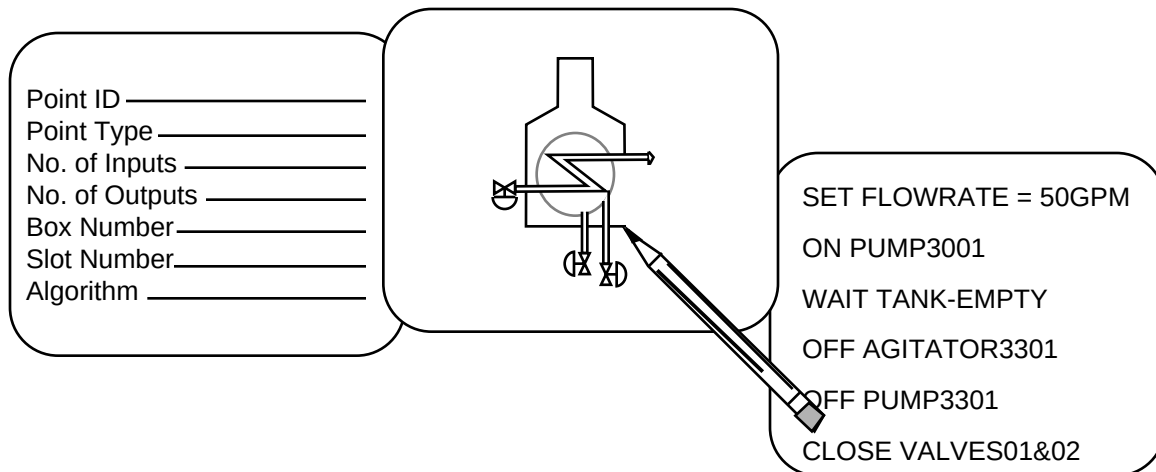
Using custom graphics, standard operating displays, and alarm displays, the operator can efficiently and effectively monitor and control the entire area of the assigned process. In addition, many process-history functions, such as logs and trends, are available to the operator.

System operation is discussed in more detail in 6— System Operation, page 113.

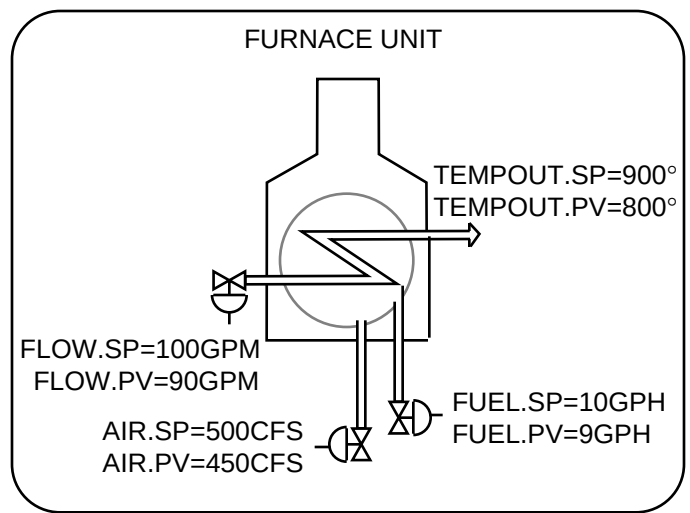
Maintenance Technician's functions

In cases where TPS System self-diagnostics are not sufficient to locate and correct a problem, the Maintenance Technician can access detailed diagnostic information by way of special engineering software routines. For those instances where problems cannot be diagnosed on site, additional help is available from the Honeywell Technical Assistance Center (TAC).

System maintenance is discussed more fully in 8— System Maintenance, page 173.



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5.2 Video Displays

Types of displays

The primary information interface for the TPS System is the high-resolution video-display monitor (VDM) on which process engineers, process operators, process supervisors, and maintenance technicians call up information for display. TPS provides a broad set of standard displays for all users. The process engineer can augment these by building custom graphics, custom reports, and help displays.

The standard displays available for the types of user differ, because their information needs are not the same. Operators, for example, have displays that help them monitor and control the area of the process for which they are responsible. Engineers, on the other hand, have displays that help them build graphic displays and construct reports. Maintenance technicians have displays that guide them through the diagnosis of a problem.

In the TPS System, displays are designed to present information to the operator in the best possible way. A combination of techniques, ranging from traditional bar graphs to contemporary pattern-recognition designs such as the Alarm Annunciator Display, is used to accomplish this goal.

Many displays have both dynamic and interactive areas. Dynamic areas contain information that is periodically updated or have graphic elements whose attributes can vary with the associated process values. For example, a furnace symbol can change color from green to red and begin to flash if the air flow falls below the low limit and goes into alarm.

Interactive areas accept input from the operator and can relay information to other parts of the TPS System. From a group display, for example, the operator can change the control mode. Interactive areas can require keyboard entries, or they can be screen targets, as described in the next paragraphs.

Calling up displays

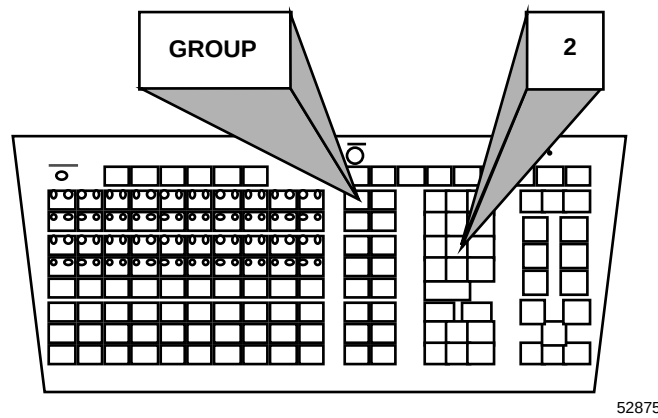
In general, displays can be called up in two ways:

- By keystroke(s), using either pre-assigned or user-assigned keys.
- By selection of a target on another display.

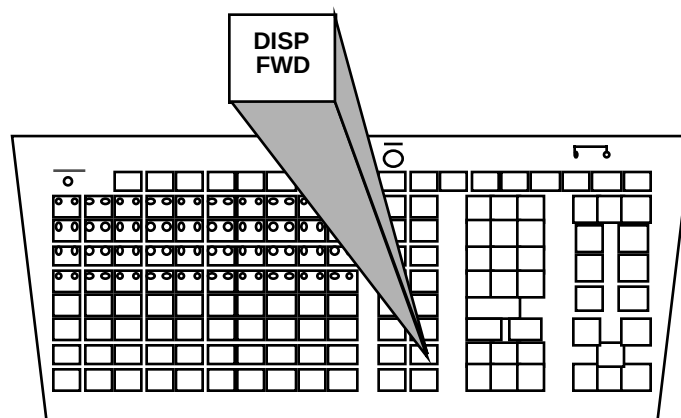
Direct keyboard call up

A number of keys are provided on the Operator's Keyboard and the Engineer's Keyboard for directly calling up displays. Some keys call up displays independent of what currently appears on the screen; others call up displays relative to what appears on the screen.

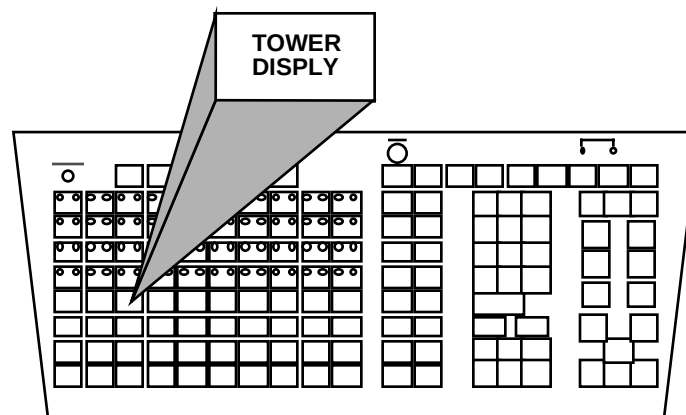
For example, pressing the <GROUP> key plus the number 2 displays the graphic representation of the eight points that make up Group 2, regardless of what was previously on the screen.



If the display for Group 1 is on the screen, however, the Group 2 display can be called up simply by pressing the <DISP FWD> key.



The number of keystrokes required to call up a display varies. Each of several standard displays is brought up using a single preassigned key. Other frequently used displays can be assigned to one of the special keys on the Operator's Keyboard, so they can also be called up with a single keystroke.



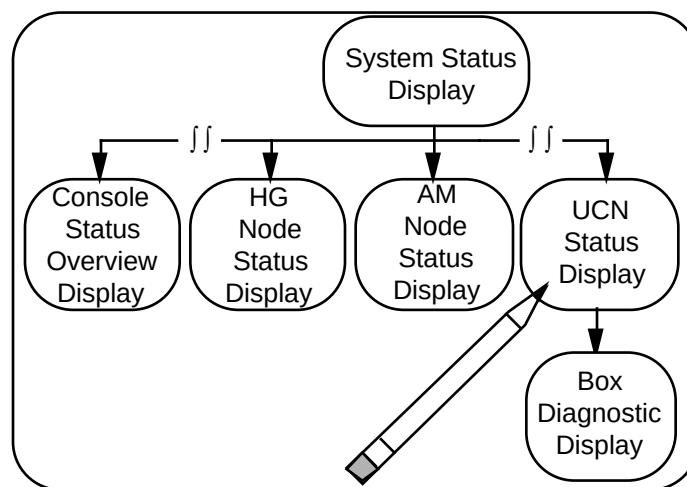
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The keys and the general functions of each keyboard are described beginning on the following page.

Call up by target selection

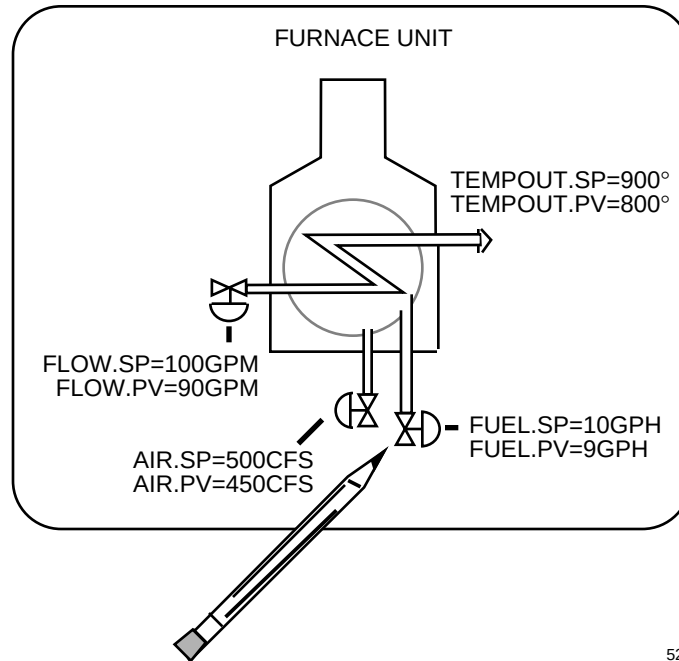
Many displays have interactive areas that allow the operator to call up another display. These "targets" can be activated by a touch if the station has the touchscreen option; otherwise the screen cursor is moved to the target, which is then selected by key or pushbutton.

The System Status Display, for example, contains a series of touch-target boxes used to call up displays showing the status of the components of the TPS System, such as a UCN Status Display.



52878

Targets can be built into custom displays. The engineer can build a graphic display, for example, so that selecting a valve symbol calls up the detail display for the point that governs that valve.



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Security on standard displays

For security, standard displays are designed so that changes to the process cannot be made directly from a single target. A typical action, for example, is to identify a parameter to be changed, key in a change in value, and to implement the change by an EXECUTE command.

5.3 Universal Station

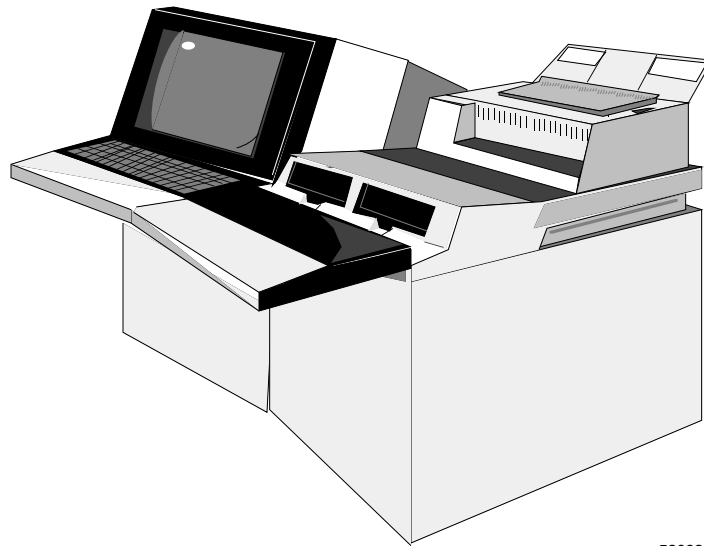
Two designs

The Universal Station has served as the primary operator interface since the introduction of TDC 3000. Today it is available in two distinct furniture designs:

- Original furniture design
- Modern ergonomic design

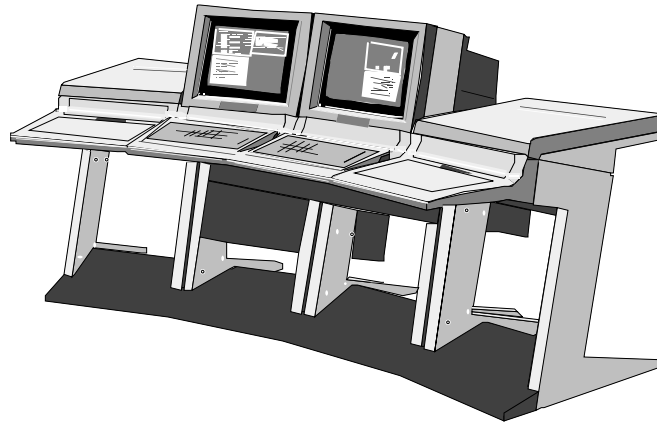
These are shown on the following two figures.

Original furniture design



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Modern ergonomic design



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Both designs accommodate operator's and engineer's keyboards, visual display terminals, electronic modules, and other peripheral devices as required. The physical design and placement of these components vary according to type of furniture and options selected. These differences are described in the paragraphs that follow.

Both types of furniture are available with additional CRT displays mounted in an upper console tier.

Universal Station Operator's Keyboard

The keyboard design has been driven by analysis of operator use. Observation of operators in action shows that keyboard activities are simple display requests over 90% of the time. Operators spend significant portions of their time moving through displays in monitoring activities. The simplest and most direct method to support this activity from the keyboard is to provide single keystroke access to frequently used displays. Dedicated keys facilitate the following operator functions, as well as others:

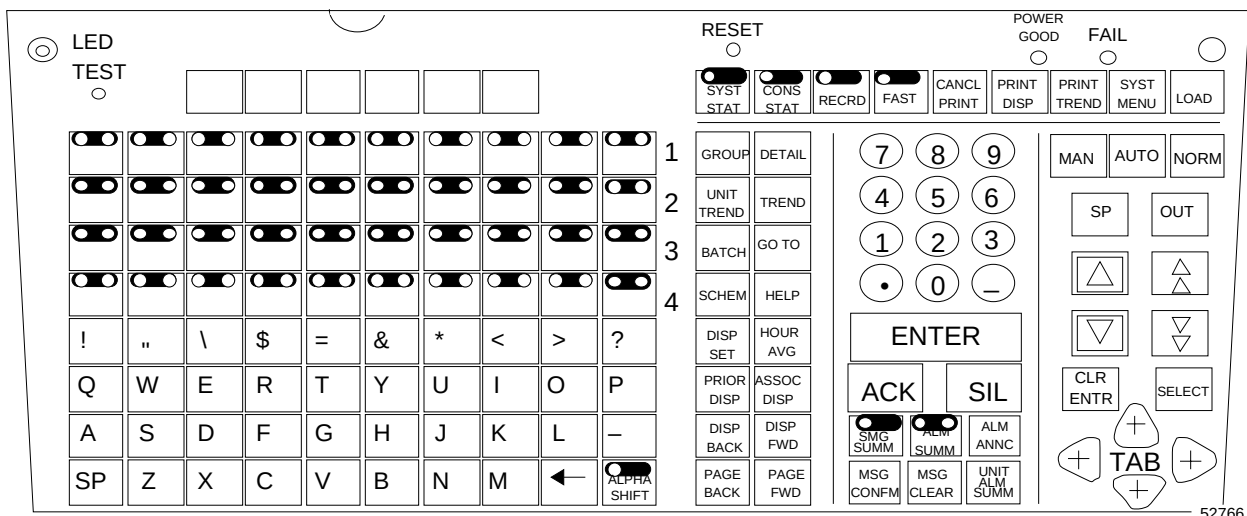
- Directly call up information about the status of the TPS System components, including the operator's own console.
- Directly call up process displays at several levels of detail.
- Move between different displays at the same level of detail.
- Select items on a particular display to be manipulated.
- Control the process at the point level.
- Enter both alpha and numeric data.

Operator's Keyboard features

The Operator's Keyboard has a number of features that make interaction with the process easy and efficient.

- **State-of-the-art panel** — The keyboard itself is a flat-plate device to provide protection against liquids and dirt that may be present in the control room. An audible "click" or "beep" provides feedback on keystrokes.

The Operator's Keyboard for the original Universal Station furniture is mounted in a recessed area to allow books and papers to be placed on the surface without interfering with the keyboard. In the newer furniture the keyboard is fitted with folding legs that enable the back to be raised for convenience during use.



- **Organization** — The new Universal Station furniture comes with a keyboard that has alpha keys arranged like those on a standard typewriter ("QWERTY"). The original Universal Station furniture is available with this same keyboard, and is also available with a keyboard that has its alpha keys arranged alphabetically.

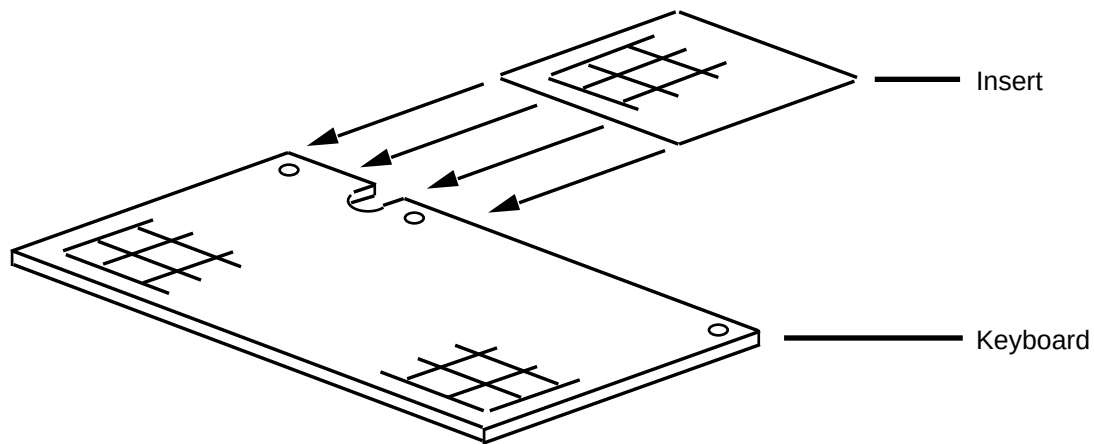
In all other respects, both keyboards are organized identically to make the operator's demands for information and interaction with the process as quick and easy as possible. Alarm and message touch keys are grouped together, for example, near the bottom center of the right half of the keyboard, and touch keys for dealing with displays are grouped into two long columns to the left of the numerical pad. Process-related touch keys for changing the operating mode (MANual, AUTOmatic, and NORMal) and for raising and lowering the setpoint (SP) and output (OUT) are grouped together at the right side of the keyboard.

The left-half of the keyboard provides a matrix of user-function keys that can be assigned to provide single-keystroke access to frequently used displays. The user can arrange this section of the keyboard, blanking out unused keys to enhance the layout, to best suit the needs of the particular area of the plant normally dedicated to the specific US. Thus, the user can implement specific key arrangements and display scenarios that best serve the operator with single-key access to the most frequent displays in a logical arrangement. This includes the ability to assign alarm lamps to these keys. It is expected that this will be the most heavily used portion of the keyboard.

- **Color coding** — Some critical touch keys are color-coded for easy recognition, like the red one to acknowledge (ACK) alarms and blue ones that raise and lower the setpoint and output values.
- **Keystroke errors** — The keyboard sounds a warning if more than one function key is pressed at a time, or if the operator makes an invalid keystroke. If the operator enters a nonexistent point name when trying to call up a point detail display, for example, an error tone is generated and a CAN NOT FIND POINT error message is displayed on the screen.
- **Process alarms** — Another aid for operators is LED indicator lamps on some touch keys that light up to indicate alarm-function status. The ALM SUMM touch key in the alarm/message bank, for example, has both a red and a yellow LED; the customer might configure the red LED to light up under severe alarm conditions and the yellow one to light up when there is a less severe alarm. On the Operator's Keyboard, 41 touch keys have both red and yellow LEDs.

The Universal Station keyboard electronics also provide three contacts for wiring external annunciators for certain user-selected types of alarms, such as all emergency, high, or low alarms.

- **User-configurable keys** — A particularly useful feature of the Operator's Keyboard is the 85 touch keys that are "configurable"—that is, they can be defined and labeled (or relabeled) by the user. These keys are defined by the process engineer during configuration (or redefined later). As shown in the following figure, a clear plastic insert containing the touch key labels slides into a pocket in the keyboard.



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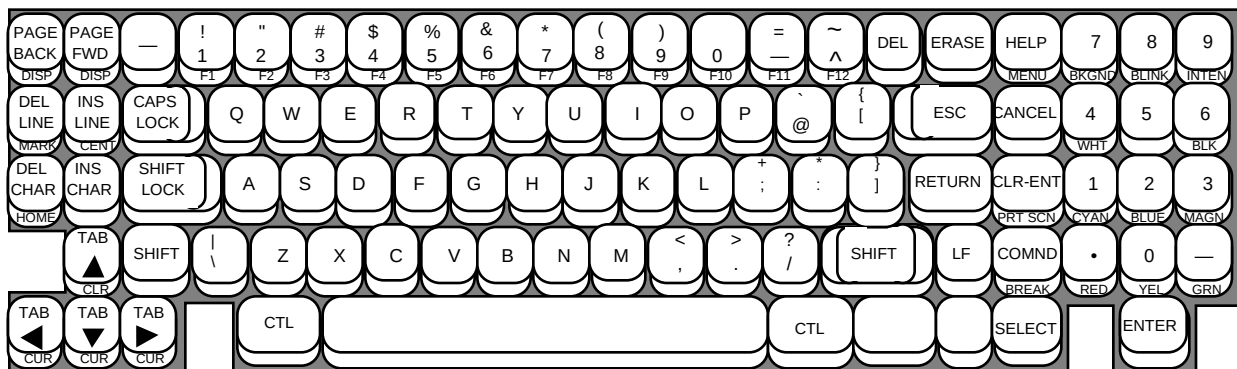
Engineer's Keyboard

Two engineering keyboard designs are available for Universal Stations. Both fixed and portable Engineer's Keyboards are used.

As shown in the following two figures, the layout is somewhat different, although the same functions are performed with both keyboards.

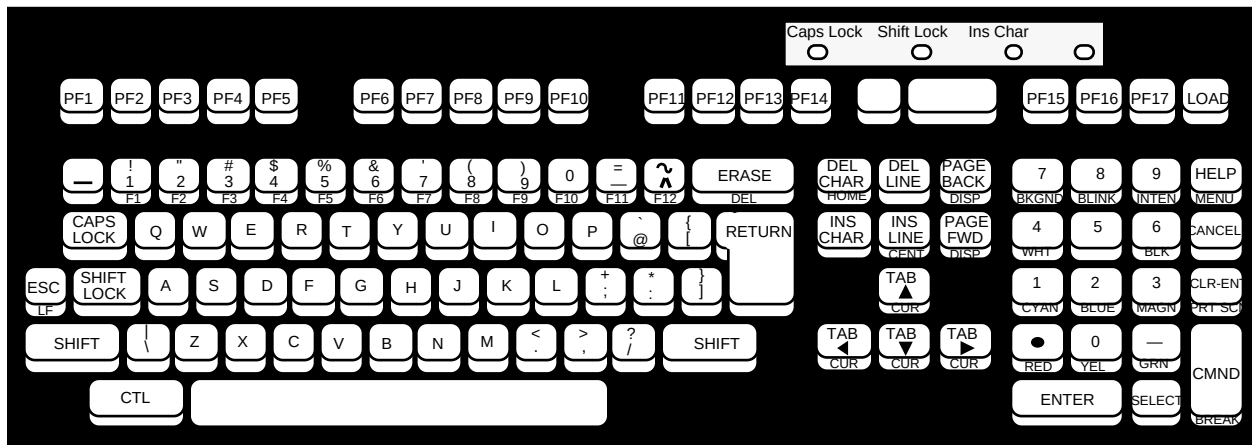
- The fixed keyboard is stored under a cover in the work surface.
- The portable keyboard is plugged in for use.

Fixed Engineer's Keyboard



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Portable Engineer's Keyboard



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Although optional on a Universal Station basis, at least one Universal Station or Universal Work Station Engineer's Keyboard is required in the system. Both types of US Engineer's Keyboard have the familiar keys of a typewriter, as well as a number of dedicated function keys to help in building and editing displays.

The keys on both keyboards are organized into groups for ease of use:

- **Data-Entry Keys**—The alpha keys have the standard QWERTY keyboard arrangement, and a separate numerical keypad is close by. Tab/cursor keys are located on the left side of the first keyboard, and in the right-center of the second keyboard.
- **Editing Keys**—Keys for inserting and deleting lines and characters are grouped together above the tab keys.
- **Function Keys**—Twelve function selections, associated with the row of numerical keys are activated by simultaneously pressing a control (CTL) key.

The second keyboard has an additional row of 17 dedicated function keys, labeled PF1 through PF17, which, when used together with the SHIFT and CTL keys, provides 51 virtual function keys. These keys are configurable using the standard Button Configuration function. The user can assign these keys to perform any existing function as well as being able to simulate existing keys on the Operator's Keyboard.

- **Attribute Keys** —While building displays, the process engineer can select directly from the keyboard the characteristics or attributes that symbols in the displays are to have. Using control keys on the right side of either keyboard, the engineer can select the color, the intensity, the blinking behavior, and whether the symbol will appear on normal or reverse background.
- **Alternate Action Keys** — Three keys lock into alternate modes when pressed and continue in the selected mode until pressed again. These are the character insert key (INS CHAR), the SHIFT LOCK key that puts all keys into upper-case mode, and the CAPS LOCK key that puts only alpha keys into uppercase mode.
- **Repeating Keys** — The tab/cursor, alpha, and numeric keys all repeat at approximately 15 characters per second, after an initial 500 ms delay.

Integrated Keyboard

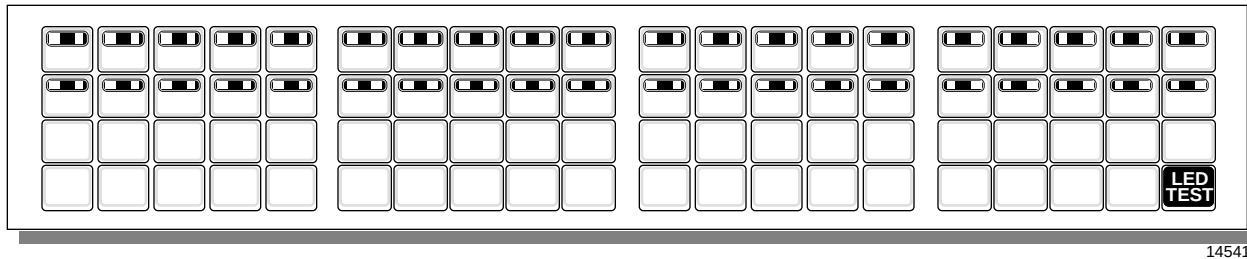
For R510, the integrated keyboard has three panels that provide the user with the ability to view the programmable buttons via the Button Configurator in a representation similar to the keyboard layout they are using (for example, ABC, QUERTY, Integrated). There is also the addition of five standard keys to invoke displays. These five keys can be simulated on the screen using key actors through a schematic or button file.

Specifically, the integrated keyboard includes:

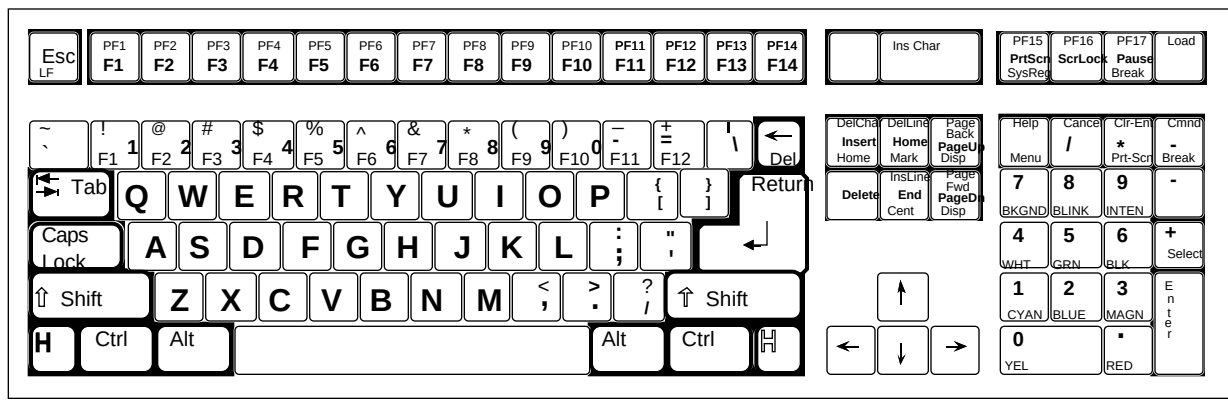
- **80 Button Pad**—The user-configurable keys are arranged in four sets of twenty, as the top, upright panel closest to the display screen.
- **Full Travel Keyboard**—This keyboard, with data entry, editing, and function keys, is the center panel of the integrated keyboard unit. The full travel keyboard offers an optional trackball.
- **Operator Control Panel**—The bottom panel is a redesign of the Operator's Keyboard. It includes six user-configurable keys. It also has five new keys to invoke standard displays:
 - AM STATS invokes the Application Modules Node Status Display
 - COMM NETWK STATS invokes the Communication Network Status Display
 - PROC NETWK STATS invokes the Process Networks Status Display
 - ORG SUMM invokes the Organizational Summary Menu
 - UNIT ASGN invokes the Unit Assignment Display

Integrated Keyboard, figure

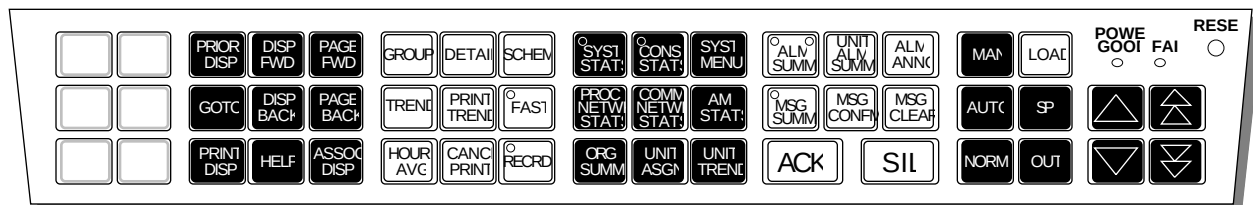
The integrated keyboard includes three keyboard panels as shown in the following figures.



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Video display monitor

Universal Stations in the original furniture use a 19-inch (48-cm) video-display monitor. Universal Stations in the new furniture use a 21-inch (53-cm) video-display monitor. Both are full-color, 640 x 448-pixel CRTs that produce high-contrast characters and graphics with crisp lines that eliminate "fuzziness," thereby minimizing operator fatigue. Action selections from a display can be made using tab and selection keys on an operator or engineering keyboard, by optional touchscreen actuation, or by optional trackball or mouse manipulation.

The video-display generator board in the Universal Station electronics module provides both a character-addressable plane and a dot-addressable plane. When graphics are being built, these planes can be configured to be transparent or opaque. Such resolution and configurability provide maximum flexibility in graphic design.

Other Universal Station Components

In addition to keyboard(s) and color CRT, an electronics module is part of every Universal Station. The electronics module can be mounted either in the console furniture or in a separate equipment rack.

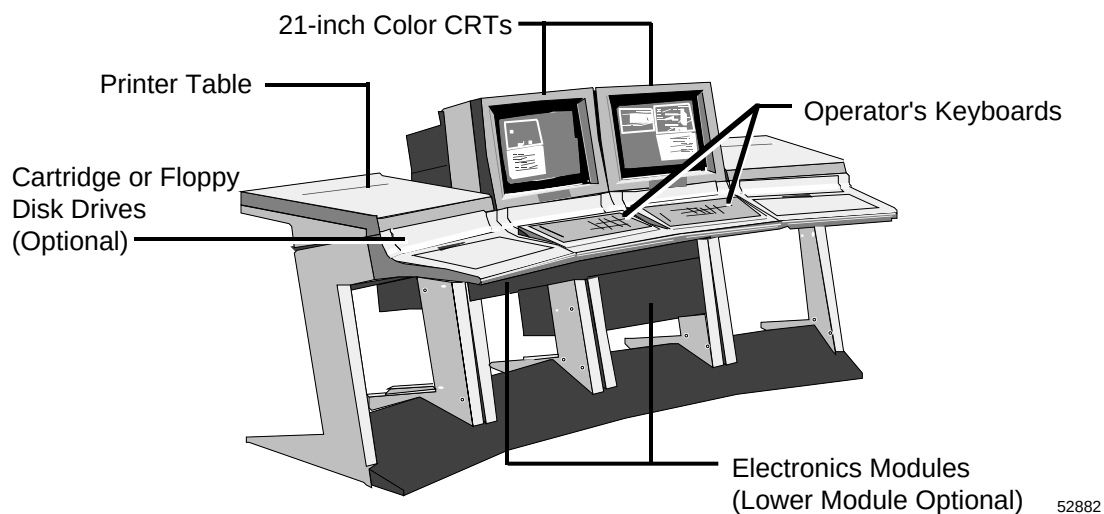
For interactive monitoring and control, additional pieces of equipment are available that add to the Universal Station functionality. These options include:

- Cartridge Disk Drives
- Floppy Disk Drives
- Dot Matrix Printers (2 models)
- Touchscreen
- Trackball
- Trend Pens

Regardless of the number of Universal Stations and the mix of options chosen, the operating procedures for the TPS System are the same.

Either type of US furniture provides space for the Universal Station electronics module, the color video-display monitor, the Operator's Keyboard, and one of the Engineer's Keyboards. Optional equipment is mounted in, or on, adjacent sections of modular furniture as shown in the following figure. The following paragraphs briefly describe each component.

Ergonomic Console Equipment Layout



Cartridge Disk Drive

The Cartridge Disk Drive is a compact, easy-to-use device for storing and retrieving data. It accepts 5.25-inch disk cartridges that have a storage capacity of 20 megabyte each.

With the Cartridge Disk Drive and cartridge disks, the user can perform these and other functions:

- Provide additional memory capacity for storage of graphic displays.
- Load module personality software when no History Module is present.
- Load sequence programs.
- Load changes in system software.
- Save and restore data bases of hiway boxes, and LCN modules.

For use with the Universal Station, the Cartridge Disk Drive is typically mounted in an enclosure that is incorporated into the console furniture. A second drive greatly facilitates the operations performed with a single drive, and at least one console should have two Cartridge Disk Drives or two Floppy Disk Drives.

Floppy Disk Drive

The Floppy Disk Drive is an alternative to the Cartridge Drive at a Universal Station, with physical dimensions that permit it to be mounted in the same console locations. It performs the identical functions, but stores just one megabyte of data on a 5.25-inch floppy diskette.

Matrix Printer

Two table top models of dot matrix printers are available. Both are output-only tractor-feed devices. One prints 132 ASCII characters-per-line at a speed of 250 characters-per-second, while the other prints 136 ASCII characters-per-line at a speed of 400 characters-per-second. On command from either the operator or the TPS System, they print:

- Logs, reports, trends, and journals
- Whatever is on the VDM screen, including graphics
- A record of system alarms

Touchscreen

With the CRT touchscreen, available for the Universal Station, placing a finger (or thin cylindrical object such as a pencil) on the screen causes the cursor to move to that point on the screen. If a target exists at that position, it will be selected when the finger is withdrawn. On appropriately configured displays, an operator can initiate a previously defined function or call up an associated display simply by touching the specified target.

Trackball

With the original Universal Station furniture, a trackball assembly can be mounted permanently in the space in front of the operator's Keyboard where an Engineer's Keyboard might otherwise be located. The trackball itself can easily be repositioned between the left and right side of the surface for the convenience of each user.

With the newer furniture, the trackball is portable and can be placed in any location convenient to the user; when in use its cable is plugged into a socket located underneath the video monitor. This trackball has three pushbuttons for use in selecting targets and entering data to the system.

Mouse

A mouse is available as an alternate option to the trackball. With either type of furniture the mouse cable plugs into a connector at the Universal Station.

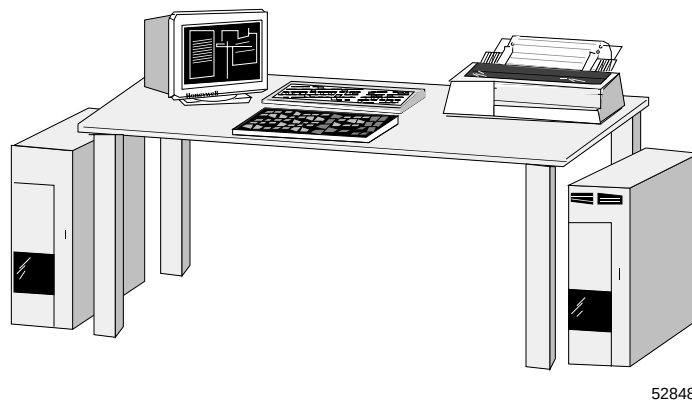
Trend Pens

To provide a continuous, real-time trend history for selected points, up to eight recorder Trend Pens can be driven by each Universal Station. The point values are accessed by the system over the LCN. The trend-pen-drive board in the Universal Station electronics module converts the digital hiway signals to analog; the Trend Pens themselves accept 1 to 5 volts or 4 to 20 milliamps as their analog input signal. The Trend Pens are housed in a VDM-like enclosure compatible with other system furniture.

5.4 Micro TDC 3000 System

Description

For those users who have processes that are relatively small, yet want the power and capability of TPS, Honeywell offers Micro TDC 3000. As shown in the following figure, Micro TDC 3000 is designed to save space and allow flexibility in the placement of its components. Its Universal Station includes a 20- inch CRT mounted on a swivel base, operator's and engineer's freestanding keyboards, two cartridge disk drives, and an ASPI 41 printer. Touchscreen or trackball is optional.



Full system functions

The human interface for Micro TDC 3000 provides the same functions and options available on the full-sized TPS.

5.5 Universal StationX

X Window capability added

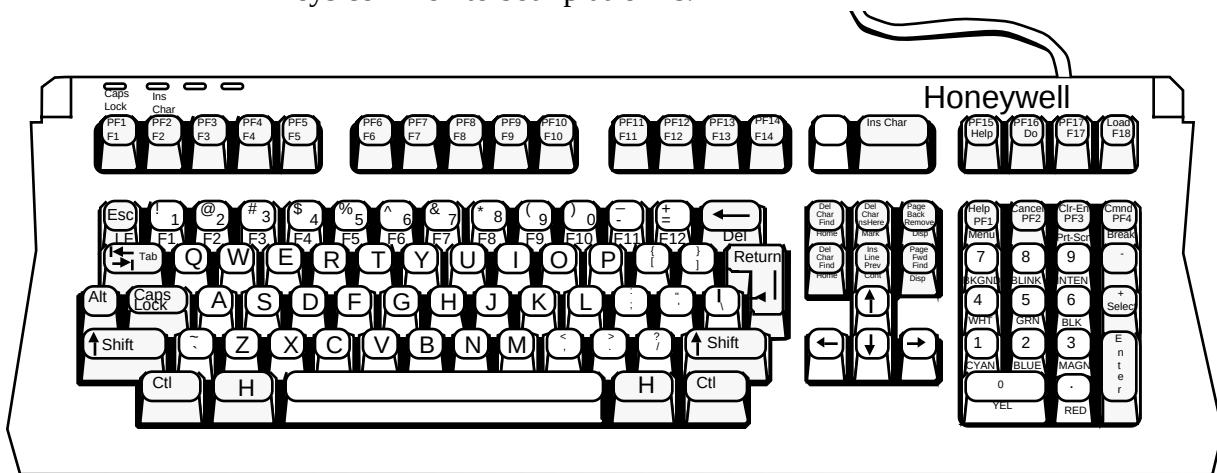
The Universal StationX performs all functions described above for the Universal Station. Universal StationX adds a new dimension to TPS by providing a direct interface between the Local Control Network and external user devices that apply MIT's X Window system to a UNIX-based platform.

Two furniture designs

Like the Universal Station, the Universal StationX can be supplied in both the original furniture design and the latest ergonomic furniture design. Touchscreen is standard. Cartridge drives, floppy drives, and US dot matrix printers are all available as options. Either a mouse or trackball is included. There are, however, some physical differences between the two types of stations. These are described in the following paragraphs.

Engineer's Keyboard

The Engineer's Keyboard has been redesigned for use with Universal StationX, and is illustrated in the following figure. While the layout differs, the available key selections, including the special function keys found on the US Portable Engineer's Keyboard, are the same. Color coding is used to distinguish between LCN control keys, X Windows control keys, and keys common to both platforms.



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Video display monitors

The video display monitors supplied with the Universal StationX are of a higher resolution, with 1280 x 1024 pixels. This produces the added sharpness needed for the reduced-size windows displayed under the X Windows environment. The original furniture uses a 19 inch "pillow" screen, while the ergonomic furniture comes with a 21 inch flat screen.

Hard disk

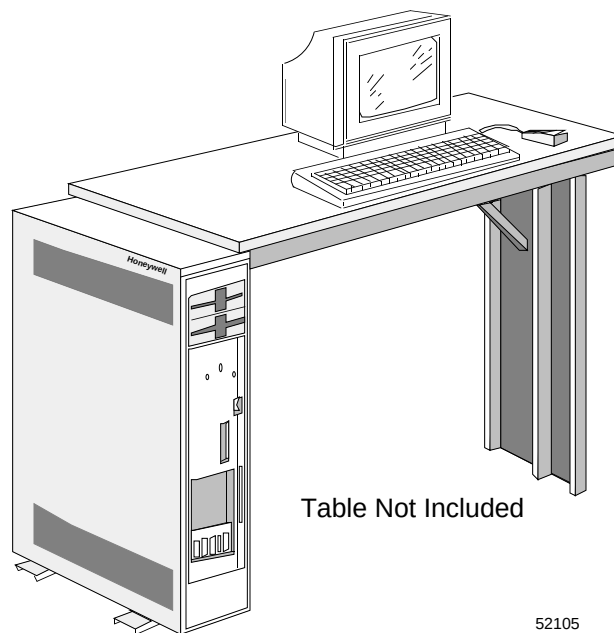
A separate 400 MB is supplied with Universal StationX. UNIX and X Windows files, along with other selected files and data are stored on this disk.

Digital audio tape drive	One Universal Station ^X also must be equipped with a digital audio tape (DAT) drive. This tape is used to bootload the UNIX coprocessor that is assembled into the Universal Station ^X electronics package.
Laser printer	A laser printer is available for producing a hard copy of any selected window displayed on the Universal Station ^X . While this printer is capable of copying a single X Window, it requires additional time to print a copy of the Native LCN Window compared with a standard Universal Station matrix printer.

5.6 Universal Work Station

Purpose

While a Universal Station or Universal Station^X is designed primarily for use by operators in a control room environment, the Universal Work Station (UWS) is intended for placement in an office for use by an engineer, process supervisor, or maintenance technician. As shown in the following figure, it consists of a monitor, keyboard, and mouse that sit on a desktop, plus a separate electronics tower. Functionally the Universal Work Station is virtually identical to the Universal Station.



Components

- Layout of the Universal Work Station Supervisor's Keyboard is identical to that of the Universal Station Operator's Keyboard.
- The Universal Work Station Engineer's Keyboard layout is identical to the Universal Station Portable Engineer's Keyboard.
- The color monitor for the Universal Work Station measures 14 inches (35.6 cm).

For the Universal Work Station, a two-button mouse is available as a cursor-positioning and target-activating device. One of the pushbuttons is used for "selecting" a target upon which the cursor has been positioned, and the second one is used for "entering" data that has been keyed onto the screen.

Section 6—System Operation

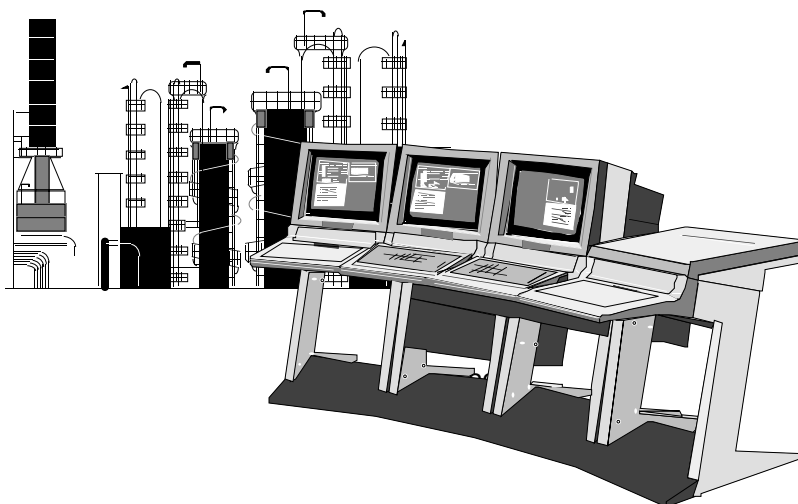
6.1 Introduction

Single window to the process

The process operator manages the process through an Operator's Console that contains one or more Universal Stations or Universal Stations^X. This console serves as a "single window" to the process, which uses the same operating procedures regardless of the size and complexity of the **TotalPlant** Solution (TPS) System or the levels of control used to accomplish the control strategies.

ATTENTION

Because a Universal Station^X performs all functions of a Universal Station, the following discussion applies equally to both types of stations.



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Universal Station functions

Each Universal Station can independently perform process and system functions, and each station in a console can be used for different functions. For example, in a 3-station console like the one in the preceding figure, the station functions might be arranged as follows:

- The Universal Station on the left displays a Graphic Display of a large portion of the process. Through such a display, the operator can review conditions with one quick look.
- The middle Universal Station is the primary working station from which the operator supervises the process and the control system, and obtains process- and system-related information.
- The Universal Station on the right is used for a display of alarm conditions in the operator's area.

Many other arrangements are possible.

6.2 Meeting the Process Operator's Needs

Full range of displays

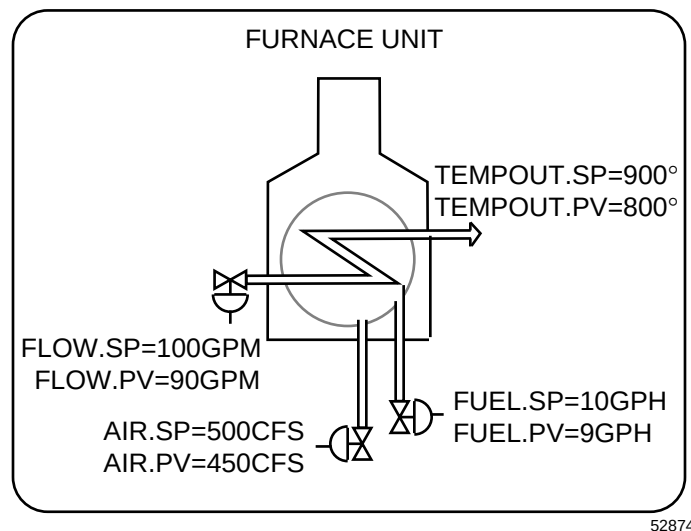
The TPS Universal Station meets the process operator's needs by providing a full range of displays that convey information about the process and the TPS System. These displays are arranged in a hierarchy that leads the operator from a broad overview with little detail to the most detailed information.

Process-operator displays can be logically grouped into three primary categories. They are outlined below, with sample displays.

Plant process displays

Plant Process displays are those displays associated directly with the user's process. They allow the operator to:

- Monitor continuous and discontinuous processes.
- Change process parameters, control modes, and sequence execution states and execution modes.
- Observe process trends.
- Handle process and sequence alarms.



TPS System displays

TPS System displays provide information about the status and operation of the Honeywell TPS System. They allow the operator to

- See the status of LCN devices, Universal Control Networks, data hiways, and process-connected boxes.
- Reassign Universal Stations, areas, units, and peripherals.

- Save essential data (checkpoint) for reloading, if necessary.
- Shut down and restart LCN devices and process-connected boxes.
- Load databases of LCN devices and process-connected boxes.
- Handle TPS System alarms.

MAKE SELECTION

DD MM YY HH:MM

SYSTEM STATUS

NCF VERS: 05Jan92 07:36:38:705

SYS VERS: R400

CONSOLE

1 - WEST AREA CONSOLE

CABLE A

CABLE B

OTHER CONSOLES

-OK

HISTORY MODULES

-OK

COMPUTING MODULES

-OK

GATEWAYS

-OK

APPLICATION MODULES

-OK

1 OK

2 OK

PROCESS NETWORK STATUS

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Related information displays

Process and system-related informational displays allow the operator to:

- Select reports and histories for viewing and printing.
- Locate what points are in units, LCN modules, and process-connected devices.
- Locate titles of areas, units, groups, graphics, etc.
- Locate points that have a common attribute (for example, all points with alarming disabled).

System Operation

6.2 Meeting the Process Operator's Needs

DD MM YY HH:MM

SYSTEM MENU

■ ORGANIZATIONAL SUMMARY MENU

■ REAL TIME JOURNAL ASSIGNMENTS

■ OVERVIEW EDIT DISPLAY

■ GROUP EDIT DISPLAY

■ CLEAR SCREEN

■ REPORT/LOG/TREND/JOURNAL MENU

■ EVENT HISTORY MENU

■ PROCESS VARIABLE RETRIEVAL

■ REMOVABLE MEDIA INITIALIZATION

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6.3 Process Control

Introduction

The process operator monitors and manipulates the process by viewing the process displays and interacting with them through the Operator's Keyboard and the optional touchscreen.

Standard process displays are provided with all TPS Systems. Users can build custom graphic displays that pictorially depict the process. Together, the combination of standard and graphic displays provide the optimum interface for operators.

Display hierarchy

The process displays are related through a hierarchy of detail, as shown in the following table. The location of each of the standard displays in the hierarchy is predefined; the user defines the hierarchy position of any graphic and help displays.

Usually, one operator at one Operator Console is given the responsibility for one process area that may contain several process units. A group is formed from up to eight points selected from a process unit.

Table 6-1 Process Display Hierarchy

Display Category	Area	Unit	Group	Detail
Operating displays	Overview display		Group display	Detail display
Operating displays	Graphic displays			
Trend displays	Area Trend display	Unit Trend display	Group Trend display	
Hourly averages display			Hourly Averages display	
Sequence displays	Module Summary display		Process Module Group display	Process Module Detail display
Alarm displays	Area Alarm display and Alarm Annunciator display	Unit Alarm Summary display		
Help displays	Help displays			
Message summary display	Message Summary display			

Graphic displays

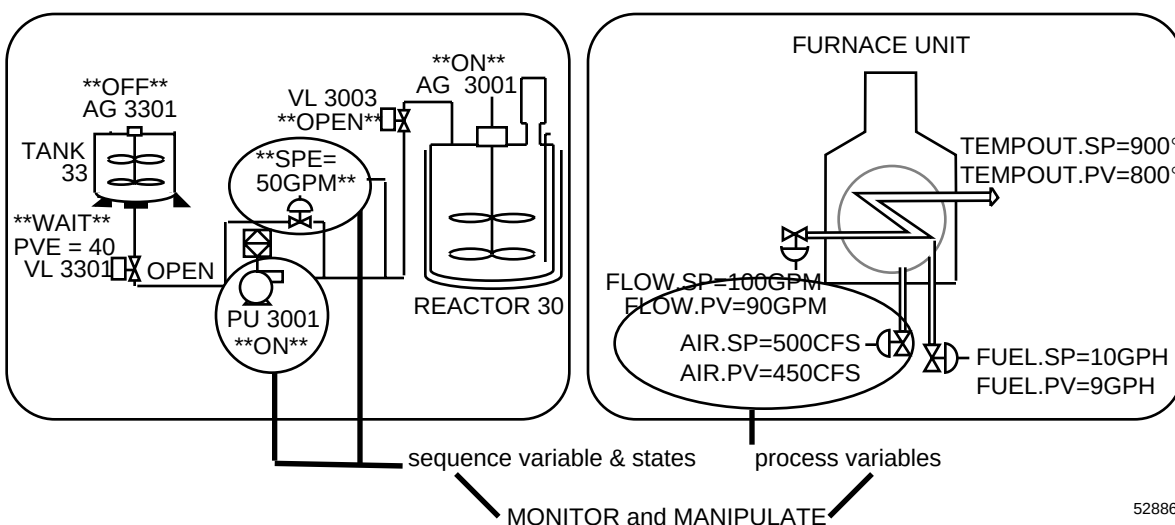
Graphic Displays are process displays that are defined by the user. Typically they are used as the primary operating displays. They can contain graphic, numeric, and textual information. They can represent a whole area, a unit, or a single point, or can be based on concepts unique to the user's plant. Graphics can be linked to many of the standard process displays and can themselves have targets that allow selection of other graphics or standard displays.

The total number of graphic displays that can be built is limited only by the combined allocated storage capacity of the available History Modules, cartridge disks, and/or floppy disks.

Graphic Displays can be built so that the operator can monitor and manipulate the process directly from them. Both discontinuous and continuous processes can be managed, as shown with the Batch Resin Unit and the Furnace Unit in the following figure.

Basically, any data point parameter or sequence that can be monitored and manipulated from any of the standard process displays can also be monitored and manipulated from the Graphic Displays, if desired. Additionally, process alarms can be annunciated and acknowledged from Graphic Displays.

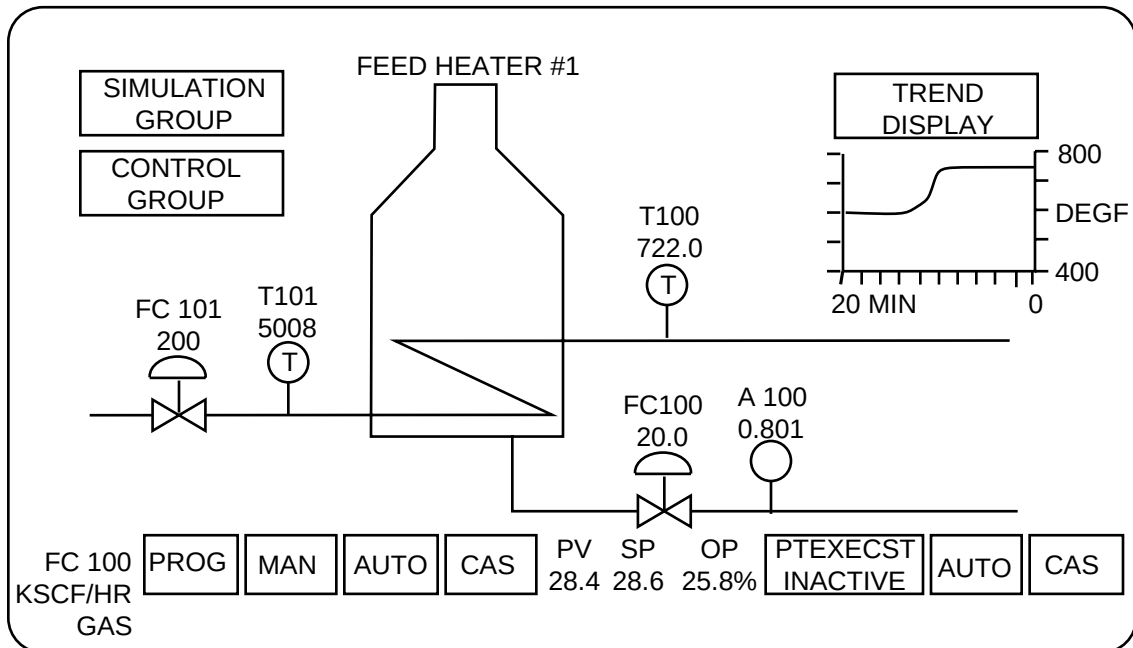
Using graphic displays



Another special feature of the Graphic Displays is the capability to bring another live display into a designated area within an existing graphic display. These process-related displays can include key information such as trends, process curves, batch totals, recipes, averages, operator messages, alarm handling information, and efficiency calculations. The operator is able to access this other process-related data without switching to another graphic.

Change zone

The operator can control the process from the Graphic Displays by way of targets contained in a designated "Change Zone." The following figure shows a graphic display that includes control capability and an active trend.



User-built help displays

Help Displays can be built by the user in support of Graphic Displays. These can contain textual and graphic information. As an example, a Help Display might contain special operating instructions for an associated Graphic Display. Help Displays are normally stored in History Modules.

Levels of display detail

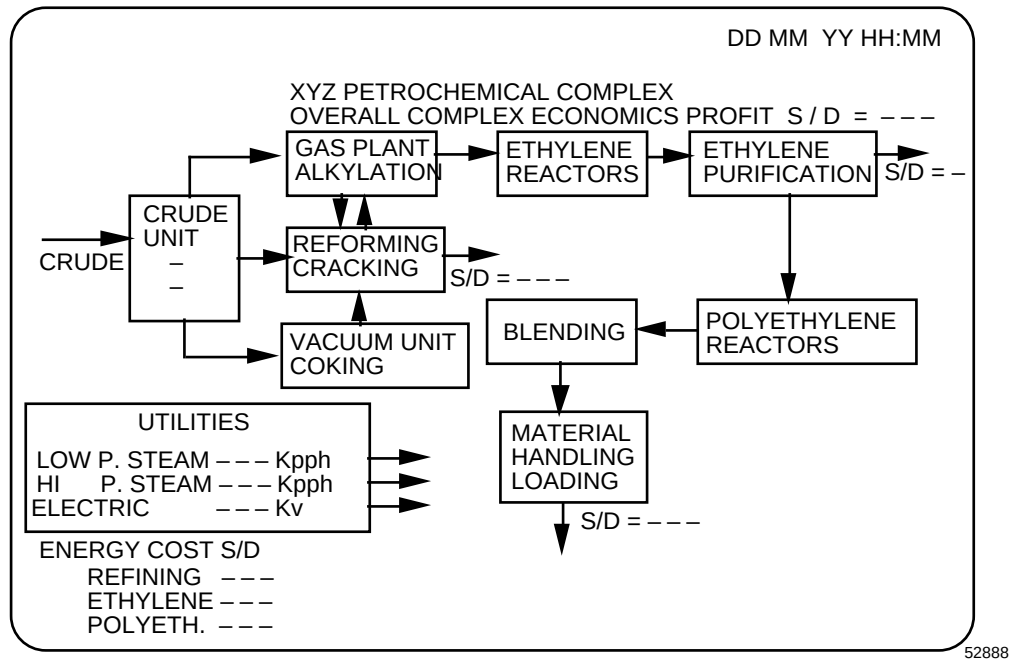
The level of detail in these displays is indicated by the level of the displays in the hierarchy:

- Area Displays summarize the operating condition and trends of the process units that make up the area assigned to the console, or of a large portion of that area.
- Unit Displays summarize the operating condition and trends of the individual discontinuous and continuous operations in the area assigned to the console. For example, a unit could be a furnace, a distillation tower, or a resin reactor train.
- Group Displays show values and status for all points in a group (up to eight), summarize operating trends within the group, or present status and sequence information for process modules.
- Detail Displays present the operating parameters and limits for individual points and process modules.

Thus, a display at the top of the hierarchy has the least detail and the broadest scope; one at the bottom has the most detail and the narrowest scope.

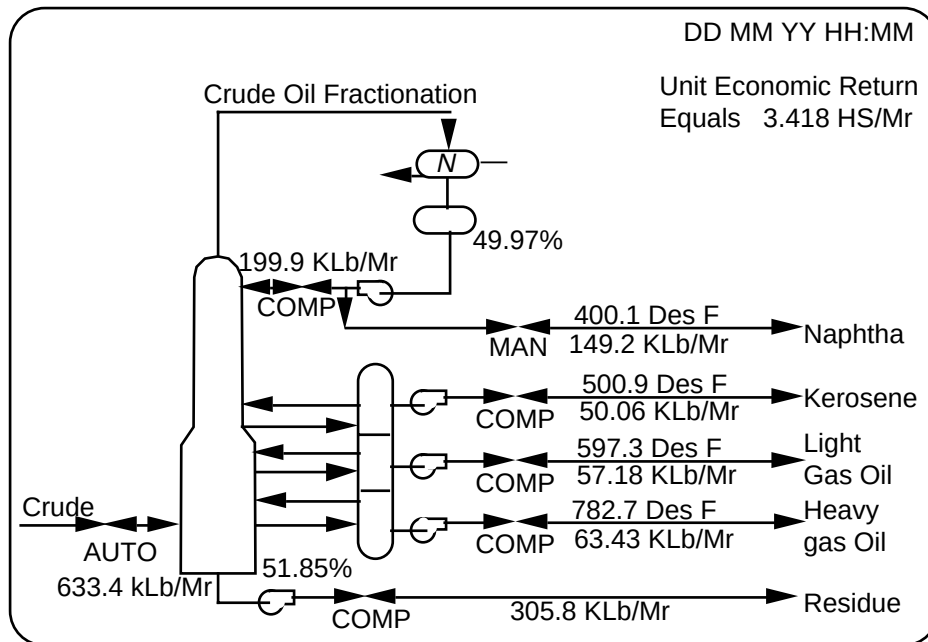
Area displays

The following is an example of an area-level graphic display from which the operator can view a wide process area. From this display, the operator can choose where to go for more information.



Unit displays

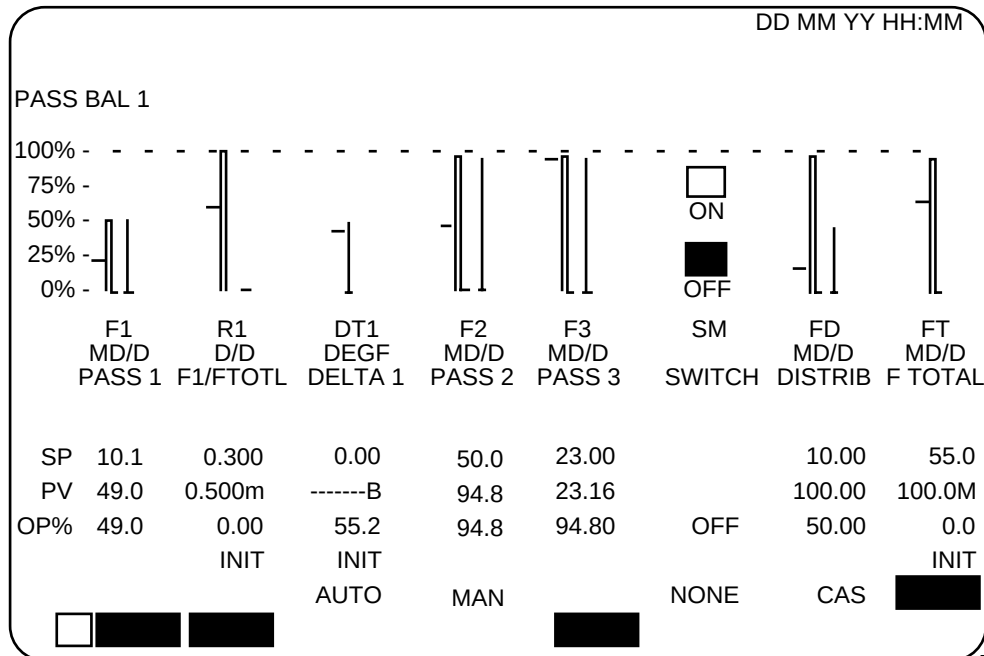
A unit-level display can also be a Graphic Display. Using a unit display, the operator can view information about a specific process unit. From this display the operator can go to a Group Display for more information.



52889

Group displays

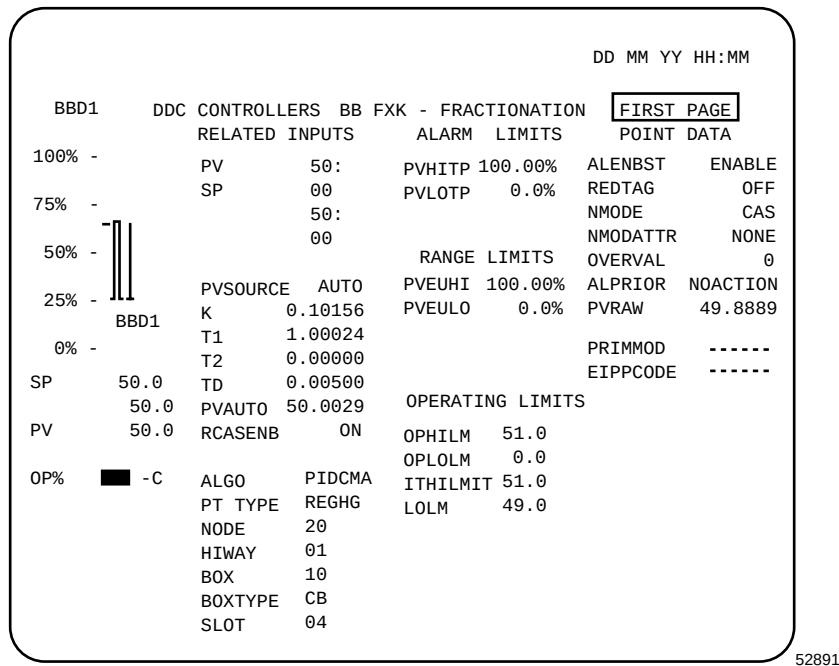
The group display is a standard display that provides operating information for a group of up to eight points. Through this display, the operator can change operating parameters and control modes. There is one such display for each of up to 400 operating groups and 50 process module groups in a system.



52890

Detail displays

Detail displays are standard displays that show all of the detail in a data point, including operating parameters, other engineer-specified parameters, and characteristics. The operator can change operating parameters through this display, and supervisors and process engineers can change other parameters and characteristics.



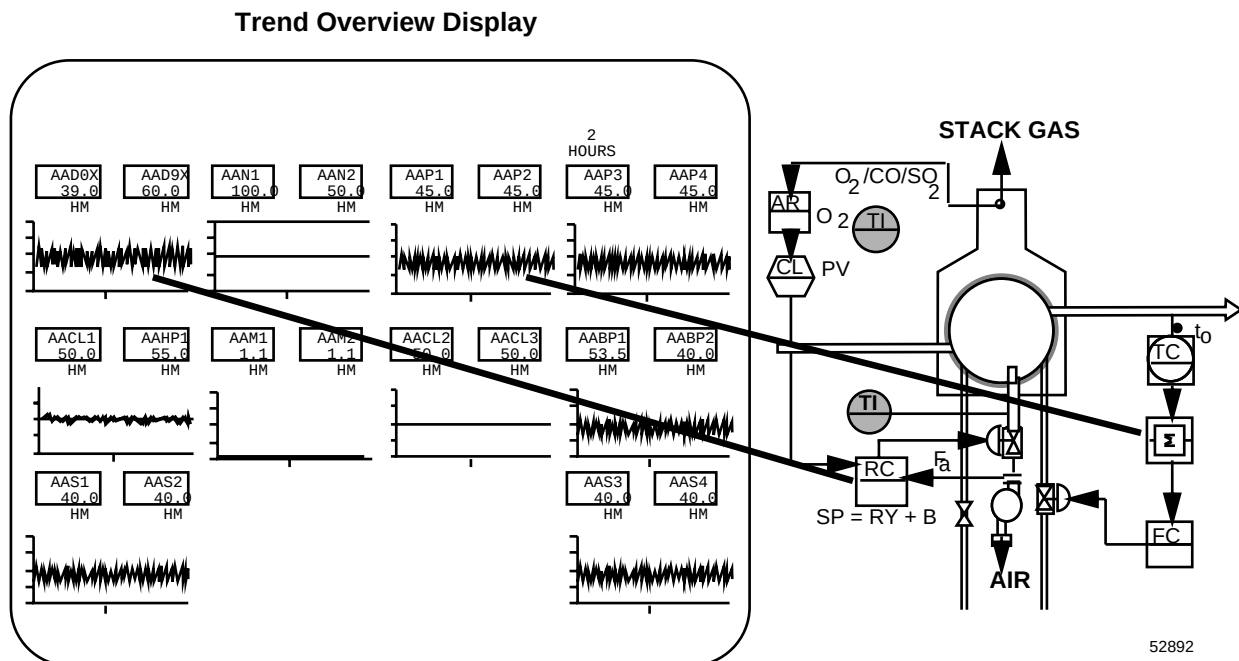
Message Summary display

The operator can observe, acknowledge, and confirm CL and SOPL messages through the Message Summary Display. This display has up to four pages and is a listing of the 96 most recent CL and SOPL messages that have been received by the Universal Station. Each message contains a message-status character, the time of the message, the unit identifier, the point identifier, and the text of the message. The message-status character indicates actions that the operator must perform, such as acknowledge and confirm the message, or remove all acknowledged and confirmed messages from the display.

Trend displays

Trend displays present recent history of key operating variables in a form similar to that of a trend pen recorder. One Trend Overview Display is available, and up to 36 Unit Trend Displays are available. While any analog variable can be assigned to any position on a trend display, it is intended that the Trend Overview Displays show key trends for an area consisting of several units, and that the Unit Trend Displays represent process units. The trends are shown over the last two hours or eight hours, and show either average values or "snapshot" (instantaneous) values.

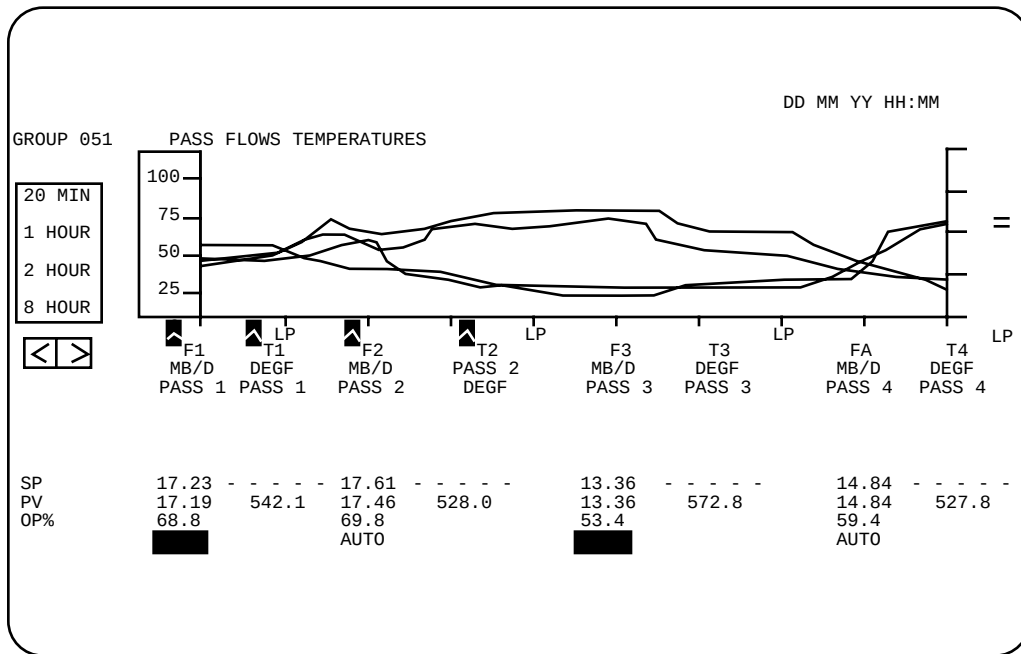
For example, the following figure shows the Trend Overview display being used to trend the fuel-inlet rate and the outlet temperature of a furnace. This allows the operator to quickly see the effect of variances in fuel-inlet rate have on the outlet temperature over time.



A paper copy of trend data can be obtained through a screen print of the trend display, through printed continuous trends, and by recording on optional trend pen recorders.

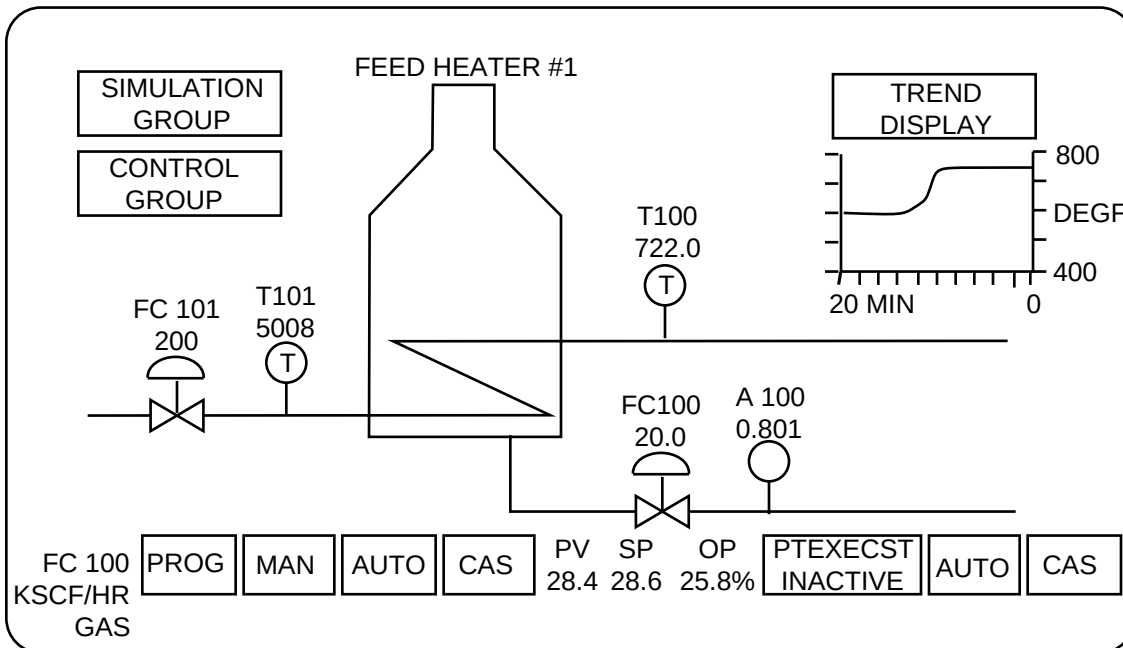
Group Trend Display

The Group Trend Display provides trending of one or more of the variables on a Group Display. The illustration below shows four PVs being trended on a single axis. Each is displayed in a different color. If more than four trends are requested, a second set of axes will appear on the bottom of the same screen. The operator can select the X-Axis time scale, data source, and step back/forward through available history.



52893

Subpicture trends Trends can be included as a subpicture in a graphic display, as shown in the example below. Up to 12 such trends, with or without axes, can be on a single display, each with four traces. The graphic can be constructed so that the trends are shown only when selected by the operator.



52887

6.4 Examples of Operating Situations

Introduction

To present a conceptual view of process operations in a TPS System, four situations are described below, using simple, generalized examples. These examples assume that the Universal Station has the touchscreen option.

The four situations are:

- Normal operation—everything is running smoothly.
- Changes occur, then everything settles down.
- Startup and shutdown.
- Alarms occur, and the operator reacts to them.

Following these examples, other process and system displays not yet mentioned are described.

Situation 1: Normal operation- - Everything is running smoothly

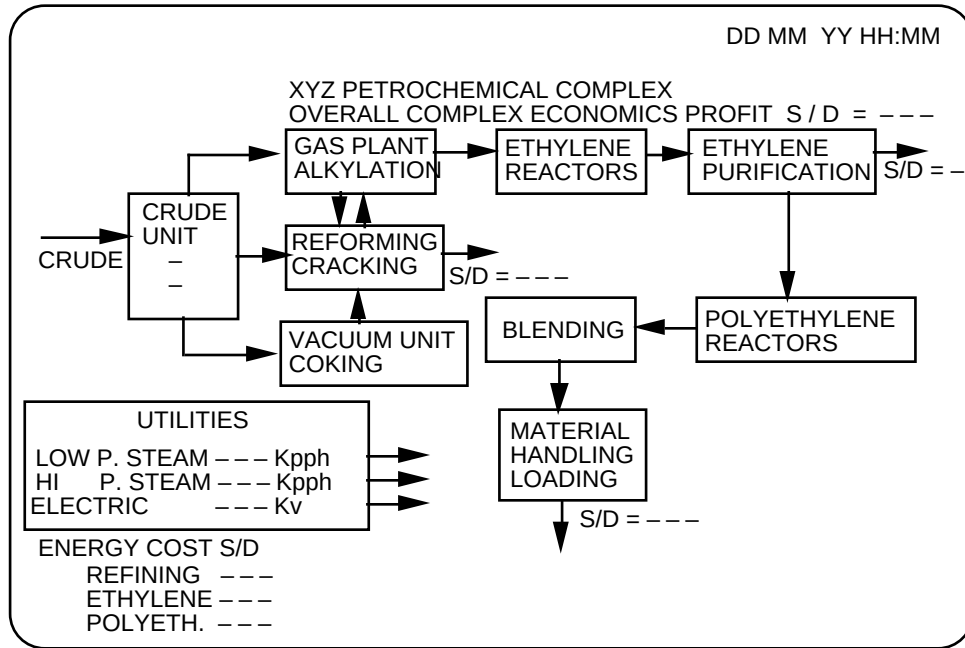
The operator is seated at a TPS Operator's Console with three Universal Stations. The Universal Stations are assigned functions as described on the first page of this section. For this example, the displays might be as follows:

- Left — A Graphic Overview Display for the area assigned to the console.
- Middle — A Group Display or a Graphic Display showing a process unit in the area assigned to the console.
- Right — The Alarm Annunciator Display or the Area Alarm Summary Display.

System Operation

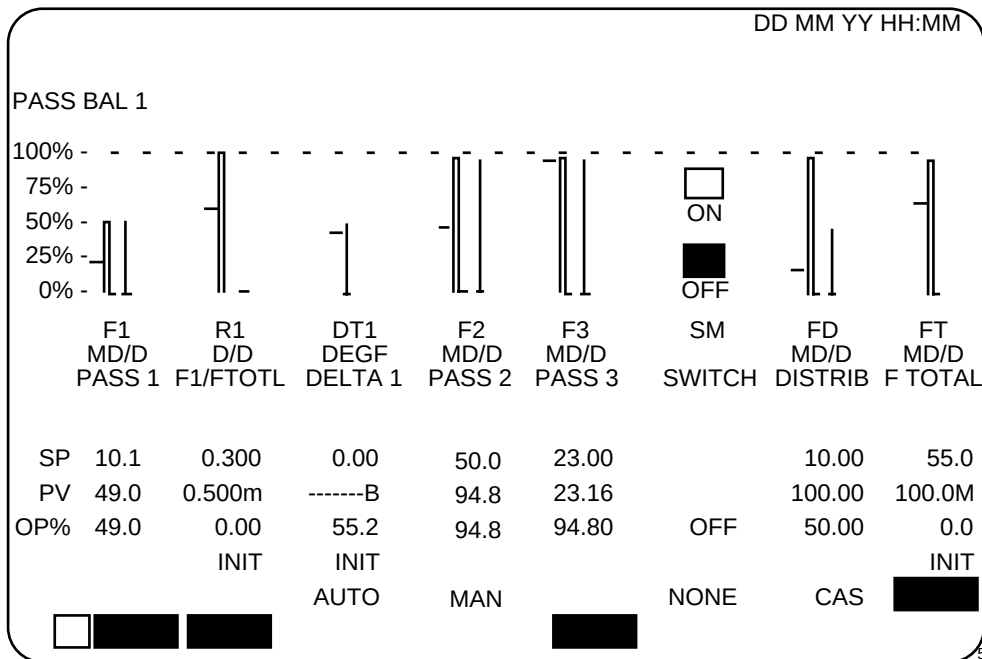
6.4 Examples of Operating Situations

With everything running smoothly, the Graphic Overview Display shows a steady system operating without alarm conditions. Digital points and process conditions can be represented and displayed in their correct state.



52888

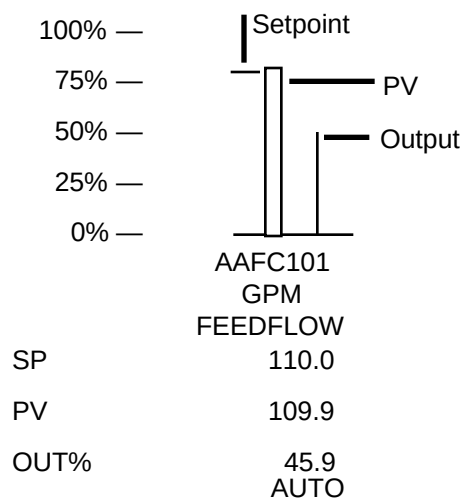
The middle display is a working display through which the operator deals with process units or specific points. The operator might have a Group Display on this screen, as illustrated below, or one of the unit-level displays.



52890

With things running smoothly, the Group Display shows all eight points operating normally. These could be eight regulatory points as in the furnace-unit example, or they could be discontinuous points as in the batch resin-unit example. The Group Display pictured here shows seven analog points and one digital point.

On the Group Display, regulatory points appear as bar graphs, below which are several text lines that provide operating information about the point. For example, for a PID point like the flow controller in the furnace unit, the bar for the PV would be level with the dash representing the SP. The PV and SP values would be almost the same, and the output value would be some reasonable percentage of full scale.



52929

For a normally operating digital point, the Group Display shows a simulation of two indicator lamps, each representing one state of the point.

For example, if the normal state of a digital input were "on," it would appear as shown.

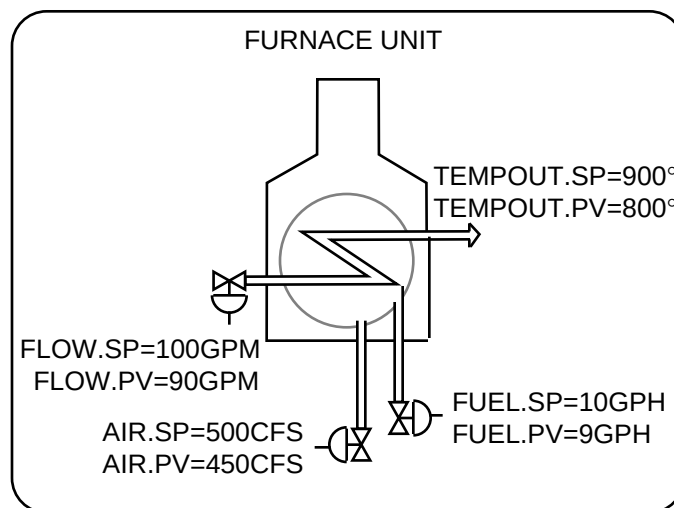
System Operation

6.4 Examples of Operating Situations



52894

Instead of a Group Display, the operator might have a process unit graphic on the middle screen. It might, for example, be a graphic of a furnace unit, similar to the one shown here. All of the loops indicated on the graphic would appear normal.



52874

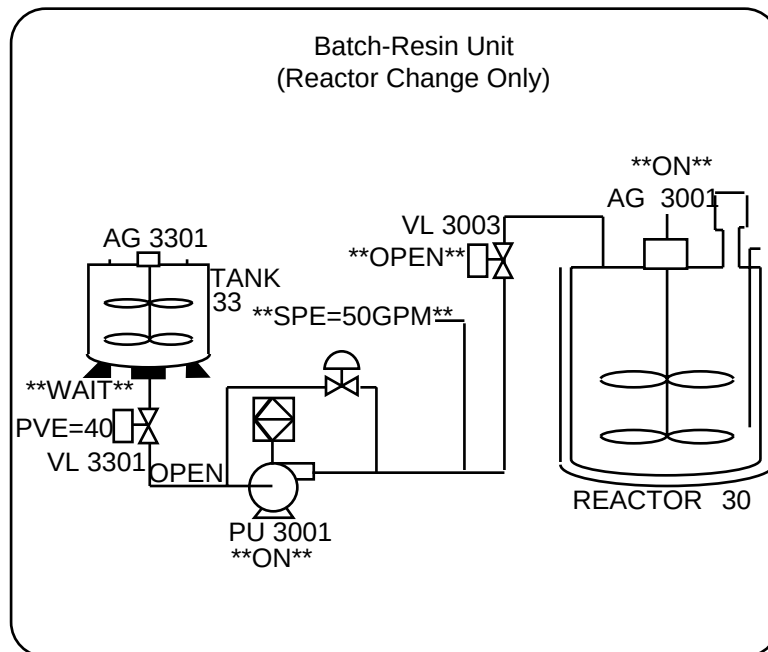
The Module Summary Display and the Process Module Group Display are the displays through which discontinuous operations are monitored and manipulated. With everything running smoothly, the operator may not need to take any action for some time. These displays would show the operations taking place as expected, with little or no operator intervention.

16 Jun 92 11:22:46 2

MODULE SUMMARY							PAGE 1	
UNIT	MODULE	SEQUENCE	OPER	EXEC	MODE	PHASE	STEP	MSG
Ø1	EM1	REACT	NORM	LOAD	AUTO	SETUP	ST2	CFM
Ø1	EM2		OFF	LOAD	AUTO			
Ø1	EM3		OFF	LOAD	AUTO			
Ø1	EM4		OFF	LOAD	AUTO			
Ø1	EM5		OFF	LOAD	AUTO			
Ø1	EM6		OFF	NL	AUTO			
Ø1	EM7		OFF	NL	AUTO			

52895

Instead of one of the batch displays, the operator could have a graphic of a batch unit on the middle display, similar to the batch resin-unit graphic shown below. It would show that the expected steps in the batch operation were taking place and any regulatory control within the batch unit would be normal.



52896

With everything running smoothly, the operator may perform routine functions with respect to the assigned process area.

To summarize this example of normal operation, the left screen displays a broad view of the operator's area, the middle screen shows more detail for a smaller portion of the process and is available to manipulate the process, if necessary, and the right screen provides instant notice of alarm conditions that may occur.

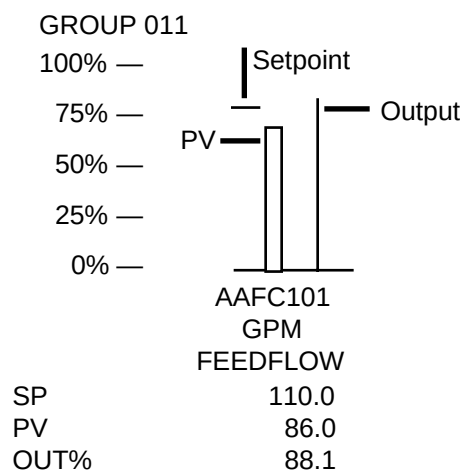
**Situation 2:
Changes occur,
then everything
settles down**

In this situation, the changes may or may not require operator intervention.

Suppose a device on the graphic display changes from its normal neutral color to red and begins to flash, indicating to the operator that a regulatory point has deviated significantly from its setpoint. If something happened in the process that was faster than the control loop's response time, sooner or later the condition will correct itself and the graphic display will return to its normal state; but if the nature or magnitude of the process change can't be handled by the control loop, the operator may have to intervene.

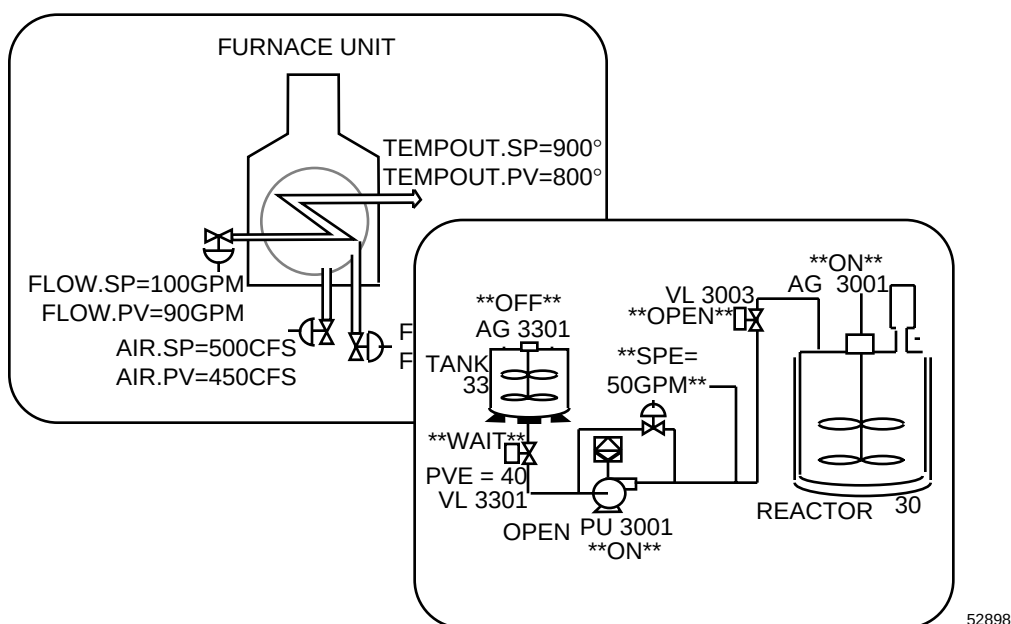
Before intervening, the operator might take a closer look, by calling the Group Display to the middle screen.

The portion of the Group Display that shows the 7th point in Group 011 might appear as shown. This point uses a PID control algorithm. Normally, the PV would be equal or almost equal to the SP. Here, however, the output is at 88% but the PV is less than the SP. If this is a flow-control point, it might be that the setpoint has been changed recently, and while the valve has opened some more the PV hasn't yet come up to the new setpoint. It could also be that the flow pressure dropped, the valve opened a little to compensate, and the PV has not yet come back to the setpoint.



52897

Instead of calling a Group Display to take a closer look at PV deviations from SP, the operator could call a unit-level graphic to the middle screen. Such graphics would probably include information about setpoints, PVs, and output values as the Group Display does. Graphics are good at showing interactions in the process that you can't see in a group display. This gives the operator the "big picture." For discontinuous operations, a graphic would probably show sequence phases and steps, and off and on conditions, as well as information for regulatory points contained in the batch unit.



If the operator does need to intervene, process changes can be made through either the Group Display or the Graphic Display.

Situation 3: Startup and shutdown

It is during startup and shutdown of a process or a portion of a process that the greatest changes take place and the most operator attention is required. For batch operations, startups and shutdowns are routine and occur frequently. For continuous operations, they occur less frequently and may not be routine at all.

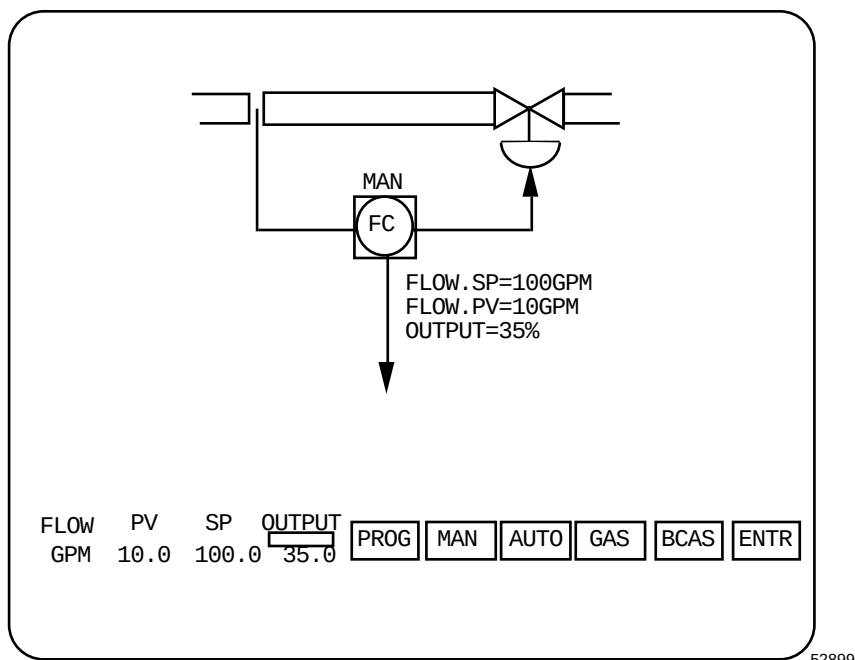
Starting up or shutting down a whole process, or a major portion of a process, may involve a broad range of operator activities, from process-parameter changes as points are brought on control, or removed from control, to status changes of process networks and process-connected boxes that affect communication with the process. Startups and rampups (grade and rate change, for example) can be automated using sequences.

Phases and steps in batch operations can be automatically started and stopped through PM/APM/HPMs and Multifunction Controllers, and the operator usually needs only to monitor them. Sequences are normally started by the operator.

The following examples illustrate startup for both continuous and discontinuous operations.

In the furnace-control scheme described in (Data Acquisition and Control) 4.2 Simple Control Example, page 47, at some point the operator must start the flow of feed into the furnace. The operator might start with the flow-control loop in the manual mode. In this mode, the operator would use a Group, Detail, or Graphic Display to manually change the output, thus controlling the valve position and the flow.

The part of a Graphic Display representing the flow-control loop might appear as shown in the illustration below.



The operator can use this display to change the control mode and the parameters. For example, the operator might select a target on the graphic, consisting of the flow controller, causing a Change Zone to appear for that device at the bottom of the screen. Then the operator could select the OUTPUT pick and enter a new output value in the port that will appear for that purpose. After verifying the new value, the operator would touch the ENTER key or ENTR target and the new value would replace the old value. The operator should then see the PV change as the valve position changes. Such parameter changes can also be made through Group Displays and Detail Displays.

When the manually adjusted output has caused the PV to move reasonably close to the setpoint, the operator might change the loop's mode from MANual to AUTOMatic. In AUTOMatic mode, the controller slot in the process-connected device would take over control of the flow and the PV would settle on or near the setpoint in a few moments.

Sequences can be started up through Graphic Displays or through the Module Summary and Process Module Group Displays.

16 Jun 92 11:22:46 2

MODULE SUMMARY								PAGE 1
UNIT	MODULE	SEQUENCE	OPER	EXEC	MODE	PHASE	STEP	MSG
01	EM1	REACT	NORM	LOAD	AUTO	SETUP	ST2	CFM
01	EM2		OFF	LOAD	AUTO			
01	EM3		OFF	LOAD	AUTO			
01	EM4		OFF	LOAD	AUTO			
01	EM5		OFF	LOAD	AUTO			
01	EM6		OFF	NL	AUTO			
01	EM7		OFF	NL	AUTO			

52895

A sequence can be monitored and manipulated through the Module Summary Display. This display allows the operator to change the status and execution mode of sequences and of process modules.

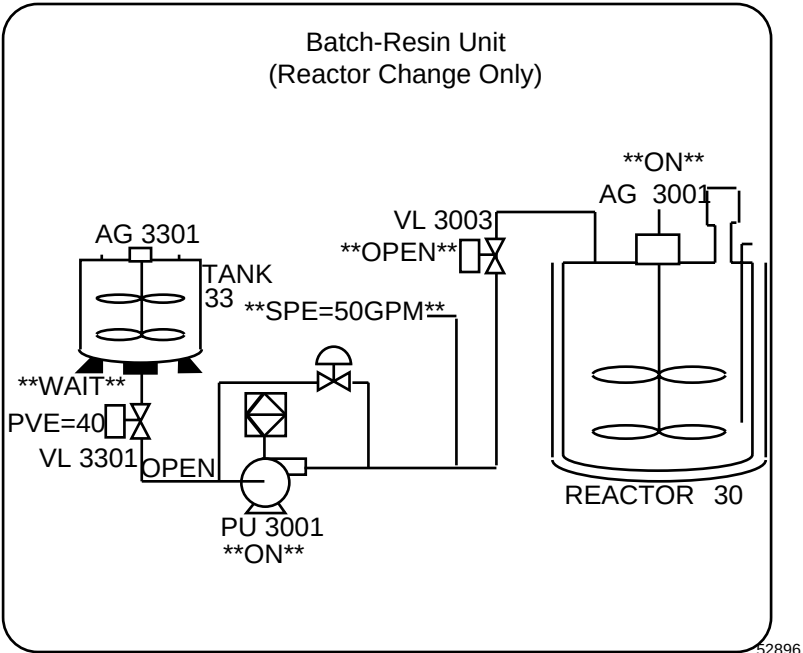
The Process Module Group Display allows the operator to control a majority of the module functions.

DD MM YY HH:MM

GROUP 401		CL3 TEST		REAC 01		REAC 02		HEAT 01	
UNIT	MODULE	SEQUENCE	OPER	EXEC	MODE	PHASE	STEP	< MSG	
01	EM1	SETT	REDY	LOAD	AUTO	SETT	OPEN		
01	EM2	REACT2	NORM	RUN	AUTO	SETUP	ST1		
01	EM3	REACT3	OFF	NL	AUTO	SETUP	ST	*	
13:17 01 EM2		GENERATE ALARMS							

52900

To start the Batch Resin Unit, for example, the operator could first start the secondary sequences from the Process Module Group Display, then use a graphic Unit Display to start the primary sequences.



52896

Situation 4:
Alarms occur—
The operator
reacts to them

The operator's immediate recognition and handling of a process or sequence alarm may prevent process upsets and the production of off-specification products. For this example we describe two situations where the operator of what was a normally operating system is notified of a process alarm through the Alarm Annunciator Display.

The Area Alarm Annunciator Display shown in the following figure is like a conventional annunciator panel. The display contains 60 annunciation boxes, each of which can be assigned up to 10 points. Colors indicate alarm priorities.

52901

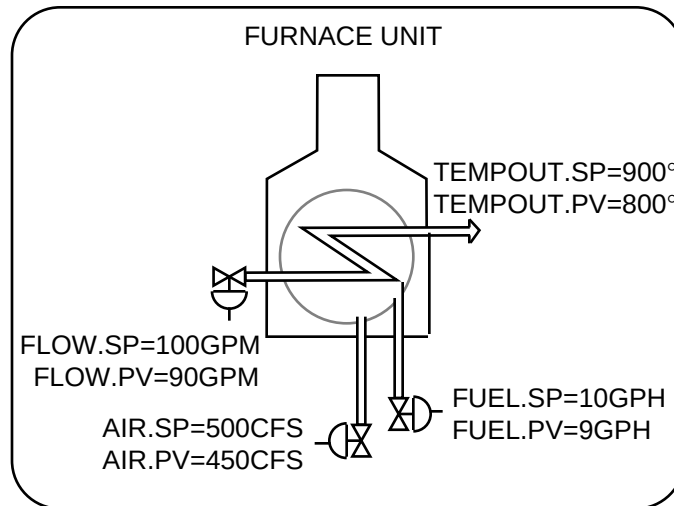
Priority Level	Universal Station Action
Emergency	Display and print alarms
High	Display and print alarms
Low	Display and print alarms
Journal Only	Print alarms
No Action	Ignored

In this example, two alarms occur at the same time:

System Operation

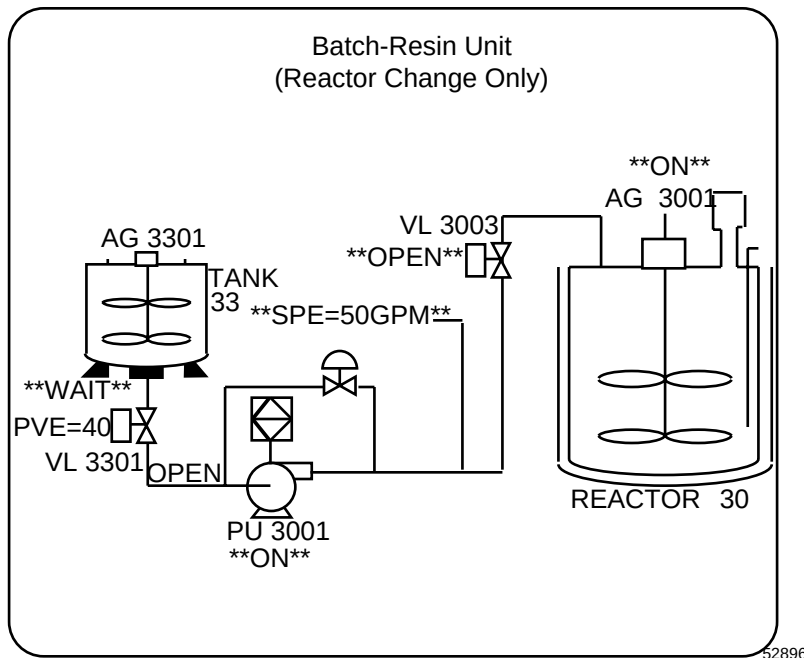
6.4 Examples of Operating Situations

- The feed-flow controller on the furnace shown can't bring the flow up to the setpoint, so the deviation of the PV from the setpoint is large enough to cause an alarm.



52874

- The sequence program is unable to turn off the pump in the resin unit shown.



52896

For the furnace alarm, the following happens:

- The background color of the box representing the point on the Alarm Annunciator Display changes according to alarm level and begins to blink. In this instance, the box might read FEEDFLOW.
- The ALM SUMM key on each Operator's Keyboard in the console begins to blink. Other keys can be assigned to light up to indicate the unit the alarm is in.
- Alarm contacts close, which can activate an external user-supplied annunciator (usually audible).
- The alarm condition is displayed on the Area Alarm Summary Display, Unit Alarm Summary Display, and operating displays (including any appropriately-configured Graphic Display), if one of these displays is shown on the console.
- The alarm is reported in the Process Alarm Journal.

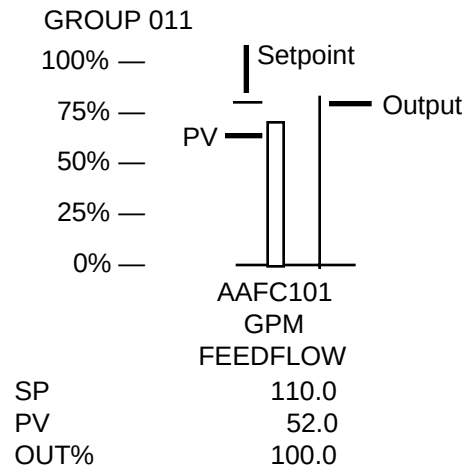
At this point, the operator would probably acknowledge the alarm, so that the audible and visual alarm indications cease to be a distraction. To silence the audible alarm, the operator pushes the SIL key. To acknowledge the visual alarm, the operator pushes the ACK key while an alarm display that shows the alarm is on the screen. This causes the keys to stop blinking and the boxes on the Annunciator Display to change to a steady light. The acknowledgment is entered in the Process Alarm Journal and is printed on a printer.

After acknowledging the alarm, the operator can handle the alarm condition.

The operator touches the FEEDFLOW box on the Alarm Annunciator Display. This calls up one of the displays that contains the flow control point. Normally, the PV would be equal or almost equal to the setpoint. Here, however, the output is at 100% but the PV is less than the setpoint. It might be that, even with the valve fully open, there isn't enough feed to reach the setpoint, or it might be that the valve is stuck at some fraction of its normal range. The operator could use the following Group Display to change the setpoint to a lower value to see if the point will come back under control, or the operator might call someone to check the feed supply and the valve.

System Operation

6.4 Examples of Operating Situations



52903

When the condition that caused an alarm is corrected (the operator having previously acknowledged the alarm), the lighted LEDs are extinguished, the external status-contact outputs are deactivated, and the return-to-normal condition is reported in the Process Alarm Journal.

There are two other alarm displays. The following Area Alarm Summary Display provides the operator with a listing of the 100 most recent emergency and high priority alarms in the area assigned to the console. Twenty alarms are listed on each of the five pages of this display.

The Unit Alarm Summary Display performs basically the same as the Area Alarm Summary Display, except that the alarm information is on a unit basis, and it also shows low-priority alarms in the unit.

Area Alarm Summary display

DD MM YY HH:MM									
AREA PLANT_E EAST AREA				ALARM SUMMARY		056 AREA ALARMS PAGE 1			
H09:56:49	DT2	PVHI	70.0	PASS2 - PASS3 DELTA TEMP	ZZ	73.4	DEGF		
E09:56:48	T2	PVHI	105.0	PASS 2 OUTLET TEMP	ZZ	110.5	DEGF		
H09:56:48	T2	BADPV		PASS 2 OUTLET TEMP	ZZ	-----	DEGF		
H09:54:18	DT1	BADPV		PASS1 - PASS2 DELTA TEMP	ZZ	-----	DEGF		
H09:54:18	DT1	PVLO	105.0	PASS1 - PASS2 DELTA TEMP	ZZ	100.1	DEGF		
H09:54:18	T1	BADPV		PASS 1 OUTLET TEMP	ZZ	-----	DEGF		
H09:54:18	T1	PVHI	80.0	PASS 1 OUTLET TEMP	ZZ	88.5	DEGF		
E09:54:13	FD	PVHI	100.0	TOTAL FLOW DISTRIBUTOR	ZZ	100.0	MB/D		
H09:52:23	MTR4	OVRDI1		SYSTEM 4 MIXER	ZZ	input 6			
H09:51:19	DT1	PVLO	110.0	PASS2 - PASS3 DELTA TEMP	ZZ	107.7	DEGF		
H09:23:18	DT1	PVLO	105.0	PASS1 - PASS2 DELTA TEMP	ZZ	106.7	DEGF		
H09:23:18	T1	PVLO	125.0	PASS 1 OUTLET TEMP	ZZ	113.0	DEGF		
H08:18:31	T100	DEVHI	30	PUMPAROUND TEMP P_00L	ZZ	50	DEGF		
E08:12:35	D023	OFF		SS14 2400V BUSS	ZZ	-----			
H08:11:15	L056	PVHI	8.0	RESID TANK 56 LEVEL	ZZ	10.54	%		

AA BA DA 11 22 33 44

Area Alarm Summary Display

52902

6.5 System Control

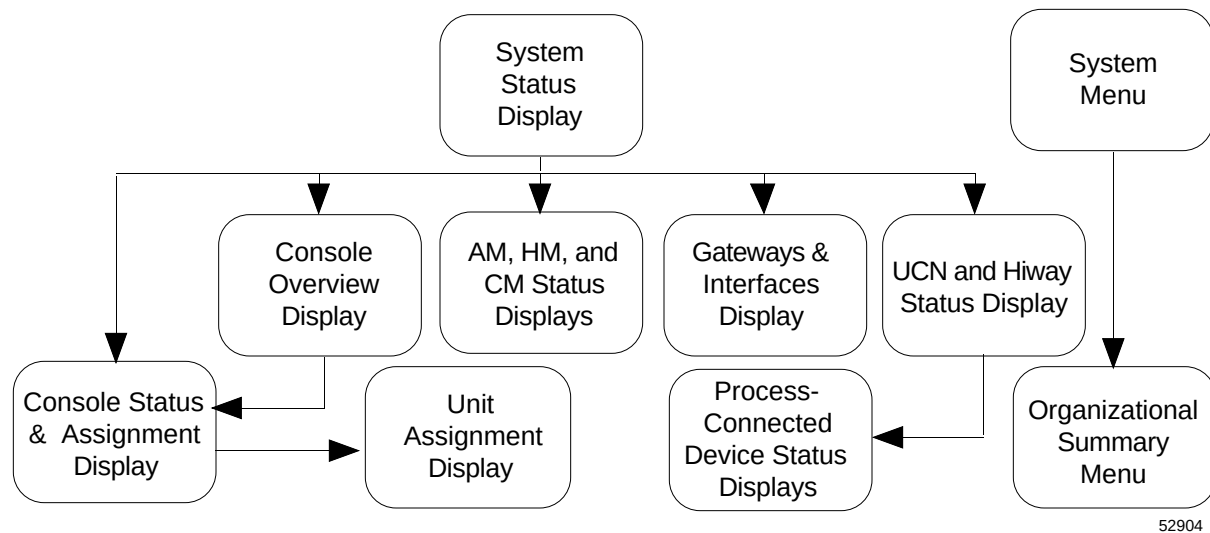
Introduction

Users monitor and change the status of the TPS System equipment through the system displays. These displays are accessed through the System Status display, page 141.

The System Menu provides access to displays that provide information about the content of the system database and that are used to invoke system-level functions related to monitoring and control of the process.

System display hierarchy

The hierarchy of the system displays is shown by the following figure.



Monitoring system status

The operator can determine the current status of the system equipment, using the System Status Display shown in the following figure. This display shows the current status of the LCN cables, Universal Stations, each LCN module, each UCN, and each Data Hiway.

System Status display

MAKE SELECTION
SYSTEM STATUS
NCF VERS: 05Jan92 07:36:38:705
DD MM YY HH:MM
SYS VERS: R400

CONSOLE

1 - WEST AREA CONSOLE

CABLE A

CABLE B

OTHER CONSOLES	-OK	HISTORY MODULES	-OK	COMPUTING MODULES	-OK
GATEWAYS	-OK	APPLICATION MODULES	-OK		

1 OK

2 OK

PROCESS NETWORK STATUS

52884

Monitoring and assigning consoles and stations

The Console Status and Assignment display shows the current status and assignments of the Universal Stations and peripherals in the console. It is through this display that process units are assigned/deassigned to consoles. Other functions are

- Universal Station Loading
- Peripheral Status Information
- Shut Down Universal Station
- Time and Date Changes
- Area Changes
- Changes to Level of Access
- Maintenance Information
- Load/Dump Enable
- Printer Assignment

**Assigning
process units**

Through the Unit Assignment Display, the operator can change the unit alarming and control status to any of 36 units assigned within the area. A process unit can be manipulated only from displays on the Universal Stations to which it is assigned. The Unit Assignment Display lists the unit name, assignment status, and alarm status for each unit within the assigned area.

**Monitoring
multiple console
status**

The Console Overview Display lists the consoles on the LCN and their current status. Through this display, the operator can call for the detailed status of any individual console to see the current status of its Universal Stations and their peripherals, and thus can determine whether or not maintenance is required at a particular Universal Station.

**Monitoring and
changing status
of control-room
equipment**

The operator can monitor and change the status of the control room equipment through the LCN Module Status Displays. A separate status display is provided for each type of LCN module. Each display lists the module address, current status, and backup.

From an individual module status display, the operator can make module restarts; shutdown a module; switch from a primary to a backup; load the personality; and perform database save/restore functions.

**Monitoring and
changing status
of UCNs and
associated
devices**

The operator can use the UCN Status Displays to manipulate UCNs and UCN-based devices, and monitor the status of the UCNs and devices. A separate UCN status display is provided for each UCN in the system. Each display shows the current UCN status and the current status of up to 32 redundant devices.

From the UCN Status Displays, the operator can perform functions such as switching to a backup UCN; issuing commands to UCN devices; and saving/restoring the database of a single device, several devices, or all devices on a UCN.

**Monitoring and
changing status
of Hiways and
boxes**

Although the display formats differ, Data Hiways and Hiway-based boxes are monitored and manipulated through the Hiway Status Displays in the same fashion as described for the UCN Status Displays above. Each Hiway Status Display shows the current status of up to 63 boxes.

**Handling TPS
System alarms**

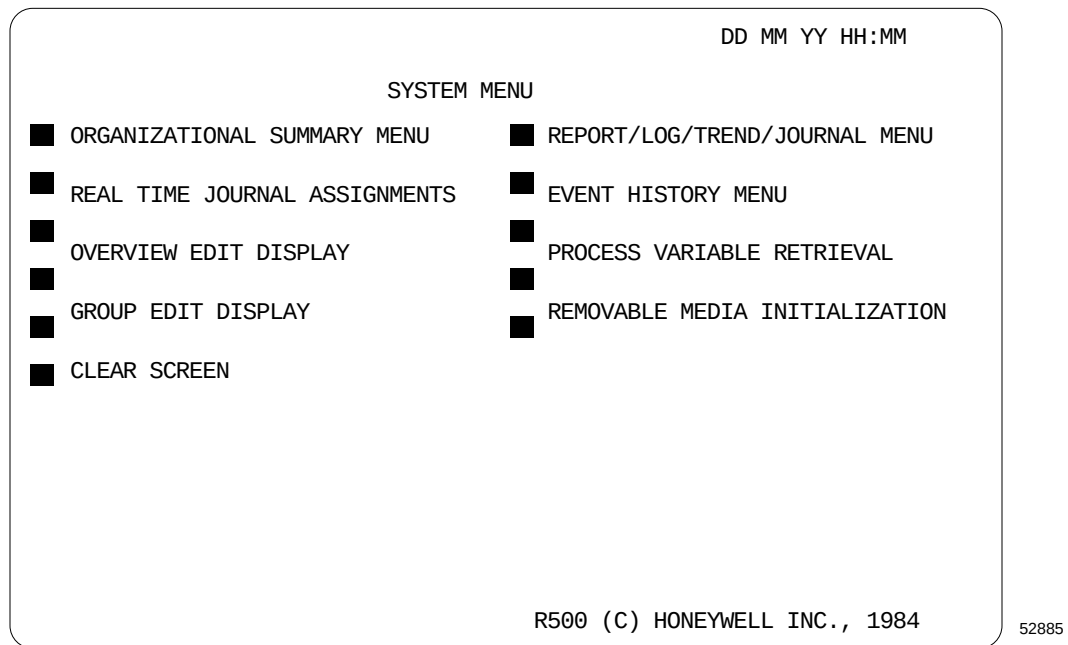
TPS System alarm conditions in the LCN modules, UCNs, Data Hiways, and process-connected devices are annunciated and displayed at the Universal Station. "Annunciation" includes closing the alarm contacts for a user-supplied external annunciator and blinking the <CONS STAT> key or <SYST STAT> key. The <CONS STAT> key blinks if the alarm condition is in the Operator Console. The <SYST STAT> key blinks if the alarm condition is in another module on the LCN, on a Process Network, or in a process-connected device. In addition, each system alarm is entered in the system Status Change journal.

TPS System alarms are acknowledged in the same way that plant process alarms are, by pushing the <ACK> key when a system alarm display is on the screen. This causes the blinking CONS STAT LED or SYST STAT LED to change to a steady light. Acknowledgment of the alarm condition is also entered in the system Status Change journal.

The LED light is extinguished when all alarms associated with that LED are corrected.

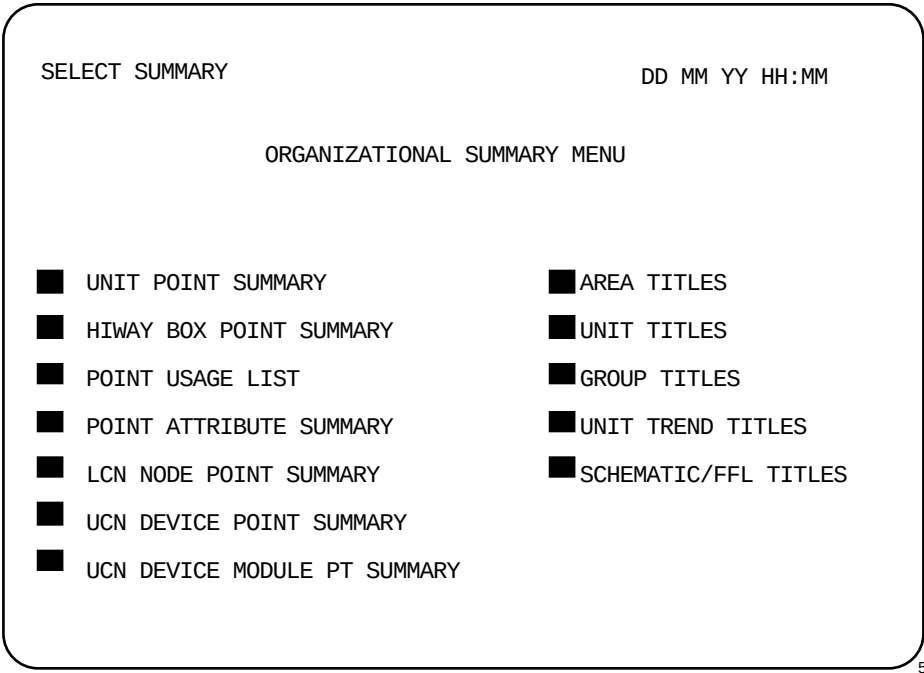
Organizational summaries

Organizational summary displays are provided at the TPS Universal Station so that the operator can easily determine point assignments, point usage, and title assignments. They are accessed from the System Menu Display, shown in the following figure.



Organizational Summary Menu

The types of organizational summaries that are available to the operator are shown in the following figure and described in the following paragraphs.



52905

**Unit Point
Summary**

The Unit Point Summary Display is a listing of all data points in a unit assigned to the Universal Station. The listing can consist of more than one page and requires that the operator enter the unit ID to display the points within the unit. For each point, the display lists the point ID, point descriptor, group number, and device address.

**Hiway Box Point
Summary**

This display provides the operator with a listing of all data points within the hiway box (device) selected by the operator. The data-point listing is arranged in slot/subslot order, and describes the data point, using the point ID, point descriptor, unit name, and group number. A separate display is provided for each hiway box, and a display can consist of more than one page.

Point Usage List

For the requested data point, this display lists all locations, such as trends, groups, and logs, to which the point has been assigned within the Universal Station's area.

**Point Attribute
Summary**

The Point Attribute Summary Display consists of a menu of display titles that describe specific point attributes, such as "Alarm Disable," "Red Tagged," or "Not Loaded." When selected, each of these subdisplays provides a "snapshot" of all the points in a specified unit having that particular attribute at that time.

LCN Node Point Summary	This display provides the operator with a listing of all points within the selected LCN module. The point listing is arranged in point ID alphabetical order and also contains the point descriptor, unit name, group number, and device address. A separate display is provided for each LCN module, and one display can consist of more than one page.
UCN Device Point Summary	Similar to the Hiway Box Point Summary, this display provides a listing of all data points within the UCN device selected by the operator. The listing shows the point ID, point descriptor, unit name, group number (if assigned), and device address. A separate display is provided for each UCN device, and a display can consist of more than one page.
UCN Device Module Point Summary	This display is identical to the UCN Device Point Summary, except it lists only those points attached to a single processor—for example, a PM High Level Analog Input Processor.
Titles	There are five titles displays: • Area, • Unit, • Group, • Unit Trend, and • Schematic, accessed under the Organizational Summary Menu. The Area Title Display lists the names and descriptors for all configured areas, the Unit Title Display lists the IDs and descriptors for all configured units, etc. The Schematic Title Display lists the titles of all graphic displays and the pathnames for all devices and volumes where these displays are stored (any graphic may be stored in more than one location).

6.6 Obtaining Process and System-Related Information

Introduction	Like the organizational summaries, the process and system-related information such as reports and organizational summaries are selected through the System Menu. Reference: Organizational summaries, page 143
Accessing logs, journals, trends, and reports	Logs, real-time journals, trends, and reports are accessed from the Report/Log/Trend/Journal Menu Display. This display lists journals, logs, printed trends, and reports that have been configured for the area assigned to the Universal Station. It shows whether they have been configured for scheduled automatic printouts, or whether they are demand only. In either case, from this display the operator can at any time print or display any journal, log, trend, or report and can also change the printer assignment.
Reports	A report consists of report header information and a grouping of one or more of the journals, logs, and/or printed trends. The Process Engineer configures all reports for the area assigned to the console.
Journals	The types of events that can appear in journals are process alarms, operator process changes, operator messages, system status changes, system errors, and system maintenance recommendations. These events can be printed in the real-time journal as the events occur, as well as being displayed or printed periodically or on demand.
Logs	A log is a set of historical values for a specified set of data point parameters. There are two types of logs: free format and fixed format. Both types are constructed at a Universal Station or Universal Work Station. Logs can be included in reports or individually called up through the Report Menu Display. If a History Module (HM) is not present, basic logs that consist of a limited number of log types and historical values are available. If an HM is present, extended logs that consist of point parameter values that are historized in the HM can be printed. Logs can be printed or displayed using a vertical format or a horizontal format.
Printed trends	Trends for up to eight points or parameters can be printed as continuous trends or trend history over a specified time interval. Historical trends can be included in reports or can be directly called up from the Report Menu Display.
Event histories	The Event History Menu Display is used to retrieve histories of process, and system-related, events. This display allows the operator to call up historized process alarms, operator process changes, operator messages, maintenance information, system status changes, system error messages, sequence-of-events messages, and system status messages.

Process value retrieval	Retrieval of process values for specified points can be done by the operator by using the Process Variable Retrieval Display. This display allows the operator to display or print process value histories, including the current value and hourly, shift, daily, and monthly averages. The period of history obtained depends on the start time and date that the operator enters.
Real-Time Journal assignments	This is a listing of all real-time journal assignments assigned to the console. From this display, the operator can activate or suspend any of the journals. Each entry in the display lists the journal type and printer identifiers.

6.7 Operational Security

Types of security Operational Security types are:

Configuration, page 148

Keyswitch access, page 148

Data-entry type checking, page 149

Configuration During system configuration, any Universal Station or Universal Work Station can be designated as a view only station. This permits a user to call up displays to monitor the process and system, but does not permit an entry that affects the database.

Keyswitch access Access to Universal Station functions is restricted by a keyswitch. Three levels of access are determined by the type of key inserted and the keyswitch position:

- **Operator Level**—permits an operator to effectively control the plant during normal operation, but does not permit changes to sensitive database parameters. The sensitive parameters of each data point are defined to some extent by the engineer, during configuration.
- **Supervisor Level**—permits supervisory personnel to alter some database parameters and permits all functions allowed at the Operator Level.
- **Engineer Level**—permits access to all functions that require access to the database, and that can be performed by the process engineer and programmer, such as display building.

Configurable levels of access are provided for many system functions, including the following:

- Saving and restoring databases of UCN or hiway devices
- Loading the database of a module
- Loading sequence programs
- Startup and shutdown of modules
- Change time and date
- Load the Engineering Personality into a Universal Station
- Access Maintenance Functions
- Assign and Deassign Units
- Enable and Disable Alarms
- Change Areas

**Data-entry type
checking**

Another security check occurs whenever an operator makes any entry. The system checks each entry to make sure it is of the right type (alpha, numeric, etc.) required for the operation. If the entry is invalid, an audible error tone is generated and an appropriate error message is displayed in the operator-entry field of the screen. For example, if when prompted to enter a unit ID, the operator entered a nonexistent ID, a message, **UNIT NOT CONFIGURED**, would appear on the screen.

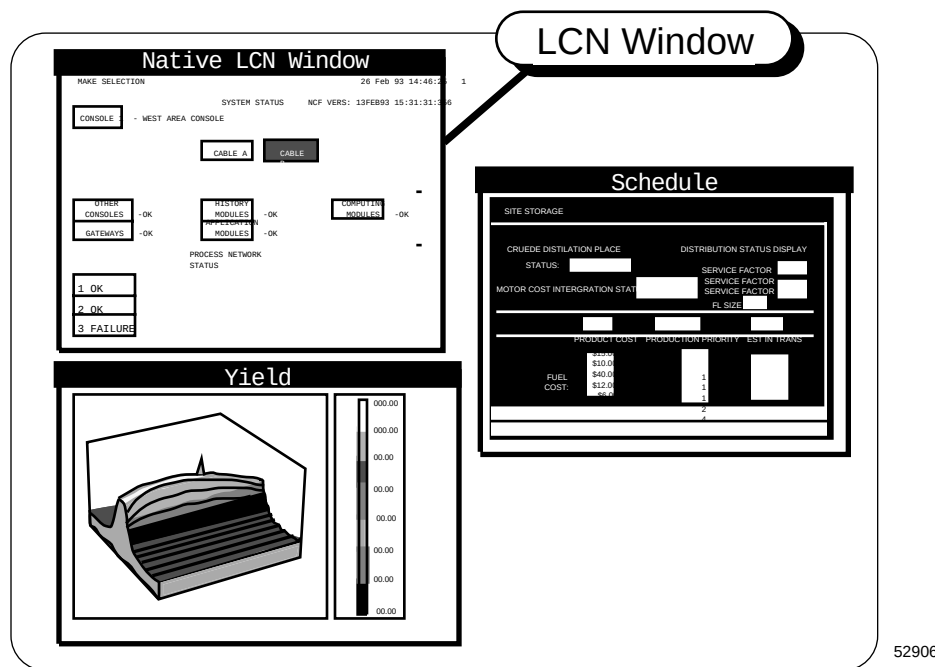
6.8 Universal StationX Operation

Integrated X Window system

Universal StationX adds a new dimension to TPS. It is a dual-ported station that fully integrates 68040 UNIX processors with standard Universal Station processors.

Usefulness of UXS

The process operator can use the Universal StationX not only to monitor and control the process using exactly the same displays and procedures already described for a Universal Station, but also can simultaneously look at pertinent data that is being collected and generated by other networked devices in your plant. The following figure illustrates how an operator at a Universal StationX can choose between the standard full-screen TPS displays or multiple displays that can be sized and positioned for optimum viewing.



Motif Window Manager

The X Window system provides the software framework that makes it possible for dissimilar computers to exchange displays. Universal StationX uses Motif Window Manager to provide a consistent appearance to all applications displayed. This software provides functions for opening, closing, moving, resizing, and highlighting the selected (“active”) window. Temporarily unneeded windows can quickly be reduced to small icons on the screen, and just as easily be restored to full size when needed again.

The Motif Window Manager applies the window interface attributes independent of the application running within the window. In addition to providing for resizing and iconifying of windows, it provides such elements as title bars, scroll bars, and control of overlapping windows.

Process system priority The process system maintains priority at a Universal Station^X. Should a failure occur in the X Windows/UNIX system, Universal Station^X is designed to revert to standard Universal Station operation.

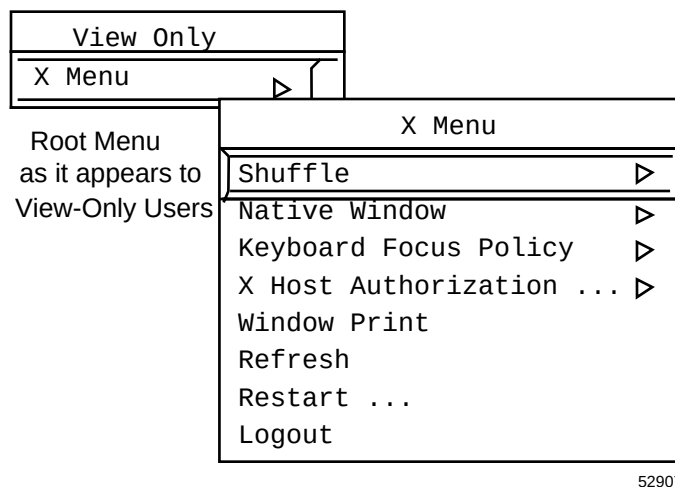
X Window access For the X Windows coprocessor to be started, a user name and password must be entered. Each person so authorized is assigned one of four possible levels of access:

- View Only
- Operator
- Supervisor
- Engineer

By clicking the mouse or trackball button with the cursor positioned in an open area of the screen, a standard root menu is displayed. Depending on the level of access, submenus can be selected that offer a list of functions.

Standard X Window menu functions are described in the following paragraphs.

Root menu with X menu In the following figure, the Root Menu is shown as it appears to view-only users.



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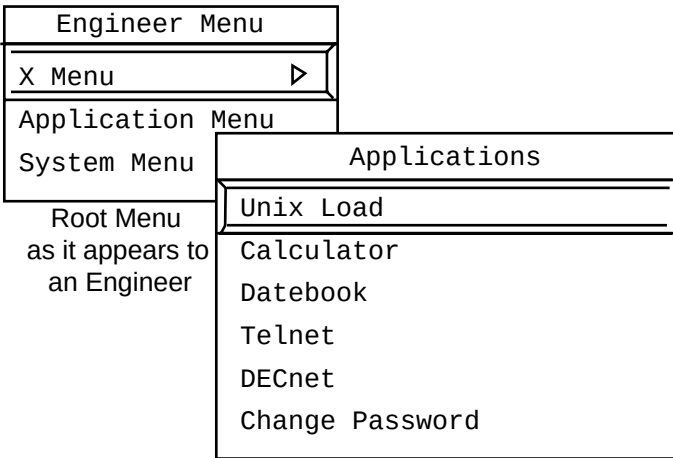
X Menu items are as follows:

- Shuffle—Moves the active X Window between foreground and background.
- Native Window—Allows the native LCN window to be partially obscured.
- Keyboard Focus Policy—Configures window behavior attribute.
- X Host Authorization—Authorizes remote computing resources to display X applications on this station.

- Window Print—Prints the content of the selected window.
- Refresh—Performs a redraw of the X application windows.
- Restart—Restarts (resets) the Motif Window Manager.
- Logout—Sends a kill message to all X hosts to close open applications, logs the user out, and idles the coprocessor.

Engineer Root Menu with Applications Menu

The following figure shows the applications menu as it appears to an engineer. Unix Load is omitted from this menu for operators and supervisors.



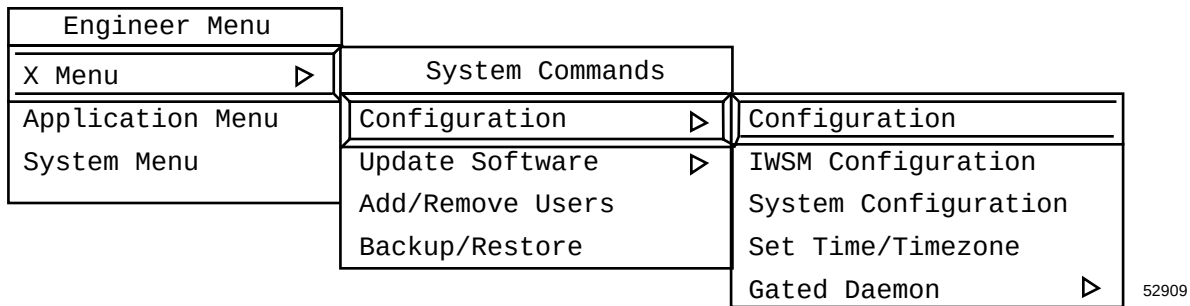
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Applications menu items are as follows:

- Unix Load—Bar chart display of UNIX processor loading.
- Calculator—Full function calculator.
- Datebook—Calendar/datebook. A window can be made to appear on the occurrence of an event.
- Telenet—Accesses the TELNET TCP/IP protocol.
- DECnet—Accesses the DECnet protocol.
- Change Password—Allows change of login password.

Engineer Root Menu with System Commands and Configuration Menus

The System Commands Menu and Configuration Menu in the following figure are available only to engineers.



- Engineer menus**
- System Commands menu
 - Update Software—Update the coprocessor software.
 - Add/Remove Users—Specify authorized users of UNIX and X Windows.
 - Backup/Restore—Backup or load the coprocessor software.
 - Configuration menu
 - IWSM Configuration—Configure the Industrial WorkSpace Manager, allowing users to shut down the workstation interface and to authorize entry of Native Mode.
 - System Configuration—Configure the coprocessor Local Area Network environment.
 - Set Time/Timezone—Set the coprocessor time, date, and time zone.
 - Gated Daemon—Turn dynamic Local Area Network routing on/off.

Custom menus Users with UNIX expertise can create their own custom menus. Such a menu might, for example, enable the operator to easily call up a graphic display that was generated by a personal computer residing on the Plant Information Network.

System Operation

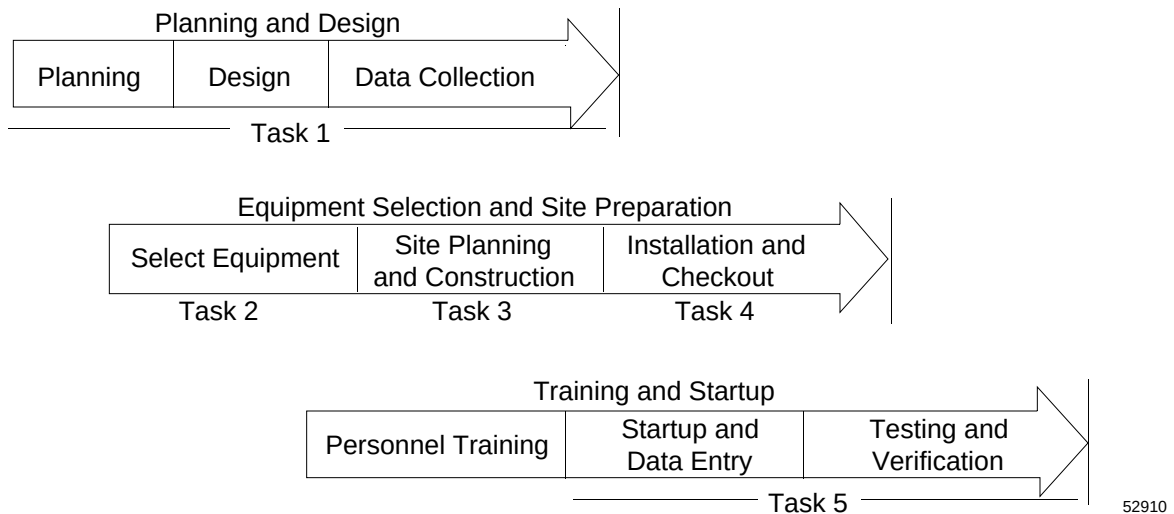
6.8 Universal StationX Operation

Section 7—System Implementation

7.1 Introduction

System implementation tasks

This section should provide a general understanding of the overlapping sets of tasks performed in implementing a **TotalPlant** Solution (TPS) System. The following figure shows an overview of these tasks.



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Task 1 – Planning and design

Planning and Design may be the most important of all. In this task all of the system implementation documents are prepared. Some of the documents are prepared by the user in the form desired. Much of the documentation is on Honeywell-supplied configuration forms.

Task 2 – Equipment selection

Equipment Selection is based on some of the early work accomplished in Task 1. The equipment selected depends on the requirements of the project. Items considered include the size of the process; the number and location of the operators, supervisors, and engineers; the need for historical data; the types of control strategies to be used; and requirements for backup and redundancy.

Task 3 – Site planning and construction

Site Planning and Construction includes planning and construction of the site(s) where the equipment will be installed. Items considered include equipment size, weight, power requirements, grounding, cabling, and air conditioning.

Task 4 – Installation and checkout

Installation and Checkout includes unloading, unpacking, and moving the equipment into its final location; cabling, grounding, applying power; and checking out the hardware.

**Task 5 – Startup,
testing, and
verification**

Startup, Testing, and Verification includes loading software, entering configuration data, testing the data and system operation, verifying data acquisition and control strategies, and bringing points, units, and areas "on control."

7.2 Planning, Design, and Data Collection

Control system planning

Control system planning begins with the transformation of the process engineer's knowledge of the process into a definition of control strategy, then moves to the details of control system design. In this decision-making process, design tradeoffs that involve a number of closely related concerns will be evaluated:

- Control-system cost
- Control-system reliability
- Control-system response characteristics

Each project needs to establish policies that control the ways in which these design tradeoffs are made. For example, there are at least a dozen possible ways to implement a cascade loop, using various combinations of TPS network nodes and process-connected devices, each way with its own impact on capability, cost, reliability, and response. The control-system designer may choose, as a matter of policy, to use only two or three of these as being most appropriate to the company's control philosophy or in reaction to site-specific situations. Control-implementation policy is affected by considerations such as:

- Whether this is a new system or an expansion
- Whether or not the control system is to be implemented in steps with advanced features to be added at a later date
- Safety, security, and reliability concerns
- Policy as established by a centralized planning organization
- Spare parts and maintenance considerations
- Training requirements

Control system design

Within the policy constraints, the control engineer makes control-design decisions based on criteria such as response requirements, loop criticality, control criticality, requirements for history retention, computing requirements, and plant geography.

After these basic decisions have been made, the process engineer needs to develop a conceptual layout that partitions the system into operational **areas** of responsibility. Areas are then subdivided into process **units** and points are combined into related **groups**, for ease of operator interpretation. Several engineers can independently work on different functions once the critical partitioning and assignment decisions have been made. For example, one engineer can work on the logs and displays, another on advanced control and a third on history data collection.

At this point, design emphasis has moved from the planning of how equipment is used to the consideration of how the operator perceives the process and its control. Information must be grouped in ways that are relevant to operational concerns. The choices made in assigning points, units, and groups to standard and custom displays and to standard and custom logs is critical in the meeting of operator needs. Attention must be paid to establishing a display hierarchy that matches the operator's need to interact with the process.

The conceptual partitioning into areas, units, and groups also leads into the specifics of network layouts that make best use of the available control system equipment.

Use of configuration forms

Information on individual areas, units, groups, and points is obtained from plant drawings and loop specifications and can be copied onto configuration forms for later entry into the system database. Complete one-for-one transcription of point data to forms is not necessary if the engineer plans to do the database building.

The configuration forms provided by Honeywell are organized to facilitate the configuration data-entry task, and are packaged to support the four general areas of Network Configuration, Point Building, File Building, and Area Configuration as summarized in the following table. Both data collection and data entry can be iterative processes, with details being added as they become known and changes to the system being accommodated as they occur.

Configuration Form Sets

Types of Configuration Data Collected	Form Types
Definitions of the network's areas, units, consoles, modules, and process-connected devices	Network Configuration Forms
Data point names and point parameter values.	Point Build Forms
Instructions for building custom graphics, logs, keyboard button action, history groups, etc.	File Build Forms
Definitions of area parameters, operating displays, log assignments and alarm annunciation policies to define the process operator's realm of responsibility.	Area Configuration Forms

The TPS configuration forms are organized to match the sequence of data entry for the specific set of TPS System functions being implemented in specific hiway boxes, LCN modules, and UCN devices. Definition of each entry is included either on the form itself or in separate form instructions. The forms provide for data to be entered by keystroke and, where appropriate, for selection of fixed choices. The instructions show ranges of permissible entries and any default values that may exist or may be applied.

Although many configuration forms are available, only those applicable to the specific mix of TPS LCN modules, UCN devices, and hiway boxes to be implemented need be used. Also, many forms facilitate the skipping of nonessential information areas, thereby reducing the scope of the data collection and data-entry tasks.

Configuration forms example

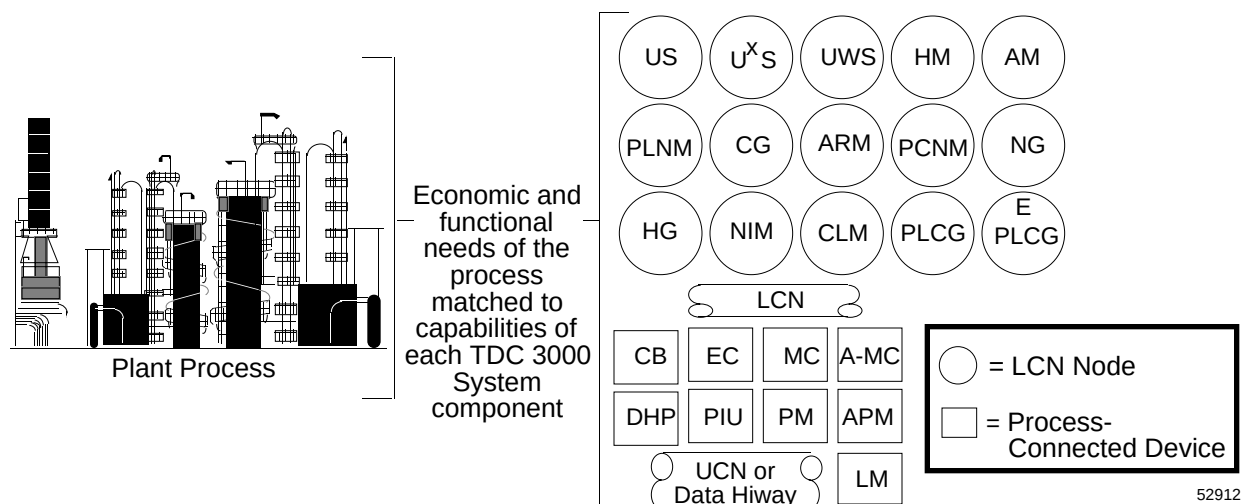
The following figure shows a portion of a configuration form that includes descriptions of each entry.

Unit:		Tagname:		Engineer:		Date:	
<i>Point Assignment Display - Begin DI Status Type configuration.</i>							
NAME				<i>range =</i>		<i>Not applicable</i>	
Tag Name				<i>default =</i>		Blank	
NODETYPE							
Node Type		PM LM				A PM Point Is Being Configured An LM Point Is Being Configured	
PNTFORM							
Point Form		Full Componnt				Point is fully displayed and alarmed Pt. is partially displayed but not alarmed	
PTDESC				<i>range =</i>		<i>Not applicable</i>	
Point Descriptor				<i>default =</i>		Blank	
EUDESC				<i>range =</i>		<i>Not applicable</i>	
Engineering Units Descriptor				<i>default =</i>		Blank	
KEYWORD				<i>range =</i>		<i>Not applicable</i>	
Keyword Descriptor				<i>default =</i>		Blank	

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7.3 Equipment Selection and Site Preparation

TPS System selection process



Selection of various TPS System components depends upon the sources of plant/process information, process management considerations, and intended use of data within the network. There are TPS System components for connecting instrumentation subsystems or outside-vendor computers to the network; for accessing the global database and/or calculating results used to provide advanced control; for maintaining historical profiles of network data values; and for providing a window to plant-wide information for display, process management, and hard copy reports.

Example: System component selection

Table 7-1 Selecting an Example TPS System

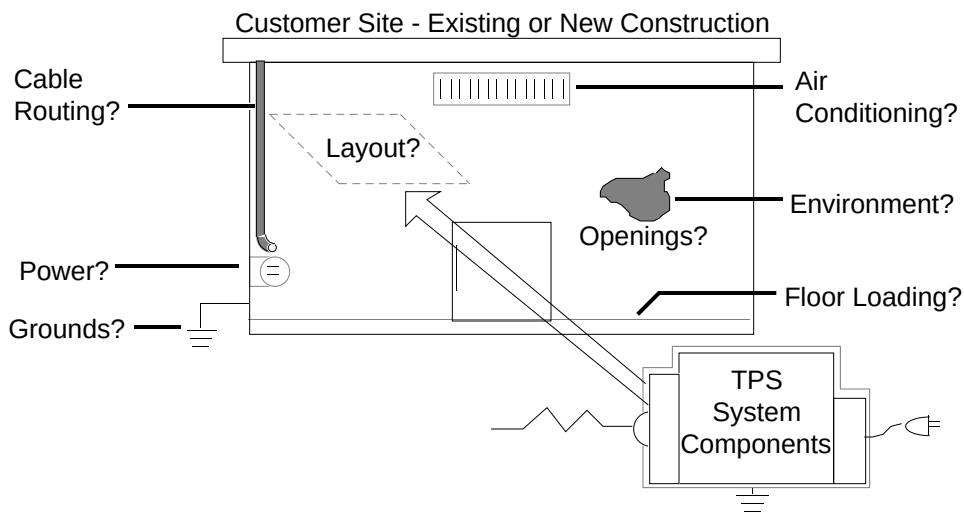
Project Requirements	TPS System Components Selected
Instrumentation subsystems are to control and monitor a continuous-type process and a batch-type process.	Three Advanced Process Managers (APMs)
The instrumentation subsystems are to be connected to the TPS network.	Redundant Network Interface Modules (NIMs)

Table 7-1 Selecting an Example TPS System (continued)

Project Requirements	TPS System Components Selected
Separate operators must independently manage three plant-process areas while viewing each area's Overview, Group, and Custom Graphic displays at the same time. In addition, there is a need for the operators to view displays created on PCs operating on a Plant Information Network under the X Windows environment.	Three TPS consoles with two Universal Stations (USS) and one Universal Station ^X in each console
A large amount of historical data must be captured and reported.	Two History Modules (HM)
A large number of noncritical process data points that could reside in instrumentation subsystems are to reside in a TPS module.	One Application Module (AM)
The R&D Department requires unhindered access to the global database for development and test of advanced control strategies.	One Universal Work Station
The Administration Department requires access to the global database as well as remote terminal and Fortran capability.	One DEC VAX computer with Plant Network Module (PLNM)

TPS site planning process

The following figure illustrates the most important planning elements to be considered when preparing the site to accommodate the TPS System.



Site planning is the set of considerations required to properly prepare the control room so that the TPS System can be installed, can be powered up, and can perform its intended tasks. Environmental considerations address room temperature and air cleanliness. Power considerations address grounding, voltage, amps, line frequency, and power integrity. Physical-equipment considerations address space, layout, cable trays, floor loading, and size of openings.

Harsh environment consideration

Corrosion is a leading cause of failure in electronic boards in harsh environments. In order to ensure the maximum possible reliability in the harshest environments, Honeywell offers two optional levels of protection:

- The first solution offers conformally coated boards, which are completely covered with a thin film of a special plastic-like polymer that is resistant to the corrosive effects of humidity and certain chemical gasses. Conformal coating makes it possible to mount APM equipment in environmentally harsh atmospheres.
- The second solution incorporates conformally coated boards in a special package which allows the user to locate their PM I/O outside the confines of a control room, near the process in environmentally severe atmospheres. This product is called Remote Hardened I/O (RHIO).

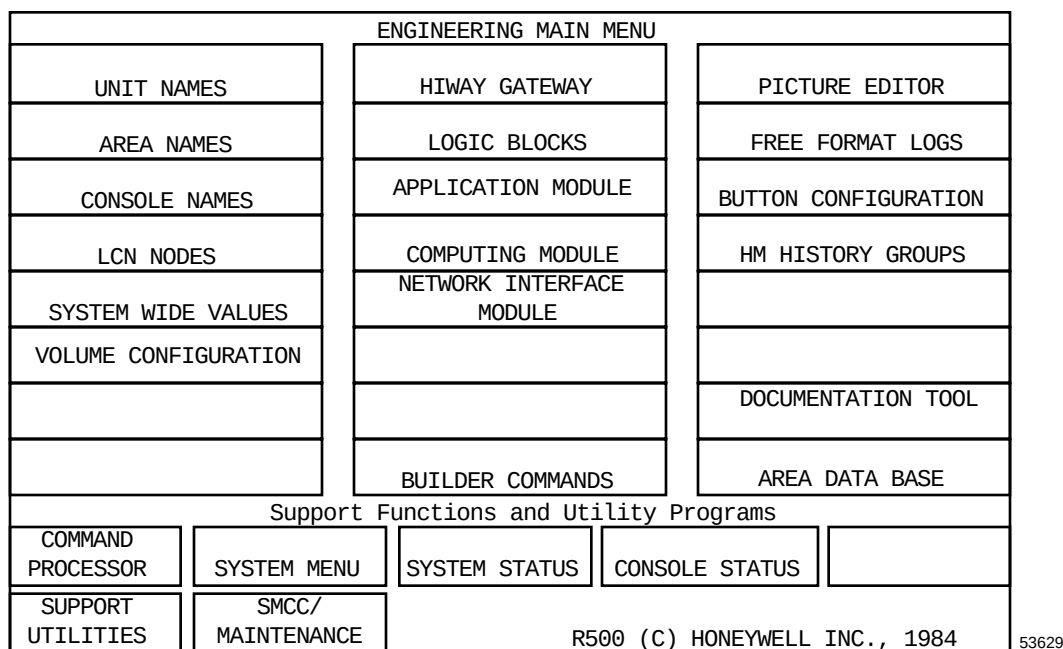
TPS System installation process

Installing the TPS System requires uncrating, inspection, move-in, and placement of equipment. Cabinet, cable, power, and ground connections are then made. After overall inspection of the installed system, each system component is powered up and unit tests and diagnostics are run on each unit. Finally, system tests and diagnostics are run to verify that the system is ready for startup and configuration data to be entered.

7.4 System Configuration

Engineering Main Menu

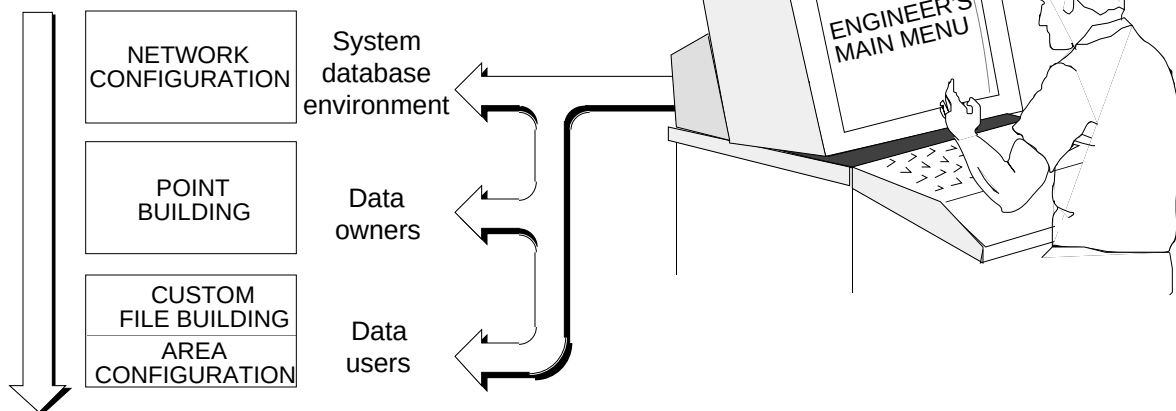
Process engineers can configure the TPS System at a Universal Station of Universal Work Station by selecting targets on the Engineering Main Menu and by entering configuration data into “fill-in-the-blanks” screen forms. The Engineering Main Menu appears in the following figure.



Configuration task summary

The tasks of TPS configuration are as follows:

Order of Database Building



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- Build a pool of units by naming all units. Name areas and consoles and define all modules, UCNs, hiways, and process-connected devices.

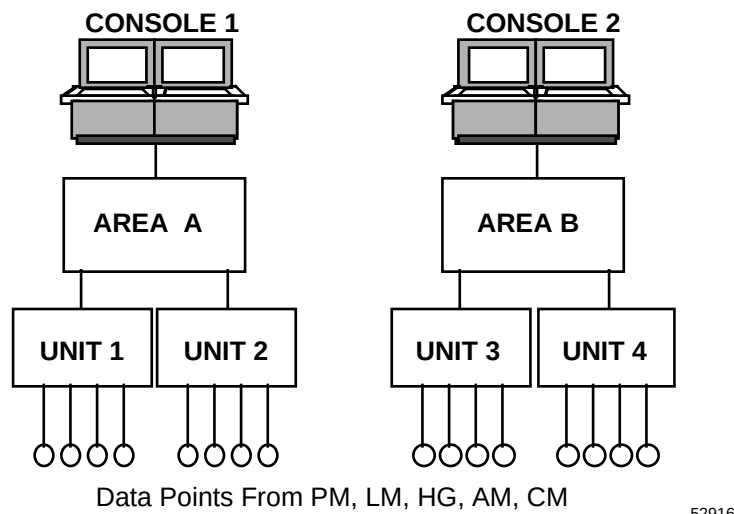
- Establish the types and sizes of files to be stored on the History Module.
- Build a pool of data points and assign each point to a unit.
- Build a pool of custom-built files (with pathnames) containing graphics, logs, history groups, button actions, descriptor libraries, and programs.
- Build area database files by choosing units, data points, and pathnames (of custom-built files) from each pool above.
- Load area database files into consoles and cause the system to form the final on-process personality for each console's Universal Stations.

Establishing the operator's realm of responsibility

Before the configuration data entry can begin, it is necessary to partition the process into conceptual groupings of equipment that can be managed by an individual working from an Operator's Console.

From the console, the operator views an area of responsibility that is composed of several process units. Each process unit is comprised of a number of related data points.

The following figure shows one such conceptual grouping for two operators, each with an assigned area comprised of two units. Either operator can observe data from any unit at his console, but can control only those units assigned to his area. Units can easily be reassigned between areas as operating conditions may require.



A TPS System, on a single LCN, can have up to 25,000 data points organized into 100 units divided between 10 areas and overseen through 10 operator consoles. There can be up to 36 units for each area, and 200 to 250 data points for each unit.

Multiple LCNs, separated by extended distances, can be interconnected through Network Gateways. This enables a single TPS System to provide integrated control over an entire large industrial complex with several physically separated control rooms and monitoring stations.

Using the Engineering Main Menu

Process engineer's use the Engineering Main Menu to enter configuration data that defines the plant process to the TPS System.

The Engineering Main Menu contains logical groups of configuration activities and activities to support the overall configuration process.

Reference: Engineering Main Menu, page 163

Activities that support items not implemented on a specific system are omitted. For example, for a system with no Data Hiway, no Application Module, and no Computing Module, the following activities are omitted:

- Hiway Gateway
- Logic Blocks
- Application Module
- Computing Module

Configuration data entry

Engineering Main Menu activities that are used to enter configuration data are summarized in the following table. Reference: Engineering Main Menu, page 163

Table 7-2 Configuration Data Entry Activities

Activity	Summary Description
Unit Names, Area Names, Console Names, LCN Nodes, System Wide Values, Volume Configuration	Network configuration—These activities enter data required by all nodes on the LCN.
Hiway Gateway, Application Module, Computing Module, Network Interface Module, Builder Commands.	Data point building—These activities build data point used in process monitoring and control.
Logic Blocks	This activity builds the logic to be executed by Logic Blocks in Multifunction Controllers on a Data Hiway.
Picture Editor	Users build custom graphic displays in this activity.
Free Format Logs	Users define the form and content of Free Format Logs in this activity.

Table 7-2 Configuration Data Entry Activities

Activity	Summary Description
Button Configuration	This activity defines the functions of user-configurable buttons on Operator's Keyboards.
HM History Groups	This activity defines continuous history data to be stored in HM History Groups on History Modules.
Area Data Base	This activity builds area database entities that define displays, reports, logs, and other process activities for each of up to ten process areas.

Support functions The support functions on the Engineering Main Menu provide tools and activities that are used to support other engineering activities and system operation. They are summarized in the following table. Reference: Engineering Main Menu, page 163

Table 7-3 Support Functions

Activity	Summary Description
Command Processor	Utility commands that create, modify, and manipulate files; print and display file, volume, and directory content; rename files; and protect files.
Support Utilities	Used to modify default volume pathnames for system and user volumes and directories, and to translate software and data to later revisions.
System Menu	Calls up the System Menu. Reference: Organizational summaries, page 143
SMCC/Maintenance	Accesses the System Maintenance Control Center portion of the Engineering Personality. This activity is used for system hardware maintenance and troubleshooting.

Table 7-3 Support Functions (continued)

Activity	Summary Description
System Status	Calls up the System Status Display. Reference: Monitoring system status, page 140
Console Status	Call up the Console Status and Assignment display. This display shows the status of the Universal Stations and other devices in a console.

Documentation Tool

The Documentation Tool is a vehicle for easy access to selected sets of data from throughout the TPS database. The particular collection of information desired can be specified, manipulated, and recorded as the need arises, or can be prespecified for retrieval at any time.

The Documentation Tool is called through the [DOCUMENTATION TOOL] target on the Engineering Main Menu, page 163

The Documentation Tool allows the user to retrieve data from

- process-connected devices
- LCN modules
- HM and disk files
- Network Interface Modules.

Once retrieved, the displayed information can be customized using available commands. Results can be output to a file and to a printer. Because of the ability to access, customize, display, and record data in a variety of different ways, the results can be beneficial to operators, maintenance personnel, and engineers.

The following table shows the data source types with some examples.

Table 7-4 Documentation Tool Data Sources and Types

Data Sources	Examples of Types of Accessed Data
System Files	Specific type of data within that file; for example: • all SEND statements used in a CL program, • Alphabetical listing of a Find Names file
Process-Connected Device or LCN node data	Point data with specific attributes; for example: • Tuning constants for all PID controllers, sorted by device address • All logic points in a device, with corresponding input connections, interlocks, and permissives • Entities of a certain type, listed alphabetically with their PVs, setpoints, and modes.

Configuring the TPS network

The configuration of the TPS network allows the process engineer to establish the environment within which process management will be implemented. This environment includes the unit names, area names, console names, LCN nodes (modules), system default values, and data storage volumes that will exist in the network.

The TPS network must be configured before the system's data owners (modules, UCN devices, and hiway boxes) can accept and validate point data built by the engineer.

The following table lists the network configuration activities on the Engineering Main Menu, page 163 with the types of data entered.

Table 7-5 Network Configuration Activities

Activity Used	Configuration Data Entered
UNIT NAMES	Unit IDs/descriptors that points will be assigned to
AREA NAMES	Descriptors for area that units will be assigned to
CONSOLE NAMES	Descriptors of consoles that will manage areas
LCN NODES	Descriptions of each network node and its peripherals
SYSTEM WIDE VALUES	Changes to preconfigured system defaults, other system information such as shift definition; purchased software options
VOLUME CONFIGURATION	Size of each HM storage volume

Building points

This is where the process engineer assigns the UCNs and/or Data Hiways and the devices that will reside thereon, and specifies the hardware content of each device. Having done this for all such process-connected devices, and having previously defined the LCN modules, the process engineer can now define to the system what pool of data points are to be available for process management. It also allows the engineer to designate the process unit for each point, and the LCN module or process-connected device each point is to reside in. (Note: the environment within which data points will exist was previously defined by the process engineer during network configuration.)

A data point name and the unit to which it is assigned are specified for each data point regardless of what point type it is or for what device it is being built.

The following table lists point-building activities on the Engineering Main Menu, page 163 with the types of data entered.

Table 7-6 Point Building Activities

Activity	Configuration Data Entered
HIWAY GATEWAY	Hiway attributes and box types; card assignments for each box; hiway points, algorithms, parameter values
LOGIC BLOCKS	If an MC is present, strings of logic blocks
APPLICATION MODULE	If an AM is present, AM points and parameter values

Table 7-6 Point Building Activities

Activity	Configuration Data Entered
COMPUTING MODULE	If a host computer is present, compute-related points and program names
NETWORK INTERFACE MODULE	UCN attributes and device types, slot assignments for each device, UCN points, algorithms, parameters, values, UCN Library

Points must be built before custom graphics and free-format logs can verify the existence of data points and access data point parameters by data point name.

Building Area databases

A process area typically consists of several process units that are monitored and controlled from one TPS console. An area database defines the process area for the Universal Stations in which it resides. A TPS system can have up to ten area databases.

Area database building begins with the Area Configuration Menu. This menu is called up by selecting the [AREA DATA BASE] target on the Engineering Main Menu, page 163

The following table summarizes the entities that process engineers build into area databases:

Table 7-7 Area Database Summary

Database Entity Type	Purpose
Unit assignment	Defines the units assigned to the area.
Display definition	Define the operating displays used in the area.
Reports, logs, and journals	Define the logs, reports, and journals printed by area Universal Stations.
Annunciation policies	Define the alarm annunciation policies for the area.
Pathname catalog	Defines the location of files used by area Universal Stations.

Configuring the Universal Station^X Coprocessor

When power is applied to a Universal Station^X, the Login Banner illustrated in the following figure appears at the bottom of the display screen. Factory preconfigured user names and passwords for all access levels enable the new user to bring the system up for initial checkout and configuration.

Login :		Password :	
OK	Native Mode	Shutdown WSI	Clear

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Information unique to each Universal Station^X must be entered into the coprocessor when that station is initially put into service. Required entries include

- Coprocessor password
- User names and passwords, and associated authorized access level
- Time, date, and timezone
- Local Area Network information
- Printer descriptions
- Industrial Workspace Manager instructions
- Remote access control authorization
- Native Window behavior
- Keyboard/mouse pointer interaction.

Once the Universal Station^X is up and configured, a user who is knowledgeable about UNIX and the X Window system can customize the X Window environment to satisfy local preferences and special requirements.

International Language Support

TPS software supports custom character sets, referred to as International Character Sets (ILS). They allow you to introduce special characters into screen displays, including Cyrillic, Japanese, and other language characters. This feature applies, with some restrictions, to R400 and R500 series software.

Any string or enumeration parameter, except tagnames, configured from the Engineering Personality can use characters from the International Character Set. ILS characters can be used as background in any custom graphic or picture entered at the Engineering Keyboard. Certain limitations may apply to your system. For example, the International Language Support feature does not extend to VAX computers through the CM50S Gateway. Ask your local Honeywell representative for more information.

Parameters that accept characters from the ILS:

Unit Descriptor	Overview Field Title
Area Descriptor	Group Title
Console Descriptor	Process Module Group Title

System Descriptor	Area Trend Title
Point Descriptor	Unit Trend Title
Point Keyword	Alarm Annunciator Box Title
Engineering Units Descriptor	Log Description
State Descriptor	Free Format Log Description
Engineering Units	Process Journal Title
CDS Value	Text File Editing
CDS Descriptor	Schematic String
CDS Parameter Name	System Journal Title
CDS Enumeration State Name	Printed Trend Title
Report Log String	Report Title
CL Message String	Sequence Phase Name
CL Comment Line	Sequence Step Label
CL Subroutine Name	Sequence Abnormal Handler Name
CL Function Name	Sequence Name

Implementation of the International Language Support function is performed locally by Honeywell International Region Engineering groups and consists of three primary tasks:

1. Developing "Font Character" definitions.
2. Assigning those defined characters to specific keys (or shifted/keys)
3. Writing and installing a syntactically correct font file.

If you wish to print out information with ILS characters, you will need a printer that has this special capability.

Additional information and specific implementation details are available to Honeywell engineers in issued drawing, 51196428.

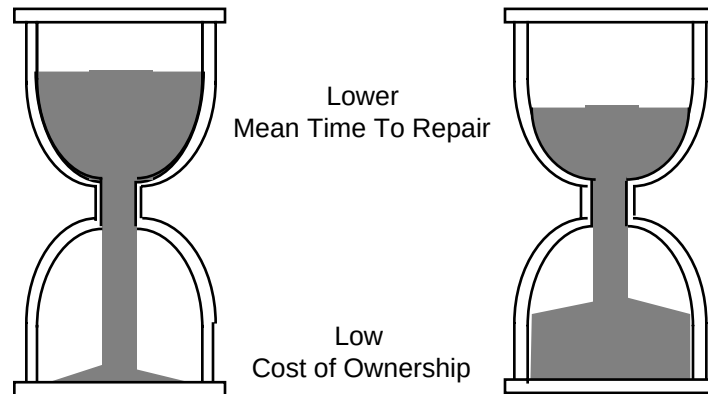
Section 8—System Maintenance

8.1 Maintenance Objectives

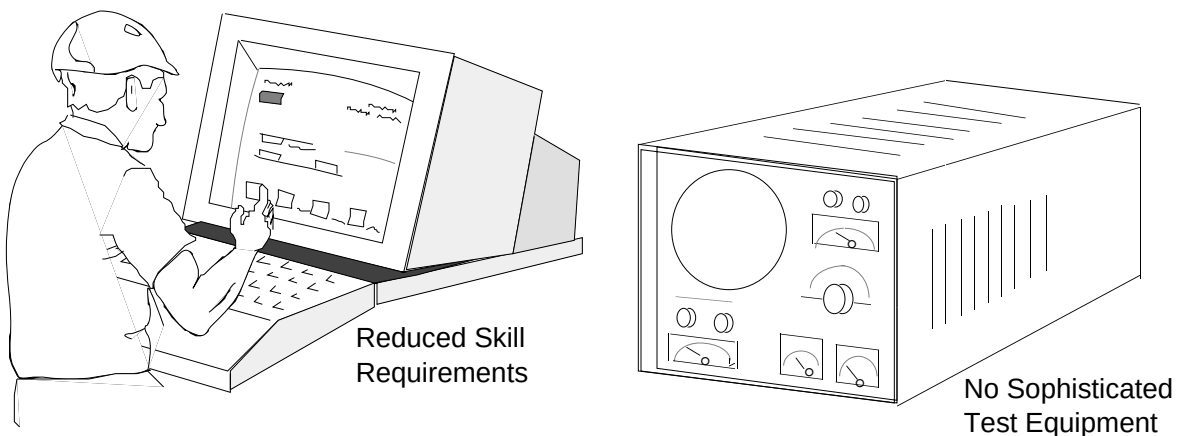
Low cost of ownership

One of the principal objectives of **TotalPlant** Solution (TPS) System design is to minimize the cost of ownership. This is accomplished by combining high dependability, and low maintenance cost.

Reference: [3— System Dependability, page 31](#)



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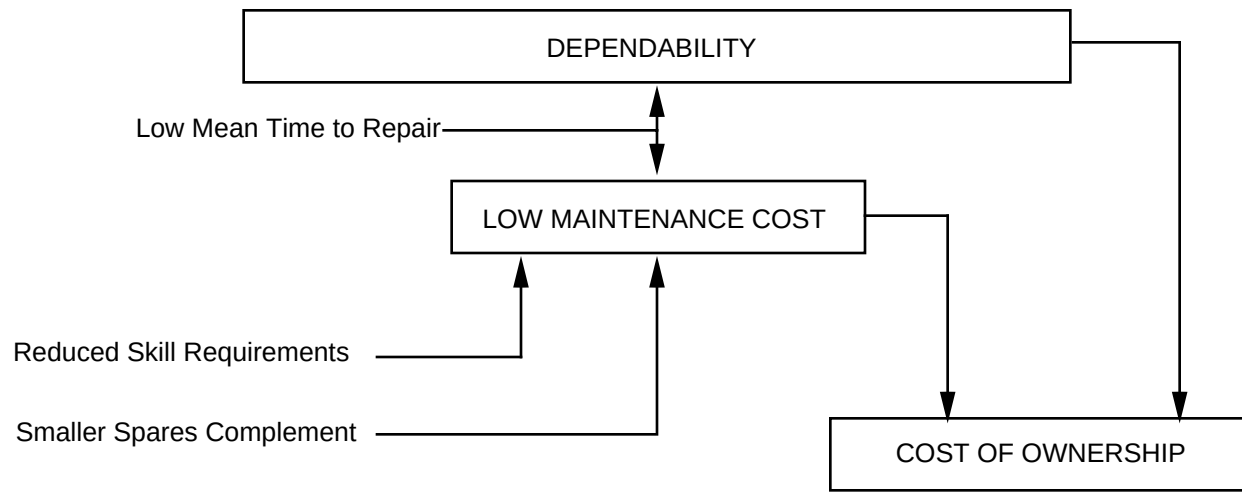
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Ownership cost factors

The following figure shows how dependability and maintainability are interrelated in the TPS System, and how they join to reduce the cost of ownership.

System Maintenance

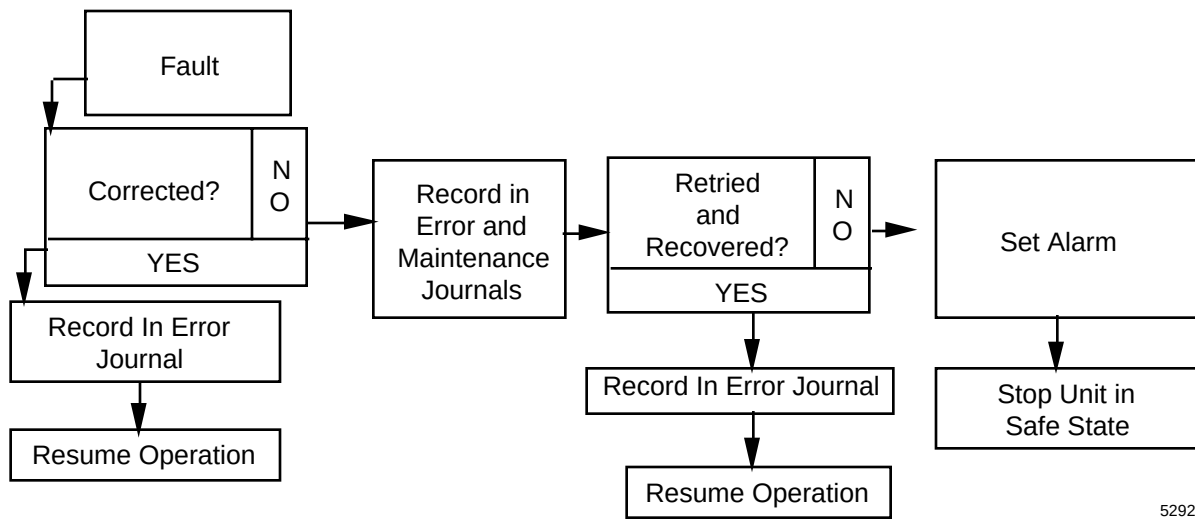
8.1 Maintenance Objectives



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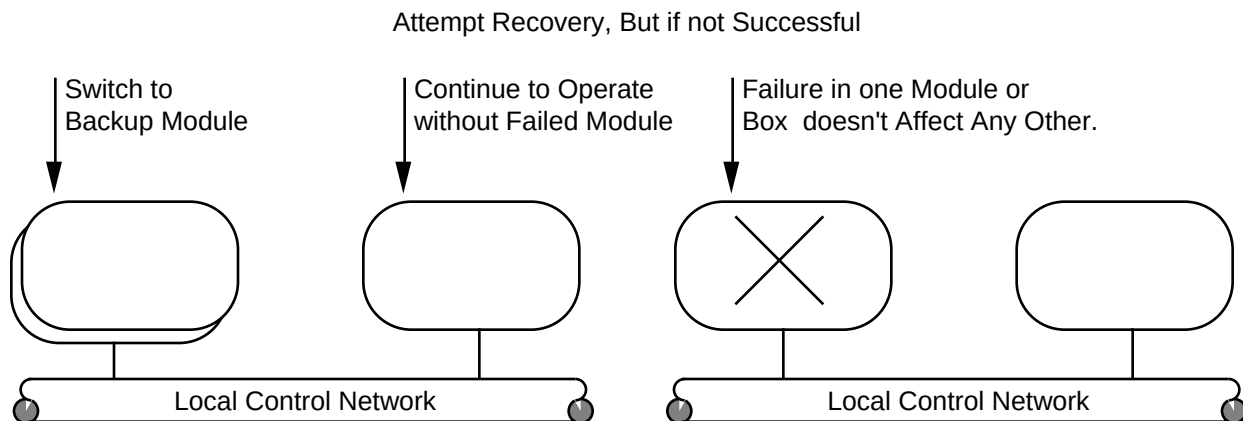
8.2 Maintenance Strategy

- Major elements** The major elements of the TPS maintenance strategy and tactics are listed below and discussed on the following pages:
- Faults are rare, and little preventive maintenance is required.
 - The faults that do occur are tolerated with little or no degradation of system performance, and they affect only the module in which they occur.
 - For most failures, the system itself determines the correct Optimum Replaceable Unit (ORU)—a board, a power supply, a peripheral device, etc.
 - The variety of ORUs is held to a minimum, so the spare parts inventory is minimized.
 - Test equipment is seldom needed, and no special or complicated test equipment is required.
- Fault containment and fault tolerance** First, some terms must be defined:
- A fault is a temporary or permanent condition in a functional unit that makes it deviate from its expected sequence of operations. A fault may exist even though it cannot be seen outside the unit, because the fault may be corrected each time it occurs.
 - An error is an abnormal or undesired result of an operation caused by a fault. An error may or may not be overcome by retrying the operation. The TPS System is designed to detect virtually all errors.
 - A failure is the inability of a functional unit to perform its expected task because of a "solid" error or an error that recurs too frequently.
- The TPS strategy for fault containment is that if a fault occurs in a module on the LCN:
- If possible, the result of the fault is corrected.
 - If the result is not corrected, the operation is retried one or more times, in the hope that the error won't recur.
 - If retries are not successful, or if too many errors recur in the module, the module has failed. The operator is notified through an alarm and the module's operation is stopped in a safe state.



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The following figure shows how the TPS architecture contributes to fault containment and fault tolerance. If a module on the LCN fails and it has a redundant module backing it up, the backup takes over in five seconds or less. If there is no backup, the system continues to operate without the failed module. Some capacity, throughput, or functions may be lost, but the system remains operational. Also, with two exceptions, a failure of one module has no effect on any other functional unit in the system. Most units in the system are independent of other units for their operation.



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The exceptions are the Network Interface Module and the Hiway Gateway. If a TPS System had a single NIM or HG linking the LCN with the process-connected devices, the process connections would be lost if this module failed. This is why NIM or HG redundant pairs are used in many installations.

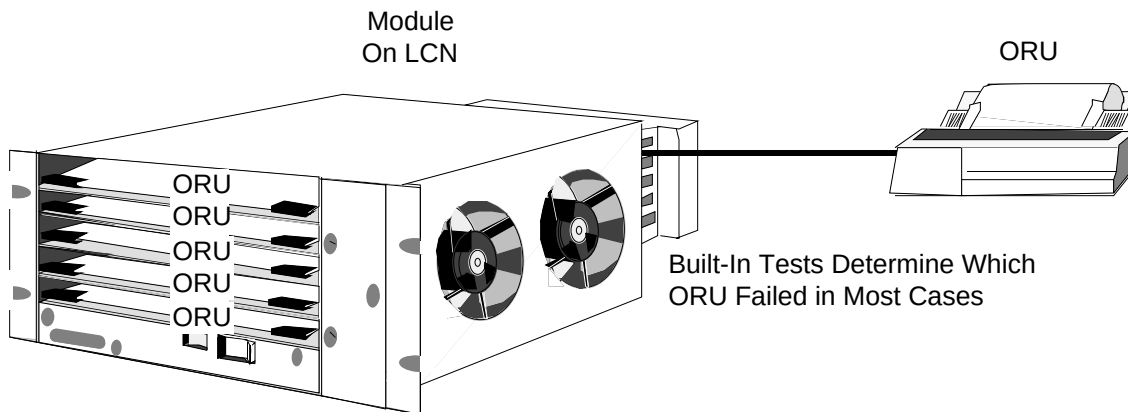
The same strategy applies both to TPS control room equipment (modules on the LCN and devices connected to the modules) and to process-connected devices. Process Managers and Advanced Process Managers on the UCN are available with redundant (Advanced) Process Manager Module control hardware and some redundant I/O Processors. Redundant Processor Files can be specified for the Logic Manager. Some boxes on the Data Hiway do not have redundant back-up units, but the Basic Controllers, Multifunction Controllers, and Extended Controllers can have redundancy through the Uninterrupted Automatic Control (UAC) option.

8.3 Maintenance Tactics

Self-diagnosis

The tests built into the TPS control room equipment are designed to diagnose a high proportion of all failures, pinpointing them to the optimum replaceable unit (ORU) without intervention by the operator or a maintenance technician while the system is online and controlling the process.

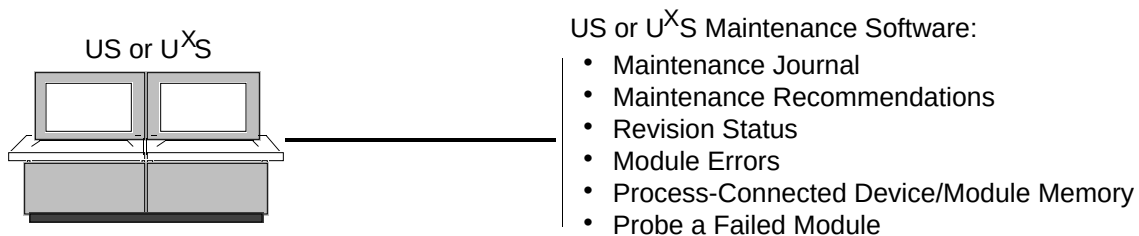
The process-connected boxes on the Data Hiway all have built-in self-tests, which either diagnose or help diagnose failures in those boxes.



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The Universal Station maintenance software

A Universal Station, or Universal Work Station, has maintenance software that is used to diagnose failures in modules on the LCN that are not diagnosed by the built-in tests, and to test the operation of the UCN, Data Hiway, and their process-connected boxes. When an on-process Universal Station is being used for maintenance, one or more of the remaining Universal Stations in the console takes over its on-process functions.



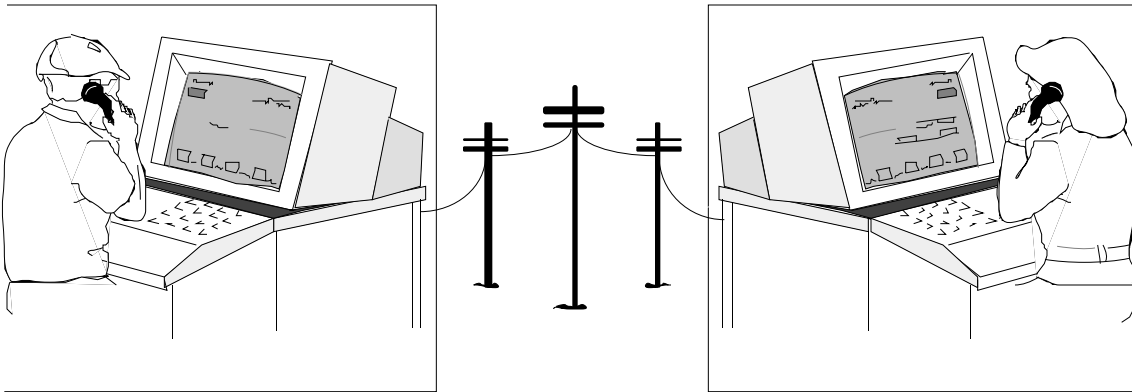
One or more of the other Universal Stations takes over process functions of the station diverted for maintenance.

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Technical Assistance Center (TAC)

Honeywell provides expert assistance in diagnosing failures that are not quickly resolved on-site. This is accomplished by telephone communications between the maintenance technician and a service specialist at Honeywell's Technical Assistance Center (TAC). The service specialist will request symptomatic information from the technician, and may request the technician to perform data-entry steps at the Universal Station (or Universal Work Station).

If necessary, the specialist will attempt to duplicate the problem on TAC-center equipment. From the information gained, the specialist then recommends remedial steps for the technician to take.



Simplified repair procedures

The key to reduced repair times for TPS Systems is the replacement of easily isolated and removed parts and their return to Honeywell for repair. The selected replacement parts are referred to as Optimum Replaceable Units or ORUs; they offer an effective tradeoff between the time-based costs of problem isolation and resolution, and the replacement costs of the parts themselves.

For some devices, such as printers and CRTs, two levels of ORU are defined, thus expanding the range of time-versus-parts costs tradeoffs. The primary ORU level is the device itself; the secondary ORU level consists of lower-level service parts that require some additional skills to isolate and replace.

The control room equipment developed for TPS is designed with the objective of minimizing the complement of ORUs. This is accomplished partly by using boards that are common throughout the LCN. The ORU concept also applies to the UCN, the Data Hiway, and the process-connected devices.

Any module on the LCN, any peripheral device, and any process-connected device on the UCN or Data Hiway can be maintained and repaired while the remainder of the system is on-process. All on-site repairs are made by replacing ORUs. ORUs themselves are repaired by Honeywell.

8.4 Built-in Maintenance Aids

Built-in self-tests	Every module on the LCN has three levels of built-in self-tests: • Startup Self-Tests — These test programs are stored in read-only memory on each board that has a microprocessor. They run after power is applied, or when the module containing the board is restarted. These tests verify the correct operation of the basic logic of each board and of the module. The self-test on the module's microprocessor board tests the memory boards and the module bus. Any detected failure lights up an LED on the failed board(s) and can also be detected over the LCN by other modules. • Quality Logic Tests — QLTs are automatically loaded and executed after the startup self-tests. QLTs verify the correct operation of the module hardware and qualifies it for loading of its on-process software. • On-Process Tests — These tests are part of the on-process software of each module and execute whether a module is primary or back-up. They run periodically to test the hardware and integrity features. After recovery from an error, these tests record the error in the system error journal for subsequent analysis. A nonrecoverable failure can cause the module to be shut down. The failure and any maintenance recommendation are reported to the operator and are recorded in the error and maintenance journals.
On-Process Analysis function	All errors and maintenance recommendations generated by the built-in tests are recorded on a system printer for later reference by maintenance personnel. Additionally, this information is accumulated by a History Module for use of the On-Process Analysis function. This History Module-resident function's primary role is the collation of error information by error type for each module, in order to perform predictive failure analysis and the issuing of maintenance recommendations. On-Process Analysis also provides a printed error history that can be returned to the factory with the ORU being replaced.
Universal Station (or UWS) maintenance software	A Universal Station's (or Universal Work Station's) maintenance capability is used when the automatic tests cannot resolve a problem. The maintenance technician communicates with the station through interactive displays and the Operator's (or Supervisor's) Keyboard. The maintenance software of the Universal Station provides these maintenance-related functions: • Interacts with the system maintenance journal; displays information in that journal, including active maintenance recommendations; and accepts maintenance activity entries.

- Reads and displays the hardware, firmware, and software **revision status** of modules on the LCN.
- Displays a snapshot of **system status**, including the status of modules on the LCN, UCNs, Data Hiways, and process-connected devices.
- Displays the **content of memory** locations in modules on the LCN and process-connected devices.
- Displays **error history** from pools kept by the modules or from accumulations kept in an assigned History Module.

**Universal Station
on-process
software**

The Operator software plays a significant role in maintenance by:

- Reporting entries in the system error journal to the operator.
- Inquiring into and displaying the status of modules, as requested.
- Stopping modules from communicating on the LCN when they have failed.
- Displaying test results.

System Maintenance

8.4 Built-in Maintenance Aids

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